Nilpotent orbits

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Chapter 1

Lie Algebras

1.1 Facts from linear algebra

1.1.1 Jordan Normal form

Here will denote $X : V \to V$ an endomorphism.

Some important facts:

- Let $f, g \in \mathbb{C}[t]$ be such that $f \cdot g(X) = 0, \gcd(f, g) = 1$. Then

$$V = \mathrm{Ker} f(X) \oplus \mathrm{Ker} g(X). \tag{1.1}$$

- Let p be the characteristic polynomial of X and

$$p(t) = \prod_{i=1}^{r} (t - a_i)^{m_i} \tag{1.2}$$

be its factorization. Then

$$V = \bigoplus_{i=1}^{r} \mathrm{Ker}(X - a_i \mathbb{1}_V)^{m_i}. \tag{1.3}$$

- $v \in V \setminus 0$ is said to be $(X - a\mathbb{1})-$cyclic with period r if $(X - a\mathbb{1})^{r-1} v \neq 0$ while $(X - a\mathbb{1})^r v = 0$. In this case the set

$$v,\ (X - a\mathbb{1}_V)v, \ldots,\ (X - a\mathbb{1}_V)^{r-1} v \tag{1.4}$$

is linearly independent.

Theorem 1.1.1. (Jordan normal form). V is the direct sum of X-cyclic subspaces, i.e., A-invariant subspaces with a basis of the form

$$v,\ (X - a\mathbb{1}_V)v, \ldots,\ (X - a\mathbb{1}_V)^{r-1}v \tag{1.5}$$

for some $v \in V \setminus 0, a \in \mathbb{C}, r \geq 1$.

Proof (sketch): Let p be the characteristic polynomial of X with unique factorization

$$p(t) = (t - a_1)^{m_1} \cdots (t - a_r)^{m_r}. \tag{1.6}$$

Then we know that

$$V = \bigoplus_{i=1}^{r} \mathrm{Ker}(X - a_i\mathbb{1}_V)^{m_i}. \tag{1.7}$$

Restriction to $V_i = \mathrm{Ker}(X - a_i\mathbb{1}_V)^{m_i}$ yields the additional hypothesis: We can assume that there exist $a \in \mathbb{C}$ and $m \geq 1$ such that

$$(X - a\mathbb{1}_V)^m = 0. \tag{1.8}$$

Now we solve the problem inductively:

- Write $B = X - a\mathbb{1}_V$ and let $m \geq 1$ be the smallest integer such that $B^m = 0$. Then there exist $v \in V \setminus 0$ such that $B^{m-1}v \in \mathrm{Ker}(B) \setminus 0$. Therefore

$$\dim(BV) < \dim(V). \tag{1.9}$$

- by induction, we have a decomposition

$$\begin{aligned} BV &= W_1 \oplus \cdots \oplus W_k, \\ W_i &= \langle w_i, \cdots B^{r_i - 1} w_i \rangle \end{aligned} \tag{1.10}$$

  into cyclic X-invariant subsapces.

- Let $v_i \in V$ be such that $Bv_i = w_i$, then

$$V_i = \langle v_i, \ldots, B^{r_i} v_i \rangle. \tag{1.11}$$

  Now we write $V' = V_1 \oplus \cdots \oplus V_k$ and complete it to a basis of V using elements of $\mathrm{Ker}(B)$.

□

Example 1.1.2. Find the Jordan Normal form of the matrix:

$$X = \begin{pmatrix} 1 & 2 & -1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix} \tag{1.12}$$

Remark 1.1.3. Let $A \in M_n(\mathbb{C})$ and $p \in \mathbb{C}[t]$ its associated characteristic polynomial. If

$$p(t) = \prod_{i=1}^{r} (t - \lambda_i)^{m_i},\ \lambda_i \neq \lambda_j \tag{1.13}$$

then the number m_i is called *algebraic multiplicity* of λ_i. It has the property

$$m_i = \dim \operatorname{Ker}(A - \lambda_i I)^{m_i}. \tag{1.14}$$

We have a general property: $\operatorname{Ker}(A - \lambda_i I)^k \overset{\neq}{\subset} \operatorname{Ker}(A - \lambda_i I)^{k+1}$ for $k = 1, \ldots, m_i - 1$ while $\operatorname{Ker}(A - \lambda_i I)^N = \operatorname{Ker}(A - \lambda_i I)^{m_i},\ N \geq m_i$.

Definition 1.1.4. If $X \in \operatorname{End}(V)$, then we define the *generalized eigenspace*

$$V_\lambda^X = \{v \in V : (X - \lambda \mathbb{1}_V)^N v = 0, \text{ for some } N \geq 1\}. \tag{1.15}$$

Of course, if $V_\lambda^X \neq 0$, then λ is an eigenvector of X and by the previous discussion

$$V_\lambda^X = \operatorname{Ker}(X - \lambda \mathbb{1}_V)^m \tag{1.16}$$

where m is the algebraic multiplicity of λ.

1.1.2 Jordan Chevalley decomposition

Remark 1.1.5. A endomorphism X is semisimple if for any $X-$invariant subspace W, we can find a X-invariant complement W'. So we have the decomposition into X-invariant subspaces:

$$V = W \oplus W'. \tag{1.17}$$

Since we are working in $\mathbb{C}$, semisimplicity is equivalent to X being diagonalizable. In fact, if X is semisimple, let $v \in V \setminus 0$ be an eigenvalue. Then $\mathbb{C}v$ is X-invariant and there exist a X-invariant subspace X such that $V = \mathbb{C}v \oplus W$. Now, induction finnnish the job. Conversely, if X is diagonalizable, let $v_1, \ldots, v_n,\ n = \dim(V)$ be a basis consisting of eigenvectors of X. Since $V = W + \langle v_1, \ldots, v_n \rangle$, we can find $v_{i_1}, \ldots, v_{i_s}$ such that $V = W \oplus \langle v_{i_1}, \ldots, v_{i_s} \rangle$.

Remark 1.1.6. The chinese remainder theorem, in the case of polynomials is the statement that if $f = g_1 \cdots g_r$, $\gcd(g_1, \ldots, g_r) = 1$, then

$$h + (f) \xrightarrow{\cong} (h + (g_1), \ldots, h + (g_r)) \tag{1.18}$$

is an isomorphism.

By the previous remark, solving a system of congruences:

$$\begin{aligned} f &= a_1 \bmod g_1 \\ &\cdots \\ f &= a_r \bmod g_r \end{aligned} \tag{1.19}$$

is just a linear algebra problem.

Example 1.1.7. Find a solution to:

$$\begin{aligned} f &= 2t + 1 \bmod t^2 + 1 \\ f &= 4 \bmod t - 1 \end{aligned} \tag{1.20}$$

First we see that $\gcd(t^2 + 1, t - 1) = 1$ and if $f = (t^2 + 1)(t - 1)$, then

$$T : h + (f) \mapsto (h + (t^2 + 1), h + (t - 1)) \tag{1.21}$$

is an isomorphism. We have the basis $1+(f), t+(f), t^2+(f)$ of $\mathbb{C}[t]/(f)$, $1+(t^2+1), t+(t^2+1)$ of $\mathbb{C}[t]/(t^2 + 1)$ and $1 + (t - 1)$ of $\mathbb{C}[t]/(t - 1)$. With respect to these basis, T has matrix expression

$$X = \begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & 0 \\ 1 & 1 & 1 \end{pmatrix}. \tag{1.22}$$

Now, to solve the system of congruences, we solve the linear system $Xv = (1, 2, 4)^T$ which gives $v = (3/2, 2, 1/2)$, or

$$f = \frac{1}{2}t^2 + 2t + \frac{3}{2}. \tag{1.23}$$

Theorem 1.1.8. (Jordan - Chevalley decomposition). Let $X \in \mathrm{End}(V)$.

(1) We have a (unique) decomposition $X = X_s + X_n$ into a semisimple and a nilpotent element such that $[X_s, X_n] = 0$.

(2) Both X_s, X_n are polynomials in X without constant terms.

Proof: As before, if p is the characteristic polynomial of X with factorization $p = \prod_{i=1}^{r}(t - a_i\mathbb{1})^{m_i}$. Then $V = \bigoplus_{i=1}^{r} V_i$, $V_i = \mathrm{Ker}(X - a_i\mathbb{1})^{m_i}$. Now we solve a system of congruences:

- If exist $a_i = 0$:

$$\begin{aligned} f &= a_1 \bmod (t - a_1)^{m_1} \\ &\cdots \\ f &= a_r \bmod (t - a_r)^{m_r} \end{aligned} \tag{1.24}$$

- If $a_i \neq 0$, $\forall i$, then we add the equation $f = 0 \bmod t$ to the previous system.

Now, it's clear that $f(X)|_{V_i} = a_i\mathbb{1}_{V_i}$, and $f(X)$ is diagonalizable. Furthermore, if $g(t) = t - f(t)$, then $g(X)|_{V_i} = X - a_i\mathbb{1}\big|_{V_i}$ which is nilpotent with index m_i. Taking $m = m_1 \cdots m_r$ we see that $g(X)^m = 0$, so $g(X)$ is a nilpotent operator. Now we just define $X_s = f(X)$ and $X_n = g(X)$.

Lemma 1.1.9. Let $X \in \mathrm{End}(V)$, then $\mathrm{ad}(X_s) = \mathrm{ad}(X)_s$ and $\mathrm{ad}(X_n) = \mathrm{ad}(X)_n$.

Prood: Recall that $\mathrm{ad}(X) = L_X - R_X$, where $L_X(Y) = XY, R_X(Y) = YX$. It's clear that L_X, R_X commute and are nilpotent if X is nilpotent. Therefore $\mathrm{ad}(X)$ is nilpotent if X is nilpotent. Furthermore, if X is semisimple, choose an eigenbasis $v_1, \ldots, v_n$ and let $E_{ij} : v_k \mapsto v_{jk}v_i$. Then we see that

$$\mathrm{ad}(X)E_{ij}v_k = \delta_{kj}a_iv_i - a_k\delta_{kj}v_i = (a_i - a_j)E_{ij}v_k. \tag{1.25}$$

So $\mathrm{ad}(X)E_{ij} = (a_i - a_j)E_{ij}$ and $\mathrm{ad}(X)$ is diagonalizable. By uniqueness of Chevalley decomposition we see prove the statement. □

1.2 Engel and Lie theorems

Theorem 1.2.1. (Engel's theorem). If $\mathfrak{g} \subset \mathrm{End}(V)$ consisting of nilpotent endomorphisms, then there exist $v \in V \setminus 0$ such that $Xv = 0$, $\forall X \in \mathfrak{g}$. Furthermore, this means that we can find a basis such that

$$\mathfrak{g} \subset \mathfrak{t}_n = \left\{ \begin{pmatrix} 0 & * \\ 0 & 0 \end{pmatrix} \right\} \tag{1.26}$$

Proof: The strategy of the proof is the following:

- Apply induction on dimension of $\mathfrak{g}$ to show that there exist a codimension 1 ideal $\mathfrak{h} \subset \mathfrak{g}$.

- Apply induction of dimension of $\mathfrak{h}$ to conclude that $W = \{v \in V : Xv = 0,\ \forall X \in \mathfrak{h}\} \neq 0$. In this case, any $Y \in \mathfrak{g}$ outside $\mathfrak{h}$ restricts to W and we finnish the proof using the fact that Y is nilpotent.

The candidate for a codimension 1 ideal in $\mathfrak{g}$ is any maximal (proper) subalgebra $\mathfrak{h} \subset \mathfrak{g}$. Now, $\mathfrak{g}/\mathfrak{h}$ is a vector space with the adjoint action of $\mathfrak{h}$. In particular $\mathrm{ad}(X)$ acts nilpotently as an element in $\mathrm{End}(\mathfrak{g}/\mathfrak{h})$ and by the induction hypothesis there exist $Y \in \mathfrak{g}$ such that $\mathrm{ad}(\mathfrak{h})Y \subset \mathbb{C}Y$. Therefore $\mathfrak{h} \oplus \mathbb{C}Y \subset \mathfrak{g}$ is a subalgebra. By the maximality assumption, we have $\mathfrak{g} = \mathfrak{h} \oplus \mathbb{C}Y$.

Now, let $W = \{v \in V : Xv = 0,\ \forall X \in \mathfrak{h}\}$ which is nonzero if we apply the induction hypothesis again for $\dim(\mathfrak{h})$. So, to finnish the proof we just need $v \in W$, $Yv = 0$. Let $v \in V$, then

$$XY(v) = YX(v) + [X,Y](v) \overset{[X,Y]\in\mathfrak{h}}{=} YX(v) \overset{X\in\mathfrak{h}}{=} 0,\ \forall X \in \mathfrak{h}. \tag{1.27}$$

Therefore $Y(v) \in W$, i.e., $Y|_W \in \mathrm{End}(W)$ is a nilpotent endomorphism. So we can find $w \in W \setminus 0$ such that $Y(w) = 0$. □

Remark 1.2.2. By just reading the definition, the condition of $\mathfrak{g}$ being nilpotent looks stronger than the condition of each $\mathrm{ad}(X)$, $X \in \mathfrak{g}$ being a nilpotent endomorphism of $\mathfrak{g}$. However, by Engel's theorem, the two conditions are the same. In fact, if every $\mathrm{ad}(X)$ is nilpotent, then $\mathrm{ad}(\mathfrak{g})$ is a nilpotent subalgebra of $\mathfrak{gl}_{\mathfrak{g}}$ and by Engel's theorem $\mathrm{ad}(\mathfrak{g}) \subset \mathfrak{n}_n$ for some choice of basis of $\mathfrak{g}$. This says that $\mathrm{ad}(\mathfrak{g})$ is nilpotent and thus $\mathfrak{g}/Z(\mathfrak{g})$ is nilpotent, which implies $\mathfrak{g}$ nilpotent.

Remark 1.2.3. (Be careful !) If $\mathfrak{g} \subset \mathrm{End}(V)$ we is a nilpotent Lie subalgebra, then we can not immediately apply Engel's theorem, because we need that $\mathfrak{g}$ be a subalgebra consisting of nilpotent endomorphisms of V.

Lemma 1.2.4. (Weight spaces are invariant). Let $\mathfrak{h} \subset \mathfrak{g}$ an ideal, V a representation of $\mathfrak{g}$ and $\lambda \in \mathfrak{h}^*$. Then $V_{\mathfrak{h}}^{\lambda} = \{v \in V : X(v) = \lambda(X)v,\ \forall X \in \mathfrak{h}\}$ is $\mathfrak{g}-$invariant.

Proof: Let $w \in W = V_{\mathfrak{h}}^{\lambda}, Y \in \mathfrak{g}$ and define $U = \langle w, Y(w), Y^2(w), \dots \rangle$. This is clearly a $Y-$invariant subspace. We also claim that this is a $\mathfrak{h}-$invariant subspace. In fact, for $X \in \mathfrak{h}$ we have $Xw = \lambda(X)w$,

$$XY(w) = YX(w) + [X,Y](w) \overset{[X,Y]\in\mathfrak{h}}{=} \lambda(X)Y(w) + \lambda([X,Y])w \in U \tag{1.28}$$

and in general, $XY^k(w) \in U$ can be proved by induction. In particular we see that $X|_W$ is triangular, with trace $\lambda(X)\dim(W)$. Finnaly, with this observation we see that

$$\lambda([X,Y])\dim(W) = \mathrm{Tr}([X,Y]|_W) = 0 \Rightarrow \lambda([X,Y]) = 0. \tag{1.29}$$

Therefore, we see that $XY(w) = \lambda(X)Y(w)$, i.e., $Y(w) \in W$. □

Theorem 1.2.5. (Lie theorem). If $\mathfrak{g} \subset \mathrm{End}(V)$ is a solvable Lie subalgebra, then there exist $v \in V \setminus 0$ fixed by $\mathfrak{g}$. This means that we can find a basis of V such that

$$\mathfrak{g} \subset \mathfrak{b}_n = \left\{ \begin{pmatrix} * & * \\ 0 & * \end{pmatrix} \right\}. \tag{1.30}$$

Proof: The strategy here is the following:

- Apply induction on $\dim(\mathfrak{g}/[\mathfrak{g},\mathfrak{g}])$ to conclude that there exist a codimension 1 ideal in $\mathfrak{g}$.
- Apply induction again to to find a nonzero vector v fixed by $\mathfrak{h}$ and finnish the proof.

Since $\mathfrak{g}$ is solvable, $[\mathfrak{g},\mathfrak{g}]$ is proper ideal in $\mathfrak{g}$. So $\mathfrak{g}/[\mathfrak{g},\mathfrak{g}]$ is a nonzero abelian Lie algebra. By induction, there exist a codimension 1 ideal $\mathfrak{a} \subset \mathfrak{g}/[\mathfrak{g},\mathfrak{g}]$. If $\pi^{-1}(\mathfrak{a}) = \mathfrak{h}$, then $\mathfrak{h}/[\mathfrak{g},\mathfrak{g}] = \mathfrak{a}$ and

$$\dim(\frac{\mathfrak{g}/[\mathfrak{g},\mathfrak{g}]}{\mathfrak{a}}) = \dim(\frac{\mathfrak{g}/[\mathfrak{g},\mathfrak{g}]}{\mathfrak{h}/[\mathfrak{g},\mathfrak{g}]}) = \dim(\mathfrak{g}/\mathfrak{h}). \tag{1.31}$$

Therefore, $\mathfrak{h}$ is a codimension 1 ideal in $\mathfrak{g}$. Let $Y \in \mathfrak{g}$ be such that $\mathfrak{g} = \mathfrak{h} \oplus \mathbb{C}Y$. By the induction hypothesis, there exist $v \in V \setminus 0$ fixed by $\mathfrak{h}$, i.e., $W = V_{\mathfrak{h}}^{\lambda} \neq 0$ for some $\lambda \in \mathfrak{h}^*$. Finnaly, by the previous lemma $Y|_W \in \mathrm{End}(W)$, so we just take w an eigenvector of $Y|_W$. □

Remark 1.2.6. Lie's theorem is easier to apply than Engel's theorem because the condition lies entirely the Lie subalgebra itsel (i.e., its solvability condition), not in its action on V.

1.3 Semisimplicity criterions

The two main theorems of this section is the Cartan criteria for semisimple and solvable Lie algebras. All necessary definitions follows after the statement of the theorem.

Theorem 1.3.1. (CC(ss)). Cartan criteria for semisimple Lie algebras: A Lie algebra $\mathfrak{g}$ is semisimple if and only if the Killing form $K^{\mathfrak{g}}$ is nondegenerate.

Theorem 1.3.2. (CC(sol)). Cartan criteria for solvable Lie algebras: A Lie algebra $\mathfrak{g}$ is solvable if and only if $K^{\mathfrak{g}}(\mathfrak{g},[\mathfrak{g},\mathfrak{g}]) = 0$.

Some key definitions that we need:

- Central series of a Lie algebra: Is the decreasing sequence of ideals $\mathfrak{g}^k$ defined by

$$\begin{cases} \mathfrak{g}^0 = \mathfrak{g}, \\ \mathfrak{g}^k = [\mathfrak{g}, \mathfrak{g}^{k-1}], k \geq 1. \end{cases} \tag{1.32}$$

- Derived series of a Lie algebra: Is the decreasing sequence of ideals $\mathfrak{g}^{(k)}$ defined by

$$\begin{cases} \mathfrak{g}^{(0)} = \mathfrak{g}, \\ \mathfrak{g}^{(k)} = [\mathfrak{g}^{(k-1)}, \mathfrak{g}^{(k-1)}], k \geq 1. \end{cases} \tag{1.33}$$

- A Lie algebra is said to be *nilpotent* if its central series

$$\mathfrak{g} \supset \mathfrak{g}^1 \supset \cdots \supset \mathfrak{g}^k = 0 \tag{1.34}$$

terminates at 0.

Remark 1.3.3. An equivalent definition of nilpotent Lie algebras is the following one: There exist a chain of ideals

$$\mathfrak{g}_0 = \mathfrak{g} \supset \mathfrak{g}_1 \supset \cdots \supset \mathfrak{g}_k = 0 \tag{1.35}$$

such that $\mathfrak{g}_i/\mathfrak{g}_{i+1} \in Z(\mathfrak{g}/\mathfrak{g}_{i+1})$.

In fact, if it's the case, then we claim that $\mathfrak{g}^i \subset \mathfrak{g}_i,\ i \geq 0$. For $i = 0$ it's just the definition. Now, in general

$$\mathfrak{g}^{i+1} = [\mathfrak{g}, \mathfrak{g}^i] \overset{\mathfrak{g}^i \subset \mathfrak{g}_i}{\subset} [\mathfrak{g}, \mathfrak{g}_i] \overset{\mathfrak{g}_i/\mathfrak{g}_{i+1} \in Z(\mathfrak{g}/\mathfrak{g}_{i+1})}{\subset} \mathfrak{g}_i \tag{1.36}$$

which shows that the central series terminates in zero.

- A Lie algebra is said to be *solvable* if its derived series

$$\mathfrak{g} \supset \mathfrak{g}^{(1)} \supset \cdots \supset \mathfrak{g}^{(k)} = 0 \tag{1.37}$$

terminates at 0.

Remark 1.3.4. An equivalent definition for solvable Lie algebras: There exist a sequence of *Lie subalgebras*

$$\mathfrak{g}_0 = \mathfrak{g} \supset \mathfrak{g}_1 \supset \cdots \supset \mathfrak{g}_k = 0 \tag{1.38}$$

such that $\mathfrak{g}_{i+1} \subset \mathfrak{g}_i$ is an ideal and $\mathfrak{g}_i/\mathfrak{g}_{i+1}$ is abelian.

In fact, since $\mathfrak{g}_i/\mathfrak{g}_{i+1}$ is abelian $[\mathfrak{g}_i, \mathfrak{g}_i] \subset \mathfrak{g}_{i+1}$. Therefore we can use an induction to prove that $\mathfrak{g}^{(i)} \subset \mathfrak{g}_i$.

- A Lie algebra $\mathfrak{g}$ is semisimple if it has no nonzero solvable ideal.
- A Lie algebra $\mathfrak{g}$ is simple, if it is nonabelian and it has no nonzero proper ideal.

Remark 1.3.5. It's important to note, that the definition of semisimple Lie algebras automaticaly exclude the case of abelian Lie algebras. However, for the definition of simple Lie algebras, the property "has no proper nonzero ideals" may include the Lie algebra $\mathbb{C}$, which is abelian. So, to make the definition consistent, we need to impose it to be nonabelian.

An important observation is that $\mathfrak{g}^k \subset \mathfrak{g}^{(k)}$, which can be proved by induction. The base case is just the definition, and if $\mathfrak{g}^{k-1} \subset \mathfrak{g}^{(k-1)}$, then

$$\mathfrak{g}^{(k)} = [\mathfrak{g}^{(k-1)}, \mathfrak{g}^{(k-1)}] \subset [\mathfrak{g}, \mathfrak{g}^{(k-1)}] \subset [\mathfrak{g}, \mathfrak{g}^{(k-1)}] = \mathfrak{g}^k. \tag{1.39}$$

Example 1.3.6. Let $H = \mathbb{C}p \oplus \mathbb{C}q \oplus \mathbb{C}c$ with commutation relations

$$[p, q] = c,\ [c, -] = 0. \tag{1.40}$$

Then H is a nilpotent Lie algebra. In fact, its central series is just $H \supset \mathbb{C}c \supset 0$.

Example 1.3.7. Let $\mathfrak{g} = \mathbb{C}a \oplus \mathbb{C}b$ with commutation relation $[a, b] = b$. It's solvable Lie algebra that is not nilpotent. In fact its derived series is:

$$\mathfrak{g} \supset \mathbb{C}b \supset 0. \tag{1.41}$$

However, its central series is $\mathfrak{g} \supset \mathbb{C}b \supset \mathbb{C}b \cdots$, which never terminates.

Lemma 1.3.8. (Facts about solvable ideals). Let $\mathfrak{g}$ be a solvable Lie algebra. We have:

(1) If $\mathfrak{h} \subset \mathfrak{g}$ is a subalgebra, then $\mathfrak{h}$ is solvable.

(2) If $f : \mathfrak{g} \to \mathfrak{h}$ is a homomorphism, then $f(\mathfrak{g})$ is solvable.

(3) If $f : \mathfrak{h} \to \mathfrak{g}$ is a monomorphism, then $f^{-1}(\mathfrak{g})$ is solvable.

(4) Now, in general we have

$$\mathfrak{g} \text{ solvable} \iff \mathfrak{h}, \mathfrak{g}/\mathfrak{h} \text{ are solvable} \tag{1.42}$$

provided $\mathfrak{h} \subset \mathfrak{g}$ is an ideal.

Proof: (1) follows from the fact that $\mathfrak{h}^{(k)} \subset \mathfrak{g}^{(k)}$. (2) and (3) follows from the identities

$$\begin{aligned} &f([I,I]) = [f(I), f(I)], \ (I \subset \mathfrak{h} \text{ and ideal}) \\ &f^{-1}([I,I]) \supset [f^{-1}(I), f^{-1}(I)], \ (I \subset \mathfrak{g} \text{ an ideal}). \end{aligned} \tag{1.43}$$

So, if I/J is an abelian Lie algebra, i.e., $[I,I] \subset J$, we get

$$[f^{-1}(I), f^{-1}(I)] \subset f^{-1}([I,I]) \subset f^{-1}(J) \tag{1.44}$$

and $f^{-1}(I)/f^{-1}(J)$ is abelian. So, if $\mathfrak{g}$ has a decreasing sequence

$$\mathfrak{g}_0 = \mathfrak{g} \supset \mathfrak{g}_1 \supset \cdots \supset \mathfrak{g}_k = 0 \tag{1.45}$$

such that $\mathfrak{g}_{i+1}/\mathfrak{g}_i, i \geq 0$ is abelian, then $f^{-1}(\mathfrak{g}_i)$ is a decreasing sequence in $\mathfrak{h}$ satisfying the same conditions and terminating in zero. For (4) we use the same reasoning applying to $\pi : \mathfrak{g} \to \mathfrak{g}/\mathfrak{h}$ to get a decreasing sequence

$$\mathfrak{g}_0 = \mathfrak{g} \supset \mathfrak{g}_1 \supset \cdots \supset \mathfrak{g}_k = \mathrm{Ker}(\pi) = \mathfrak{h}. \tag{1.46}$$

By using the fact that $\mathfrak{h}$ is solvable we finnish the proof. □

Corollary 1.3.9. Let $I, J \subset \mathfrak{g}$ be two solvable ideals, then $I + J$ is solvable.

Proof: We have the isomorphism:

$$\frac{I+J}{J} = \frac{I}{I \cap J}. \tag{1.47}$$

The right hand side of the previous equation is the quotient of two solvable ideals, so it's solvable. Therefore, since J is solvable, the item (4) of the previous lemma yields the result. □

Definition 1.3.10. (Radical of a Lie algebra). Since the sum of two solvable ideals is solvable, we can define the following solvable ideal of a (finite dimensional) Lie algebra $\mathfrak{g}$:

$$\mathrm{Rad}(\mathfrak{g}) = \sum_{I \subset \mathfrak{g} \text{ solvable}} I \tag{1.48}$$

which is called the radical of a Lie algebra.

Lemma 1.3.11. (ab ideal criteria for ss Lie algebras). A Lie algebra $\mathfrak{g}$ is semisimple if it has no nonzero abelian ideals.

Proof: The proof use negation of two the two sides. If $\mathfrak{g}$ has a nonzero abelian ideal, the it has a nonzero solvable ideal, so it's not semisimple. Conversely, if $\mathfrak{g}$ is not semisimple, it has a nonzero solvable ideal $\mathfrak{h}$. Then, there exist a smallest integer $k \geq 1$ such that $\mathfrak{h}^{(k)} \neq 0$. In this case

$$\mathfrak{h}^{k+1} = [\mathfrak{h}^{(k)}, \mathfrak{h}^{(k)}] = 0, \tag{1.49}$$

so $\mathfrak{h}^{(k)}$ is abelian. □

Lemma 1.3.12. (Killing form restricts to ideals). Let $\mathfrak{h} \subset \mathfrak{g}$ be an ideal. Then

$$K^{\mathfrak{h}} = K^{\mathfrak{g}}\big|_{\mathfrak{h}\times\mathfrak{h}}. \tag{1.50}$$

Proof: Let $\{v_1, \ldots, v_k, v_{k+1}, \ldots, v_n\}$ be a basis of $\mathfrak{g}$ such that $\mathfrak{h} = \langle v_1, \ldots, v_k \rangle$. If $h_i \in \mathfrak{h},\ i = 1, 2$, then we have matrix expressions

$$\mathrm{ad}(h_i) = \begin{pmatrix} H_i & A_i \\ 0 & 0 \end{pmatrix},\ H_i \in M_k(\mathbb{C}),\ A_i \in M_{k,n-k}(\mathbb{C}). \tag{1.51}$$

Therefore

$$\begin{aligned} K^{\mathfrak{g}}(h_1, h_2) &= \mathrm{Tr}_{\mathfrak{g}}(\mathrm{ad}(h_1) \circ \mathrm{ad}(h_2)) \\ &= \mathrm{Tr}_{\mathfrak{g}} \begin{pmatrix} H_1 & A_1 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} H_2 & A_2 \\ 0 & 0 \end{pmatrix} \\ &= \mathrm{Tr}_{\mathfrak{g}} \begin{pmatrix} H_1 H_2 & H_1 A_2 \\ 0 & 0 \end{pmatrix} \\ &= \mathrm{Tr}_{\mathfrak{h}}(H_1 H_2) \\ &= \mathrm{Tr}_{\mathfrak{h}}(\mathrm{ad}(h_1) \circ \mathrm{ad}(h_2)) \\ &= K^{\mathfrak{h}}(h_1, h_2). \end{aligned} \tag{1.52}$$

□

Next, we will use the CC(sol) to prove CC(ss).

Proof of theorem (1.3.1): Assume $K^{\mathfrak{g}}$ is nondegenerate and $\mathfrak{h}$ is an abelian ideal. As in the proof of the previus lemma, write $\mathfrak{h} = \langle v_1, \dots, v_k \rangle$ and $\mathfrak{g} = \langle v_1, \dots, v_k, v_{k+1}, \dots, v_n \rangle$. Let $X \in \mathfrak{h}$ and $Y \in \mathfrak{g}$, then with respect to this basis we have

$$\mathrm{ad}(X) = \begin{pmatrix} 0 & * \\ 0 & 0 \end{pmatrix}, \ \mathrm{ad}(Y) = \begin{pmatrix} * & * \\ 0 & * \end{pmatrix}. \tag{1.53}$$

Therefore $K^{\mathfrak{g}}(X,Y) = \mathrm{Tr}(\mathrm{ad}(X) \circ \mathrm{ad}(Y)) = \mathrm{Tr} \begin{pmatrix} 0 & * \\ 0 & 0 \end{pmatrix} = 0$ and $X = 0$ by nondegeneracy of the Killin form.

Conversely, if $\mathfrak{g}$ is semisimple, we have $\mathfrak{g}^{\perp}$ an ideal such that

$$0 = K^{\mathfrak{g}}(X, [Y,Z]) = K^{\mathfrak{g}^{\perp}}(X, [Y,Z]), \ \forall X, Y, Z \in \mathfrak{g}^{\perp}. \tag{1.54}$$

Therefore $\mathfrak{g}^{\perp}$ is solvable by CC(sol), so $\mathfrak{g}^{\perp} = 0$ which means that $K^{\mathfrak{g}}$ is nondegenerate.

Lemma 1.3.13. (Orthogonal decomposition by a ss ideal). Let $\mathfrak{h} \subset \mathfrak{g}$ be an ideal. If $(\cdot,\cdot) : \mathfrak{g} \times \mathfrak{g} \to \mathbb{C}$ is an invariant, symmetric bilinear form and $\mathfrak{h}^{\perp} = \{X \in \mathfrak{g} : (X,Y) = 0, \ \forall Y \in \mathfrak{h}\}$. Then, if $(\cdot,\cdot)\big|_{\mathfrak{h}\times\mathfrak{h}}$ is nondegenerate, we have

$$\mathfrak{g} = \mathfrak{h} \oplus \mathfrak{h}^{\perp}. \tag{1.55}$$

Proof: We claim that $\mathfrak{h}^{\perp}$ is an ideal. In fact, let $X \in \mathfrak{h}^{\perp}, Y \in \mathfrak{g}, Z \in \mathfrak{h}$, then

$$([X,Y], Z) = (X, [Y,Z]) \overset{[Y,Z]\in\mathfrak{h}}{=} 0. \tag{1.56}$$

Now, since $(\cdot,\cdot)\big|_{\mathfrak{h}\times\mathfrak{h}}$ is nondegenerate, we have an induced isomorphism:

$$\rho : \mathfrak{h} \overset{\cong}{\to} \mathfrak{h}^* : X \mapsto (\rho(X) : Y \mapsto (X,Y)). \tag{1.57}$$

Therefore, for any $X \in \mathfrak{g}$, there exist $Y \in \mathfrak{h}$ such that $\rho(Y) = (X,\cdot)|_{\mathfrak{h}}$. In particular, this means: $(X - Y, \cdot)|_{\mathfrak{h}} = 0$, i.e., $X - Y \in \mathfrak{h}^{\perp}$. This proves that $\mathfrak{g} = \mathfrak{h} + \mathfrak{h}^{\perp}$.

Finally, nondegeneracy of $(\cdot,\cdot)|_{\mathfrak{h}\times\mathfrak{h}}$ implies $\mathfrak{h} \cap \mathfrak{h}^{\perp} = 0$. □

Theorem 1.3.14. A Lie algebra $\mathfrak{g}$ is semisimple if and only if it is a direct sum of simple Lie algebras:

$$\mathfrak{g} = \mathfrak{g}_1 \oplus \cdots \oplus \mathfrak{g}_k. \tag{1.58}$$

Proof: Assume $\mathfrak{g}$ not simple, so there exist a nonzero proper ideal $\mathfrak{h} \subset \mathfrak{g}$. We claim that $K^{\mathfrak{h}}$ is nondegenerate. In fact, if $X, Y, Z \in \mathfrak{a} = \mathfrak{h} \cap \mathfrak{h}^{\perp}$, then

$$K^{\mathfrak{a}}(X, [Y,Z]) = K^{\mathfrak{g}}(X, [Y,Z]) = 0. \tag{1.59}$$

So $\mathfrak{a}$ is solvable by CC(sol) and $\mathfrak{a} = 0$. Therefore $K^{\mathfrak{g}}\big|_{\mathfrak{h}\times\mathfrak{h}}$ is nondegenerate and $\mathfrak{h}$ is semisimple. By the previous lemma $\mathfrak{g} = \mathfrak{h} \oplus \mathfrak{h}^{\perp}$ and induction on dimension concludes this direction of the proof.

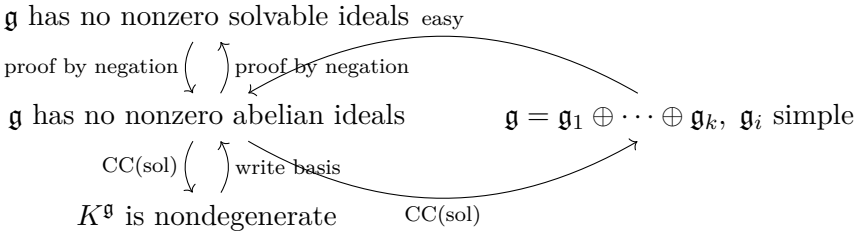
Conversely if $\mathfrak{g} = \mathfrak{g}_1 \oplus \cdots \oplus \mathfrak{g}_k$, $\mathfrak{g}_i$ simple. Then if $\mathfrak{a} \subset \mathfrak{g}$ is an ideal, we have a Lie algebra isomorphism

$$\bigoplus_j i_j^{-1}(\mathfrak{a}) \to \mathfrak{a} : (X_1, \ldots, X_k) \mapsto \sum_j i_j(X_j) \tag{1.60}$$

where $i_j : \mathfrak{g}_j \to \mathfrak{g} : X \mapsto (0, \ldots, \hat{X}, \ldots, 0)$, $\hat{} = j$th position, is the natural inclusion. Under this condition each $i_j^{-1}(\mathfrak{a})$ is an abelian ideal, so it must be zero. Therefore $\mathfrak{a} = 0$. □

Remark 1.3.15. It's a consequence of the previous proof that on a semisimple Lie algebra, the Killing form restricted to ideals is nondegenerate, i.e., any nonzero ideal is semisimple on a semisimple Lie algebra. The proof deppends strongly on CC(sol), which is a strong result.

A summary of this chapter:



To conclude anything for $\mathfrak{g}$ "having no nonzero abelian/solvable ideals" we need CC(sol).

Remark 1.3.16. From the previous result, we see that every semisimple Lie algebra $\mathfrak{g}$ is perfect, i.e., $[\mathfrak{g}, \mathfrak{g}] = \mathfrak{g}$. However there exist perfect Lie algebras that are not semisimple. For example, let $\mathfrak{h}$ be a Lie algebra and V a $\mathfrak{g}-$module. We construct a Lie algebra

$$\mathfrak{g} = \mathfrak{h} \oplus V,\ [(X, u), (Y, v)] = ([X, Y], Xv - Yu). \tag{1.61}$$

This Lie algebra is not semisimple, in fact $\mathrm{Rad}(\mathfrak{g}) = V$. However, if V is irreducible and $\mathfrak{h}$ is semisimple, then $\mathfrak{g}$ is perfect. In fact, for any $v \in V \setminus 0$, the subspace $\mathfrak{h}v = \{X_1 \cdots X_k(v) : X_i \in \mathfrak{h}, k \geq 1\}$ is $\mathfrak{h}-$invariant, so it's must be V. However

$$\begin{aligned} &[(X_1, 0), [(X_2, 0), \ldots, [(X_k, 0), (Z, u)] \ldots]] \\ &= ([X_1, [X_2, \ldots, [X_k, Z] \ldots]], X_1 X_2 \cdots X_k(u)\ ) \end{aligned} \tag{1.62}$$

If $\mathfrak{g} \subset \mathfrak{gl}_V$ is a Lie subalgebra, we have an useful criterion to decide if $\mathfrak{g}$ is semisimple.

Lemma 1.3.17. Let (π, V) the tautological representation of $\mathfrak{g}$ on V and assume:

1. $I_V \notin \mathfrak{g}$
2. π is irreducible

Then $\mathfrak{g}$ is semisimple.

Proof: Suppose that there exist a nonzero $\mathfrak{a} \subset \mathfrak{g}$ solvable ideal. Then $\pi(\mathfrak{a}) \subset \mathfrak{gl}_V$ is solvable and by Lie's theorem, there exist $v \in V \setminus 0$ such that $\pi(a)v = \lambda(a)v, \forall a \in \mathfrak{a}$. Therefore the weight space $V_\lambda^{\mathfrak{a}} \neq 0$. But since $\mathfrak{a}$ is an ideal, the space $V_\lambda^{\mathfrak{a}}$ is $\mathfrak{g}-$invariant. Since V is irreducle, we see that $V = V_\lambda^{\mathfrak{a}}$, which means that $\pi(a) = \lambda(a)I_V$. Since the tautological representation π is faithful, if $a \in \mathfrak{a} \setminus 0$, then $0 \neq \pi(a) = \lambda(a)I_v \in \mathfrak{g}$, i.e., $I_v \in \mathfrak{g}$ which is a contradiction. □

1.4 Cartan subalgebra and weight space decomposition

1.4.1 Generalized eigenspaces vs generalized weight spaces

Let $\mathfrak{g}$ a Lie algebra, $\mathfrak{h} \subset \mathfrak{g}$ an ideal and (π, V) a representation of $\mathfrak{g}$. We have the table of definitions:

Eigenspace E_λ^A	$v \in V : Av = \lambda v$
Weight space $V_\lambda^{\mathfrak{h}}$	$v \in V : \pi(X)v = \lambda(X)v,\ \forall X \in \mathfrak{h}$
(Generalized) Eigenspace E_λ^A	$v \in V : (A - \lambda \mathbb{1}_V)^N v = 0,\ N \gg 0$
(Generalized) Weight space $V_\lambda^{\mathfrak{h}}$	$v \in V : (\pi(X) - \lambda(X)\mathbb{1}_V)^N v = 0,\ \forall X \in \mathfrak{h},\ N \gg 0$

Lemma 1.4.1. The following relation

$$\pi(E_\alpha^{\mathrm{ad}(X)})E_\lambda^{\pi(X)} \subset E_{\alpha+\lambda}^{\pi(X)} \tag{1.63}$$

is satisfied for generalized eigenspaces.

Proof: Let $Y \in E_\alpha^{\mathrm{ad}(X)}$ and $v \in E_\lambda^{\pi(X)}$. So

$$\begin{aligned}(\pi(X)-\alpha-\lambda)^N\pi(Y)v &= (\mathrm{ad}(\pi(X))+R_{\pi(X)}-\alpha-\lambda)^N\pi(Y)v \\ &= \sum_{k=0}^{N}\binom{N}{k}(\mathrm{ad}(\pi(X))-\alpha)^{N-k}(R_{\pi(X)}-\lambda)^k\pi(Y)v \\ &= \sum_{k=0}^{N}\binom{N}{k}(\mathrm{ad}(\pi(X))-\alpha)^{N-k}\pi(Y)(\pi(X)-\lambda)^k v \\ &= \sum_{k=0}^{N}\binom{N}{k}\pi((\mathrm{ad}(X)-\alpha)^{N-k}Y)(\pi(X)-\lambda)^k v\end{aligned} \tag{1.64}$$

which is zero for $N \gg 0$. □

Lemma 1.4.2. Let $\mathfrak{g}_\alpha^{\mathfrak{h}}$ be the generalized weightspace for the representation $(\mathrm{ad}, \mathfrak{g})$ of $\mathfrak{h}$. Then if (π, V) is a representation of $\mathfrak{g}$, we have

$$\pi(\mathfrak{g}_\alpha^{\mathfrak{h}})V_\lambda^{\mathfrak{h}} \subset V_{\alpha+\lambda}^{\mathfrak{h}}. \tag{1.65}$$

Proof: If $\alpha, \lambda \in \mathfrak{h}^*$, we have the equalities:

$$\mathfrak{g}_\alpha^{\mathfrak{h}} = \bigcap_{X\in\mathfrak{h}} E_{\alpha(X)}^{\mathrm{ad}(X)},\ V_\lambda^{\mathfrak{h}} = \bigcap_{X\in\mathfrak{h}} E_{\lambda(X)}^{\pi(X)}. \tag{1.66}$$

So if $Y \in \mathfrak{g}_\alpha^{\mathfrak{h}}$ and $v \in V_\lambda^{\mathfrak{h}}$, then $Y \in E_{\alpha(X)}^{\mathrm{ad}(X)},\ \forall X \in \mathfrak{h}$ and $v \in E_{\lambda(X)}^{\pi(X)},\ \forall X \in \mathfrak{h}$. Therefore

$$\pi(Y)v \in \pi(E_{\alpha(X)}^{\mathrm{ad}(X)})E_{\lambda(X)}^{\pi(X)} \subset E_{\alpha(X)+\lambda(X)}^{\pi(X)},\ \forall X \in \mathfrak{h}. \tag{1.67}$$

□

1.4.2 Spectral decomposition for nilpotent subalgebras

Lemma 1.4.3. We have the following fundamental lemma:

(1) $\mathfrak{h}$ nilpotent Lie algebra and (π, V) a representation, then we have a decomposition into generalized weight spaces

$$V = \bigoplus_{\lambda \in \mathfrak{h}^*} V_\lambda^{\mathfrak{h}} \tag{1.68}$$

as a direct sum of representations of $\mathfrak{h}$.

(2) If $\mathfrak{h} \subset \mathfrak{g}$ is a nilpotent Lie subalgebra and (π, V) a representation of $\mathfrak{g}$. We have (π, V) and $(\mathrm{ad}, \mathfrak{g})$ as representations of $\mathfrak{h}$ with respective decompositions:

$$V = \bigoplus_{\lambda \in \mathfrak{h}^*} V_\lambda^{\mathfrak{h}}, \ \mathfrak{g} = \bigoplus_{\alpha \in \mathfrak{h}^*} \mathfrak{g}_\alpha^{\mathfrak{h}} \tag{1.69}$$

and $\pi(\mathfrak{g}_\alpha^{\mathfrak{h}}) V_\lambda^{\mathfrak{h}} \subset V_{\alpha+\lambda}^{\mathfrak{h}}$.

Proof: To prove (1) we divide into cases:

Case 1. For any $X \in \mathfrak{h}$, $\pi(X)$ has only one eigenvalue $\lambda(X)$. If such is the case, we can apply Lie theorem ($\mathfrak{h}$ is nilpotent and thus, solvable) to conclude that there exist $v \in V \setminus 0$ such that $\pi(X)v = \lambda(X)v, \forall X \in \mathfrak{h}$. So, by linear algebra we see that $V = E_{\lambda(X)}^{\pi(X)}$, $\forall X \in \mathfrak{h}$, so

$$V = \bigcap_{X \in \mathfrak{h}} E_{\lambda(X)}^{\pi(X)} = V_\lambda^{\mathfrak{h}}. \tag{1.70}$$

Case 2. There exist $X \in \mathfrak{h}$ such that $\pi(X)$ has at least two eigenvalues. If such is the case, we have $V = \bigoplus_{s \in \mathbb{C}} E_s^{\pi(X)}$ where any $E_s^{\pi(X)} \neq 0$ must have dimension smaller than $\dim(V)$. The result follows by induction on $\dim(V)$ provided we show that $E_s^{\pi(X)} \neq 0$ is $\mathfrak{h}-$invariant. However

$$\pi(E_0^{\mathrm{ad}(X)}) E_s^{\pi(X)} \subset E_s^{\pi(X)} \tag{1.71}$$

while $\mathfrak{h} = E_0^{\mathrm{ad}(X)}$, because $\mathfrak{h}$ is nilpotent.

Finally (2) is a direct consequence of (1) and the previous sections. □

Remark 1.4.4. The base case for the induction on case (2) is $V = \mathbb{C}v$. In this case, it's clear that $\pi(X)v = \lambda(X)v$ for some $\lambda \in \mathfrak{h}^*$, i.e., $V = V_\lambda^{\mathfrak{h}}$.

1.4.3 Regular elements of a Lie algebra

Definition 1.4.5. Let V be a vector space, a map $c : V \to V$ is said to be a polynomial in V if for some choice of basis $v_1, \ldots, v_n$, there exist a polynomial $p \in \mathbb{C}[t]$ such that

$$c(x_1 v_1 + \cdots + x_n v_n) = p(x_1, \ldots, x_n). \tag{1.72}$$

It's clear from the previous definition that if $c : V \to \mathbb{C}$ is a polynomial, then for any choice of basis, the corresponding map $(x_1, \ldots, x_n) \mapsto c(x_1 v_1 + \cdots + x_n v_n)$ is a polynomial. Furthermore, it's clear that the space of polynomials in V is a ring.

Lemma 1.4.6. Let $\mathfrak{g}$ be a Lie algebra and define $\chi_X(t) = \det(tI - \mathrm{ad}(X))$. Then

$$\chi_X(t) = t^d + c_{d-1}(X)t^{d-1} + \cdots + c_0(X) \tag{1.73}$$

where $c_i : \mathfrak{g} \to \mathbb{C}$ is a polynomial.

Proof: To prove this we just need to recall the formula

$$\det(t\mathbb{1}_V - A) = \sum_{k=0}^{n} (-1)^k \mathrm{Tr}(\bigwedge^k A) t^{n-k}. \tag{1.74}$$

To see this, use the Jordan decomposition $A = A_s + A_n$ with respecto to some basis $\{v_1, \ldots, v_n\}$ of V. Then

$$\det(t\mathbb{1}_V - A) = \det(t\mathbb{1}_V - A_s) = \prod_{j=1}^{n} (t - \lambda_j) = \sum_{k=0}^{n} (-1)^k e_k(\lambda_1, \ldots, \lambda_n) t^{n-k} \tag{1.75}$$

where $e_k(X_1, \ldots, X_n) = \sum_{1 \le i_1 < \cdots < i_k \le n} X_{i_1} \cdots X_{i_k}$. However, in this basis

$$\bigwedge^k A_s (v_{i_1} \wedge \cdots \wedge v_{i_k}) = \lambda_{i_1} \cdots \lambda_{i_k} (v_{i_1} \wedge \cdots \wedge v_{i_k}) \tag{1.76}$$

so $\mathrm{Tr}(\bigwedge^k A) = e_k(\lambda_1, \ldots, \lambda_n)$. To finnish the proof it's sufficient to check that $\mathrm{Tr}(\bigwedge^k A) = \mathrm{Tr}(\bigwedge^k A_s)$ which is a consequence of the fact that $\bigwedge^k A_s = (\bigwedge^k A)_s$.

Now we just apply this formula to see that

$$\chi_X(t) = \sum_{k=0}^{n} (-1)^k \mathrm{Tr}(\bigwedge^k \mathrm{ad}(X)) t^{n-k}. \tag{1.77}$$

Since $\mathrm{Tr}(\bigwedge^k \mathrm{ad}(X))$ is clearly a polynomial in X, we are done. □

Definition 1.4.7. The rank of a Lie algebra is defined to be the smallest $r \geq 0$ such that $c_r \neq 0$. In this case a regular element $X \in \mathfrak{g}$ is an element satisfying $c_r(X) \neq 0$.

This definition makes sense because if $c : V \to \mathbb{C}$ equal to 0 in some basis, then $c = 0$, i.e., being different from zero is a property that does not deppend on the choice of basis.

Example 1.4.8. $\mathfrak{sl}_2$ is a rank 1 Lie algebra. In fact, if

$$F = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, H = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \text{ and } E = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \tag{1.78}$$

then, for $X = \beta F + \alpha H + \gamma E$ we have

$$\begin{aligned} \chi_X(t) &= \det \begin{pmatrix} t + 2\alpha & -2\beta & 0 \\ -\gamma & 0 & \beta \\ 0 & 2\gamma & t - 2\alpha \end{pmatrix} \\ &= t^3 - 4(\alpha^2 - \beta\gamma)t \\ &= t^3 - 4\det(X)t. \end{aligned} \tag{1.79}$$

Example 1.4.9. $\mathfrak{gl}_n$ is a rank n Lie algebra. In fact if $A \in \mathfrak{gl}_n$ and $A = D + N$, $D = \mathrm{Diag}(\lambda_1, \ldots, \lambda_n)$ is the corresponding Jordan decomposition, then $\mathrm{ad}(D)E_{ij} = (\lambda_i - \lambda_j)E_{ij}$. In particular

$$\begin{aligned} \chi_A(t) = \chi_D(t) &= \prod_{i,j}(t - (\lambda_i - \lambda_j)) \\ &= t^n \prod_{i \neq j}(t - (\lambda_i - \lambda_j)). \end{aligned} \tag{1.80}$$

So $\mathrm{Rank}(\mathfrak{gl}_n) \geq n$. Furthermore, by selecting a matrix A with distinct eigenvalues, we get $c_n(A) = \prod_{i \neq j}(\lambda_i - \lambda_j)$, i.e., $\mathrm{Rank}(\mathfrak{gl}_n) = n$.

Lemma 1.4.10. Let $\mathfrak{g}$ be a rank r Lie algebra and $X \in \mathfrak{g}$. Then

$$\dim(\mathfrak{g}_0^X) \geq r \tag{1.81}$$

and the equality happens if and only if X is regular.

Proof: Let $\chi_X(t) = \prod_{i=1}^{s}(t - \lambda_i)^i$ be the characteristic polynomial of $\mathrm{ad}(X) \in \mathrm{End}(\mathfrak{g})$.

Then $\dim(E_{\lambda_i}^{\mathrm{ad}(X)}) = m_i$. Furthermore, by definition $\mathfrak{g}_0^X = E_0^{\mathrm{ad}(X)}$, so $\dim(\mathfrak{g}_0^X) = k$, the multiplicity of m_i, $\lambda_i = 0$, so $\chi_X(t) = t^k \prod_{\lambda_i \neq 0}(t - \lambda_i)^{m_i}$. So, in this case $c_k(t) = \prod_{\lambda_i \neq 0}(-\lambda_i)^{m_i} \neq 0$ and $k \geq r$. The equality happens if and only if X is regular by definition. □

1.4.4 Existence of Cartan subalgebras

Definition 1.4.11. Given a subalgebra $\mathfrak{h} \subset \mathfrak{g}$, we define its normalizer by

$$\mathcal{N}(\mathfrak{h}) = \{X \in \mathfrak{g} : [X, \mathfrak{h}] \subset \mathfrak{h}\}. \tag{1.82}$$

The normalizer $\mathcal{N}(\mathfrak{h})$ is a subalgebra itself: $X, Y \in \mathcal{N}(\mathfrak{h})$ implies

$$[Z, [X, Y]] = [\underset{\in \mathfrak{h}}{[Z, X]}, Y] + [X, \underset{\in \mathfrak{h}}{[Z, Y]}] \in \mathfrak{h}, \ \forall Z \in \mathfrak{h}. \tag{1.83}$$

Definition 1.4.12. A cartan subalgebra $\mathfrak{h} \subset \mathfrak{g}$ is a subalgebra satisfying:

1. $\mathfrak{h}$ is nilpotent
2. $\mathcal{N}(\mathfrak{h}) = \mathfrak{h}$

The next lemma gives a variety of examples for Cartan subaglebras.

Lemma 1.4.13. If $\mathfrak{g} \subset \mathfrak{gl}_n$ is a Lie subalgebra containing $D = \mathrm{Diag}(\lambda_1, \ldots, \lambda_n)$ with $\lambda_i \neq \lambda_j$, $i \neq j$, then

$$\mathfrak{h} = \mathfrak{g} \cap \{\text{diagonal matrices}\} \tag{1.84}$$

is a Cartan subalgebra of $\mathfrak{g}$.

Proof: Suppose that exists some $D = \mathrm{Diag}(\lambda_1, \ldots, \lambda_n)$, $\lambda_i \neq \lambda_j$ in $\mathfrak{g}$ and define $\mathfrak{h} = \mathfrak{g} \cap \{\text{diagonal matrices}\}$. It's clear that $\mathcal{N}(\mathfrak{h})$ is abelian and thus, nilpotent. So it remains to show that $\mathcal{N}(\mathfrak{h}) \subset \mathfrak{h}$. If $X \in \mathcal{N}(\mathfrak{h})$, then $[X, D] \in \mathfrak{h}$. Furthermore, if $X = \sum_{i,j} X_{ij} E_{ij}$, then

$$[X, D] = \sum_{i,j} X_{ij}[E_{ij}, D] = -\sum_{i,j} X_{ij}(\lambda_i - \lambda_j) E_{ij} \in \mathfrak{h} \tag{1.85}$$

which forces $X_{ij} = 0$ whenever $i \neq j$, i.e. $X \in \mathfrak{h}$. Therefore $\mathcal{N}(\mathfrak{h}) = \mathfrak{h}$.

Lemma 1.4.14. If $\mathfrak{h} \subset \mathfrak{g}$ is a Cartan subalgebra. Consider the decomposition of the representation $(\mathrm{ad}|_{\mathfrak{h}}, \mathfrak{g})$

$$\mathfrak{g} = \bigoplus_{\alpha \in \mathfrak{h}^*} \mathfrak{g}_\alpha^{\mathfrak{h}} \tag{1.86}$$

into generalized *weight spaces*. Then $\mathfrak{h} = \mathfrak{g}_0^{\mathfrak{h}}$.

Proof: Since $\mathfrak{h}$ is nilpotent it's clear that $\mathfrak{h} \subset \mathfrak{g}_0^{\mathfrak{h}}$. Therefore, we have the nilpotent action $\mathrm{ad}(\mathfrak{h})$ on $\mathfrak{g}_0^{\mathfrak{h}}/\mathfrak{h}$. Suppose that $\mathfrak{g}_0^{\mathfrak{h}}/\mathfrak{h} \neq 0$, then by Engel's theorem, there exist $Y \in \mathfrak{g}_0^{\mathfrak{h}} \setminus \mathfrak{h}$ such that $\mathrm{ad}(X)Y \in \mathfrak{h}$, $\forall X \in \mathfrak{h}$. Therefore $Y \in \mathcal{N}(\mathfrak{h}) = \mathfrak{h}$, which is a contradiction. So $\mathfrak{g}_0^{\mathfrak{h}}/\mathfrak{h} = 0$, i.e., $\mathfrak{h} = \mathfrak{g}_0^{\mathfrak{h}}$. □

The next theorem proves the existence of Cartan subagebras.

Theorem 1.4.15. (Cartan). Let $X \in \mathfrak{g}$ be a regular element, and consider the decomposition

$$\mathfrak{g} = \bigoplus_{s \in \mathbb{C}} E_s^{\mathrm{ad}(X)} \tag{1.87}$$

into generalized *eigenspaces*. Then $\mathfrak{h} = E_0^{\mathrm{ad}(X)}$ is a Cartan subalgebra.

Proof: It's immediate that $\mathrm{ad}(E_0^{\mathrm{ad}(X)})E_0^{\mathrm{ad}(X)} \subset E_0^{\mathrm{ad}(X)}$, i.e., $[\mathfrak{h}, \mathfrak{h}] \subset \mathfrak{h}$. So $\mathfrak{h}$ is a subalgebra. Next we will show that $\mathfrak{h}$ is nilpotent.

Define $V = \bigoplus_{s \in \mathbb{C}^*} E_s^{\mathrm{ad}(X)}$ and the sets

$$\begin{aligned} U &= \{X \in \mathfrak{h} : \mathrm{ad}(X)|_{\mathfrak{h}} \text{ is non nilpotent}\}, \\ R &= \{X \in \mathfrak{h} : \mathrm{ad}(X)\big|_V \text{ is invertible}\}. \end{aligned} \tag{1.88}$$

These two sets are Zariski open. In fact

$$U^C = \{X \in \mathfrak{h} : \mathrm{ad}(X)^{\dim(\mathfrak{h})+1} = 0\}. \tag{1.89}$$

But, the equation $\mathrm{ad}(X)^{\dim(\mathfrak{h})+1} = 0$ is a system of polynomial equations in X. Also we have

$$R^C = \{X \in \mathfrak{h} : \det(\mathrm{ad}(X)|_V) = 0\}. \tag{1.90}$$

Again, the equation $\det(\mathrm{ad}(X)|_V) = 0$ is a polynomial in X.

Furthermore we observe:

- $X \in \mathfrak{h}$ because $\mathrm{ad}(X)X = 0$,
- $X \notin U$ because $\mathrm{ad}(X)|_{\mathfrak{h}}$ is nilpotent by definition,
- $X \in R$ because $\mathrm{ad}(X)\big|_V$ has no zero eigenvalue by construction, so $\det(\mathrm{ad}(X)|_V) \neq 0$.

• We see that $\mathfrak{h}$ is a nilpotent subalgebra if and only if $U = \varnothing$.

We claim $U = \varnothing$: If not, then $U \cap R \neq \varnothing$ (any Zariski open set is Zariski dense). So there exist $Y \in U \cap R$. Now, we have decomposition

$$E_0^{\mathrm{ad}(Y)} = E_0^{\mathrm{ad}(Y)|_{\mathfrak{h}}} \oplus E_0^{\mathrm{ad}(Y)|_V}. \tag{1.91}$$

But, since $Y \in R$, we see that $E_0^{\mathrm{ad}(Y)|_V} = 0$ ($\mathrm{ad}(Y)$ can not have 0 eigenvalues). Therefore

$$E_0^{\mathrm{ad}(Y)} = E_0^{\mathrm{ad}(Y)|_{\mathfrak{h}}} \subset \mathfrak{h}. \tag{1.92}$$

But we know that $\dim(E_0^{\mathrm{ad}(Y)}) \geq \dim(E_0^{\mathrm{ad}(X)})$, because X is regular. So we conclude that $E_0^{\mathrm{ad}(Y)} = \mathfrak{h}$, i.e., $\mathrm{ad}(Y)|_{\mathfrak{h}}$ is nilpotent, which is a contradiction.

It remains to show that $\mathcal{N}(\mathfrak{h}) \subset \mathfrak{h}$. Let $Y \in \mathcal{N}(\mathfrak{h})$, then $\mathrm{ad}(X)Y \in \mathfrak{h}$ and since $\mathrm{ad}(X)|_{\mathfrak{h}}$ is nilpotent we see that $\mathrm{ad}(X)^{\dim(\mathfrak{h})+1}Y = 0$, i.e., $Y \in E_0^{\mathrm{ad}(X)} = \mathfrak{h}$. □

1.5 Structural theory of semisimple Lie algebras

1.5.1 Triangular decomposition of semisimple Lie algebras

Lemma 1.5.1. Let $\mathfrak{g}$ be a semisimple Lie algebra. We claim:

1. Let $\mathrm{Der}(\mathfrak{g}) \subset \mathfrak{gl}_{\mathfrak{g}}$ be the set of derivations of $\mathfrak{g}$, i.e.,

$$D[X,Y] = [DX,Y] + [X,DY],\ \forall D \in \mathrm{Der}(\mathfrak{g}),\ X,Y \in \mathfrak{g}. \tag{1.93}$$

   Also, denote $\mathrm{Inn}(\mathfrak{g}) = \mathrm{ad}(\mathfrak{g}) \subset \mathrm{Der}(\mathfrak{g})$. We claim that

$$\mathrm{Der}(\mathfrak{g}) = \mathrm{Inn}(\mathfrak{g}). \tag{1.94}$$

2. (abstract Jordan decomposition). Let $X \in \mathfrak{g}$, then there exist commuting elements $X_s, X_n \in \mathfrak{g}$ such that $\mathrm{ad}(X_s) = \mathrm{ad}(X)_s$ and $\mathrm{ad}(X_n) = \mathrm{ad}(X)_n$.

Proof: Denote by $K = (\cdot,\cdot)$ the trace form on $\mathfrak{gl}_{\mathfrak{g}}$, i.e., for $D, D' \in \mathfrak{gl}_{\mathfrak{g}}$ we have

$$(D, D') = \mathrm{Tr}(D \circ D'). \tag{1.95}$$

Then we see that $(\mathrm{ad}(X), \mathrm{ad}(Y)) = K^{\mathfrak{g}}(X,Y)$, so $(\cdot,\cdot)\big|_{\mathrm{Inn}(\mathfrak{g})\times\mathrm{Inn}(\mathfrak{g})}$ is nondegenerate by CC(ss). Therefore, by lemma (1.3.13), we have

$$\mathrm{Der}(\mathfrak{g}) = \mathrm{Inn}(\mathfrak{g}) \oplus \mathrm{Inn}(\mathfrak{g})^{\perp}. \tag{1.96}$$

However, if $D \in \mathrm{Inn}(\mathfrak{g})^{\perp}$, then $[D, \mathrm{ad}(X)] = \mathrm{ad}(D(X)) = 0, \forall X \in \mathfrak{g}$ which implies that $D(X) \in Z(\mathfrak{g}) = 0$, i.e., $D(X) = 0,\ \forall X \in \mathfrak{g}$. Therefore $D = 0$ and $\mathrm{Der}(\mathfrak{g}) = \mathrm{Inn}(\mathfrak{g})$.

For the second statement, let $X \in \mathfrak{g}$ and consider the decomposition

$$\mathfrak{g} = \bigoplus_{s \in \mathbb{C}} E_s^{\mathrm{ad}(X)} \tag{1.97}$$

into generalized eigenspaces. We claim that $D = \mathrm{ad}(X)_s$ is a derivation of $\mathfrak{g}$. In fact let $X \in E_\lambda^{\mathrm{ad}(X)}, Y \in E_\mu^{\mathrm{ad}(X)}$, then $[X, Y] \in E_{\lambda+\mu}^{\mathrm{ad}(X)}$ and

$$D([X,Y]) = (\lambda + \mu)[X,Y] = [\lambda X, Y] + [X, \mu Y] = [D(X), Y] + [X, D(Y)]. \tag{1.98}$$

So $D \in \mathrm{Der}(\mathfrak{g}) = \mathrm{Inn}(\mathfrak{g})$ and there exist $X_s \in \mathfrak{g}$ such that $\mathrm{ad}(X_s) = \mathrm{ad}(X)_s$. Now we define $X_n = X - X_s$ so it's clear that $\mathrm{ad}(X_n) = \mathrm{ad}(X)_n$. It remains to show that $[X_s, X_n] = 0$. But $\mathrm{ad}([X_s, X_n]) = [\mathrm{ad}(X_s), \mathrm{ad}(X_n)] = [\mathrm{ad}(X)_s, \mathrm{ad}(X)_n] = 0$. □

Lemma 1.5.2. Let $\mathfrak{g}$ be a semisimple Lie algebra and $\mathfrak{h} \subset \mathfrak{g}$ a Cartan subalgebra. Then we have the decomposition into generalized weight spaces:

$$\mathfrak{g} = \bigoplus_{\alpha \in \mathfrak{h}^*} \mathfrak{g}_\alpha \tag{1.99}$$

for the adjoint action $(\mathrm{ad}|_{\mathfrak{h}}, \mathfrak{g})$ (Here we are writing $\mathfrak{g}_\alpha$ instead of $\mathfrak{g}_\alpha^{\mathfrak{h}}$).
Under this condition we have:

1. $K(\mathfrak{g}_\alpha, \mathfrak{g}_\beta) = 0$ if $\alpha + \beta \neq 0$,
2. $K\big|_{\mathfrak{g}_\alpha \times \mathfrak{g}_{-\alpha}}$ is nondegenerate,
3. The Cartan algebra $\mathfrak{h}$ $(= \mathfrak{g}_0)$ is abelian,
4. Each generalized weight space $\mathfrak{g}_\alpha$ is in fact just a weightspace (this is equivalent to say that $\mathrm{ad}(X)$ is ss, for all $X \in \mathfrak{g}$).

Proof:

1. The first item is clear by the property $\mathrm{ad}(\mathfrak{g}_\alpha)\mathfrak{g}_\beta \subset \mathfrak{g}_{\alpha+\beta}$. So, if $\alpha + \beta \neq 0$, all elements in the diagonal of the matrix representation $\mathrm{ad}(\mathfrak{g}_\alpha) \circ \mathrm{ad}(\mathfrak{g}_\beta)$ are zero.
2. Suppose that there exist $X \in \mathfrak{g}_\alpha$ such that $K(X, Y) = 0$ for all $Y \in \mathfrak{g}_{-\alpha}$. Then, by the previous item we have $K(X, Y) = 0,\ \forall Y \in \mathfrak{g}$. Therefore, by CC(ss) we have $X = 0$.
3. Here we will use CC(sol) again: We know that $\mathfrak{h}$ is nilpotent, thus solvable. Therefore, by CC(sol) we have
$$K(\mathfrak{h}, [\mathfrak{h}, \mathfrak{h}]) = 0. \tag{1.100}$$
Since $K\big|_{\mathfrak{h} \times \mathfrak{h}}$ is nondegenerate (by the second item and the fact $\mathfrak{h} = \mathfrak{g}_0$), we see that $[\mathfrak{h}, \mathfrak{h}] = 0$.

4. Let $X \in \mathfrak{g}$, by lemma (1.5.1) we know that there exist a decomposition $X = X_s + X_n$, $(X_s, X_n \in \mathfrak{g}\ [X_s, X_n] = 0)$ such that $\mathrm{ad}(X_s) = \mathrm{ad}(X)_s$ (as an endomorphism of $\mathfrak{g}$!). We claim that $X_s \in \mathfrak{h}$. In fact, if $Y \in \mathfrak{h}$, then $[\mathrm{ad}(Y), \mathrm{ad}(X)] = \mathrm{ad}([Y, X]) = 0$, because $\mathfrak{h}$ is abelian. Therefore, since $\mathrm{ad}(X_s) = \mathrm{ad}(X)_s$ is a polynomial in $\mathrm{ad}(X)$ we get $[\mathrm{ad}(Y), \mathrm{ad}(X_s)] = 0$, i.e., $[Y, X_s] = 0 \in \mathfrak{h}$, $\forall Y \in \mathfrak{h}$ and $X_s \in \mathcal{N}(\mathfrak{h}) = \mathfrak{h}$.

   > Since $\mathfrak{h}$ is nilpotent, we are tempted to apply Engel's theorem to $\mathrm{ad}(\mathfrak{h}) \subset \mathfrak{gl}_{\mathfrak{g}}$. However we do not know if $\mathrm{ad}(X)$, $X \in \mathfrak{h}$ is nilpotent as an endomorphism of $\mathfrak{g}$!

   Now we claim that $X_n = 0$.

   We will use CC(sol) again: Since $\mathfrak{h}$ is abelian, then it's solvable and thus $\mathrm{ad}(\mathfrak{h})$ is a solvable subaglebra of $\mathfrak{gl}_{\mathfrak{g}}$. Therefore, by Lie's theorem, we know that there exist a basis of $\mathfrak{g}$ such that $\mathrm{ad}(\mathfrak{h}) \subset \mathfrak{t}_n$. In particular, a nilpotent element in $\mathfrak{t}_n$ is contained in $\mathfrak{n}_n$. So, if $\mathrm{ad}(Y), Y \in \mathfrak{h}$ is a nilpotent endomorphism of $\mathfrak{g}$, then the trace form $(\mathrm{ad}(X), \mathrm{ad}(Y))_{\mathfrak{g}} = 0$ for all $X \in \mathfrak{h}$ (in fact $(\mathfrak{t}_m, \mathfrak{n}_n)_{\mathfrak{g}} = 0$). This means that $K(X, Y) = 0, \forall X \in \mathfrak{h}$, but $K|_{\mathfrak{h}\times\mathfrak{h}}$ is nondegenerate and $Y = 0$.

   Therefore we see that $\mathrm{ad}(X_n) = \mathrm{ad}(X)_n = 0$ and $X_n \in Z(\mathfrak{g}) = 0$. □

Now we know that if $\mathfrak{g}$ is a semisimple Lie algebra, then it has a Cartan subalgebra $\mathfrak{h}$ and a weightspace decomposition

$$\mathfrak{g} = \bigoplus_{\alpha \in \mathfrak{h}^*} \mathfrak{g}_\alpha \tag{1.101}$$

where $\mathfrak{g}_\alpha$ is the *weight space* relative to the adjoint action of $\mathfrak{h}$ on $\mathfrak{g}$.

Definition 1.5.3. Define the weights Δ as the set of linear functionals $\alpha \in \mathfrak{h}^*$ such that $\mathfrak{g}_\alpha \neq 0$.

Since $K\big|_{\mathfrak{h}\times\mathfrak{h}}$ is nondegenerate, we have an induced isomorphism

$$\nu : \mathfrak{h} \to \mathfrak{h}^* \tag{1.102}$$

and we define a bilinear form $K : \mathfrak{h}^* \times \mathfrak{h}^* \to \mathbb{C}$ by:

$$K(\lambda, \mu) = K^{\mathfrak{g}}(\nu^{-1}(\lambda), \nu^{-1}(\mu)). \tag{1.103}$$

We have two important lemmas before the next section.

Remark 1.5.4. For now on we will denote the Killing form just by K also.

Lemma 1.5.5. If $e \in \mathfrak{g}_\alpha, f \in \mathfrak{g}_{-\alpha}$, then

$$[e, f] = K(e, f)\nu^{-1}(\alpha). \tag{1.104}$$

Proof: If $h \in \mathfrak{h}$, then we have the sequence of equalities

$$\begin{aligned} K(h, [e, f]) &= K([h, e], f) \quad \text{(because } K \text{ is invariant)} \\ &= K(\alpha(h)e, f) \quad \text{(because } e \in \mathfrak{g}_\alpha) \\ &= K(e, f)\alpha(h). \end{aligned} \tag{1.105}$$

So $\nu([e, f])(h) = K(e, f)\alpha(h), \forall h \in \mathfrak{h}$ and we complete de proof. □

Lemma 1.5.6. If $\alpha \in \Delta \setminus 0$, then $K(\alpha, \alpha) \neq 0$.

Proof: Since $K\big|_{\mathfrak{g}_\alpha \times \mathfrak{g}_{-\alpha}}$ is nondegenerate, there exist $(e, f) \in \mathfrak{g}_\alpha \times \mathfrak{g}_{-\alpha}$ such that $K(e, f) = 1$. So, by the previous lemma

$$[e, f] = \nu^{-1}(\alpha) = h \in \mathfrak{h}. \tag{1.106}$$

We see that

$$\alpha(h) = \nu(h)(h) = K(h, h) = K(\alpha, \alpha). \tag{1.107}$$

So, if $K(\alpha, \alpha) = 0$, we get the Lie algebra $\mathfrak{a} = \mathbb{C}\langle e, f, h\rangle$:

$$[e, f] = h, \ [h, e] = 0, \ [h, f] = 0. \tag{1.108}$$

So $\mathfrak{a} \cong H_3$ is a copy of the Heisenberg algebra, which is a nilpotent subalgebra of $\mathfrak{g}$. In particular, since $\mathfrak{a}$ is solvable, its image $\mathrm{ad}(\mathfrak{a}) \subset \mathfrak{gl}_\mathfrak{g}$ is solvable and by Lie's theorem there is a choice of basis of $\mathfrak{g}$ such that the adjoint action of $\mathfrak{a}$ has matrix representation in the set of upper triangular matrices $\mathfrak{t}_n$. Therefore $\mathrm{ad}(h) = \mathrm{ad}([e, f])$ lies in the set $[\mathfrak{t}_n, \mathfrak{t}_n] = \mathfrak{n}_n$. So $\mathrm{ad}(h)$ is nilpotent. But we know that the adjoint action of each element in the Cartan subalgebra is semisimple, so we conclude that $\mathrm{ad}(h) = 0 \Rightarrow h = 0$, which is a contradiction with the fact that $\alpha \neq 0$. □

1.5.2 Maximal toral subalgebras in semisimple Lie algebras

The purpose of this subsection is to give another characterization of Cartan subalgebras in semisimple Lie algebras.

Definition 1.5.7. A toral subalgebra of a Lie algebra $\mathfrak{g}$ is an abelian Lie algebra consisting only of semisimple elements. Let E be the set of all toral subalgebras of $\mathfrak{g}$, then E has a partial order given by inclusion. We define a maximal toral subalgebra (MTS) to be a maximal element in this set.

From the previous section, we know that in a semisimple Lie algebra $\mathfrak{g}$, a Cartan subalgebra $\mathfrak{h}$ is abelian, consisting only of semisimple elements, so it's a toral subalgebra. Moreover, if $\mathfrak{t}$ is a toral subalgebra of $\mathfrak{g}$ containing $\mathfrak{h}$, then $\mathrm{ad}(\mathfrak{t})\mathfrak{h} = 0$ and $\mathfrak{t} \subset N(\mathfrak{h}) = \mathfrak{h}$. Therefore $\mathfrak{h} = \mathfrak{t}$ and $\mathfrak{h}$ is a maximal toral subalgebra. We will show that the converse is also true.

Theorem 1.5.8. A maximal toral subalgebra $\mathfrak{t}$ of a semisimple Lie algebra $\mathfrak{g}$ is a Cartan subalgebra.

Notation 1.5.9. We will write $Z^{\mathfrak{g}}(\mathfrak{h}) = \{Y \in \mathfrak{g} : \mathrm{ad}(Y)|_{\mathfrak{h}} = 0\}$.

To show this, we will need the following lemma.

Lemma 1.5.10. Let $\mathfrak{g}$ be a Lie algebra and $\mathfrak{t}$ an abelian Lie subalgebra containing only semisimple elements. Then a Cartan subalgebra of $Z^{\mathfrak{g}}(\mathfrak{t})$ is a Cartan subalgebra of $\mathfrak{g}$ containing $\mathfrak{t}$.

Proof: Let $\mathfrak{c} = Z^{\mathfrak{g}}(\mathfrak{t})$, then if $\mathfrak{h} \subset \mathfrak{c}$ is a Cartan subalgebra, then $\mathfrak{t} \subset Z^{\mathfrak{c}}(0) \subset \mathfrak{h}$, the last inclusion is true because $\mathrm{ad}_{\mathfrak{c}}(X)\mathfrak{h} = 0 \Rightarrow X \in N^{\mathfrak{c}}(\mathfrak{h}) = \mathfrak{h}$. Now define $\mathfrak{n} = N^{\mathfrak{g}}(\mathfrak{h})$. Then

$$[\mathfrak{t}, \mathfrak{n}] \subset [\mathfrak{h}, \mathfrak{n}] \subset \mathfrak{h}. \tag{1.109}$$

Therefore if $X \in \mathfrak{t}$, then we get the restriciton $\mathrm{ad}_{\mathfrak{n}}(X) \in \mathrm{End}(\mathfrak{n})$. In particular, this map is semisimple, because it's the restriction of the semisimple map $\mathrm{ad}_{\mathfrak{g}}(X)$ to $\mathfrak{n}$. Now, let $X \in \mathfrak{t}$, then we get an eigenspace decomposition

$$\mathfrak{n} = \bigoplus_{i=1}^{t} E^{\mathrm{ad}(X)}_{\lambda_i(X)},\ \lambda_i(X) \in \mathbb{C}. \tag{1.110}$$

Furthermore, since $\mathrm{ad}_{\mathfrak{n}}(\mathfrak{t}) \subset \mathrm{End}(\mathfrak{n})$ is abelian, the eigenspace decomposition becomes a *weight space* decomposition

$$\mathfrak{n} = \bigoplus_{i=1}^{t} \mathfrak{n}_{\lambda_i},\ \lambda_i \in \mathfrak{t}^*. \tag{1.111}$$

Remember that $\mathfrak{n}_{\lambda_i} = \bigcap_{X \in \mathfrak{t}} E^{\mathrm{ad}_{\mathfrak{n}}(X)}_{\lambda_i(X)}$. Now, it's clear that $\mathfrak{h} \subset \mathfrak{n}_0$. So if we take a basis of $\mathfrak{h}$, complete to a basis of $\mathfrak{n}_0$ and then take a basis of the $\mathfrak{t}-$invariant subspace $\bigoplus_{\lambda_i \neq 0} \mathfrak{n}_{\lambda_i}$, we conclude that there exist a $\mathfrak{t}-$invariant subspace $\mathfrak{d} \subset \mathfrak{n}$ such that $\mathfrak{n} = \mathfrak{h} \oplus \mathfrak{d}$.

Now $[\mathfrak{t}, \mathfrak{d}] \subset \mathfrak{h} \cap \mathfrak{d} = (0)$ and $\mathfrak{d} \subset Z^{\mathfrak{g}}(\mathfrak{t}) = \mathfrak{c}$. So we conclude that $\mathfrak{n} \subset N^{\mathfrak{g}}(\mathfrak{h}) \subset \mathfrak{c}$. Therefore $N^{\mathfrak{g}}(\mathfrak{h}) = N^{\mathfrak{c}}(\mathfrak{h}) = \mathfrak{h}$ and $\mathfrak{h}$ is a Cartan subalgebra of $\mathfrak{g}$ containing $\mathfrak{t}$. □

Proof of the theorem: Now to prove the theorem, let $\mathfrak{t}$ be a maximal toral subalgebra of a semisimple Lie algebra $\mathfrak{g}$. In the previous section we proved the existence of Cartan subalgebras for complex Lie algebras, so let $\mathfrak{h} \subset Z^{\mathfrak{g}}(\mathfrak{t})$ be a Cartan subalgebra. We know

from the previous lemma that it's also a Cartan subalgebra of $\mathfrak{g}$ which contains $\mathfrak{t}$. However, since $\mathfrak{g}$ is semisimple, we know that $\mathfrak{h}$ is a maximal toral subalgebra. Therefore, we must have $\mathfrak{h} = \mathfrak{t}$. □

1.5.3 Another definition of Rank of a Lie algebra

Here we will fix a semisimple Lie algebra $\mathfrak{g}$.

Recall the definition of the rank of a Lie algebra that we saw on the previous section. The rank of a Lie algebra $\mathfrak{g}$ is the smallest integer $r \geq 1$ such that $C_r \neq 0$, where

$$C(X) = \det(t - \mathrm{ad}(X)) = t^r C_r(X) + t^{r+1} C_{r+1}(X) + \cdots . \tag{1.112}$$

Elements $X \in \mathfrak{g}$ such that $C_r(X) \neq 0$ were called *regular*. We saw that in a semisimple Lie algebra, these elements are necesseraly semisimple. Here, we change terminology and call them "*Kac-regular*". It's clear that if $\mathrm{ad}(X)$ has eigenvalues $\{\lambda_i\}$, then

$$C(X) = t^{\dim(\mathfrak{g}_0^X)} \prod_{\lambda_i \neq 0} (t - \lambda_i)^{m_i} \tag{1.113}$$

where $\mathfrak{g}_0^X$ is the generalized eigenspace associated to the eigenvalue 0 for $\mathrm{ad}(X)$. From the previous equation we have

$$\dim(\mathfrak{g}_0^X) \geq \mathrm{Rank}(\mathfrak{g}) \tag{1.114}$$

and the equality happens only if X is Kac-regular. Therefore we have

$$\mathrm{Rank}(\mathfrak{g}) = \min\{\dim(\mathfrak{g}_0^X) : X \in \mathfrak{g}\}. \tag{1.115}$$

Now we will give another definition of rank of a Lie algebra:

$$\mathrm{Rank}'(\mathfrak{g}) = \min\{\dim(Z^{\mathfrak{g}}(X)) : X \in \mathfrak{g}\}. \tag{1.116}$$

and we will call a X with centralizer in $\mathfrak{g}$ of minimal dimension, a *regular* element of X. Since $Z^{\mathfrak{g}}(X) \subset \mathfrak{g}_0^X$, it's clear that $\mathrm{Rank}(\mathfrak{g}) \geq \mathrm{Rank}'(\mathfrak{g})$. We claim that we actualy have equality, i.e., the two definitions of rank coincide. To see this, observe that if $\ell = \mathrm{Rank}'(\mathfrak{g})$ and $n = \dim(\mathfrak{g})$, then

$$\begin{aligned} \dim(Z^{\mathfrak{g}}(X)) \text{ is not minimal } = \ell &\iff \mathrm{Rank}(\mathrm{ad}(X)) \text{ is not maximal } = n - \ell \\ &\iff \text{all } (n-\ell) \times (n-\ell) - \text{minors of } \mathrm{ad}(X) \text{ vanishes.} \end{aligned} \tag{1.117}$$

This last condition is equivalent to the vanishing of a set of polynomials in X. So

$$\left\{ \begin{array}{c} \dim(Z^{\mathfrak{g}}(X)) \\ \text{is minimal } = \ell \end{array} \right\} \bigcap \{C_r \neq 0\} \neq \varnothing \tag{1.118}$$

because Zariski open sets are also dense. Moreover, we proved before that if X is Kac-regular, then it's semisimple, so $\mathfrak{g}_0^X = Z^{\mathfrak{g}}(X)$. Therefore, if X is in the intersection, then

$$\text{Rank}(\mathfrak{g}) = \dim(\mathfrak{g}_0^X) = \dim(Z^{\mathfrak{g}}(X)) = \text{Rank}'(\mathfrak{g}). \tag{1.119}$$

From here we will write $\text{Rank}(\mathfrak{g}) = r$.

Theorem 1.5.11. Let $\mathfrak{g}^{\text{sr}}$ be the set of regular semisimple elements of $\mathfrak{g}$, then this set coincides with the set of Kac-regular elements of $\mathfrak{g}$.

Proof: Let $X \in \mathfrak{g}$, remember that

$$C(X) = t^{\dim(\mathfrak{g}_0^X)} \prod_{\lambda_i \neq 0} (t - \lambda_i)^{m_i} = t^{\dim(\mathfrak{g}_0^X)} \prod_{\lambda_i \neq 0} (-\lambda_i)^{m_i} + \text{ higher order monomials.} \tag{1.120}$$

We have two cases

- (X_s is regular). In this case $C_r(X) = C_r(X_s) \neq 0$ by the above equation. So X is Kac-regular, so it's also semisimple. Then $X = X_s \in \mathfrak{g}^{\text{rs}}$.
- (X_s is not regular). In this case we have $C_r(X) = C_r(X_s) = 0$, so X is not Kac-regular.

This proves the assertion. □

1.6 Representations of $\mathfrak{sl}_2\mathbb{C}$

We will give here the main facts concerning the irreducible representations of $\mathfrak{sl}_2$.

- Let $\mathfrak{sl}_2 = \mathbb{C}Y \oplus \mathbb{C}H \oplus \mathbb{C}X$ with the standard relations:

  $$[H, X] = 2X,\ [H, Y] = -2Y,\ [X, Y] = H. \tag{1.121}$$

  So H is semisimple and span the Cartan subalgebra of $\mathfrak{sl}_2$. If (π, V) is an irreducible representation of $\mathfrak{sl}_2$, then we have a decomposition into $\pi(H)-$eigenspaces (this can be done because of the preservation of Jordan decomposition):

  $$V = \bigoplus_{\alpha \in \mathbb{C}} E_\alpha^{\pi(H)} = \bigoplus_{\alpha \in \mathbb{C}} V_\alpha. \tag{1.122}$$

- (Fundamental calculation). We have $X\mathfrak{g}_\alpha \subset \mathfrak{g}_{\alpha+2}$. In fact, if $v \in V_\alpha$, then

  $$HX(v) = XH(v) + [H, X](v) = \alpha X(v) + 2X(v) = (\alpha + 2)X(v). \tag{1.123}$$

  In a similar way we see that $Y\mathfrak{g}_\alpha \subset \mathfrak{g}_{\alpha-2}$.

- By the previous calculation we see that if $\alpha \in \mathbb{C}$ is such that $V_\alpha \neq 0$, then

$$V' = \bigoplus_{n\in\mathbb{Z}} V_{\alpha+2n} \subset V \tag{1.124}$$

is an invariant subspace and $V = V'$. Therefore, all $\mathrm{ad}(H)-$eigenvectors are mod 2 related. Moreover, since V is finite dimensional, the $\mathrm{ad}(H)-$eigenvalues form an (unbroken) string of numbers:

$$\alpha_0 - 2k, \alpha_0 - 2(k-1), \dots, \alpha_0 - 2, \alpha_0 \tag{1.125}$$

where $\alpha_0 \in \mathbb{C}$ is the greater complex number such that $V_{\alpha_0} \neq 0$. By unbroken, we mean $V_{\alpha_0-2(k-n)} \neq 0, n = 0, \dots, k$. In fact, if not, then we would have a proper invariant subspace in V.

- Let n be the biggest complex number such that $V_n \neq 0$ and let $v \in V_n \setminus 0$, then $V' = \langle v, Y(v), Y^2(v), \dots \rangle$ is an invariant subspace. It's clear that V' is $H, Y-$invariant. Moreover, it's $X-$invariant:

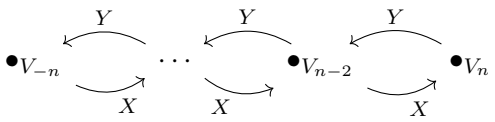
$$\begin{aligned} XY^k(v) &= YXY^{k-1}(v) + HY^{k-1}(v) \\ &= Y^2XY^{k-2}(v) + YHY^{k-2}(v) + HY^{k-1}(v) \\ &\dots \\ &= Y^{k-1}H(v) + Y^{k-2}HY(v) + \cdots YHY^{k-2}(v) + HY^{k-1}(v) \\ &= \underbrace{(n + n - 2 + \cdots + n - 2(k-1))}_{k \text{ times}} Y^{k-1}(v) \\ &= (kn - 2\frac{(k-1)k}{2})Y^{k-1}(v) \\ &= k(n-k+1)Y^{k-1}(v). \end{aligned} \tag{1.126}$$

Therefore $V = V'$. In particular we see that if $k \geq 0$ is the greatest integer such that $Y^k(v) \neq 0$ while $Y^{k+1}(v) = 0$, then $(k+1)(n-k) = 0$. So $n = k \in \mathbb{Z}_{\geq 0}$ and the unbroken string of eigenvalues must be of the form:

$$n, n-2, \dots, n-2(n-1), n-2n = -n. \tag{1.127}$$

So the eigenvalues form a symmetric string of $n+1$ integers around 0.

- An important observation about the previous calculation is that $V_{n-2k} = \mathbb{C}Y^k(v)$, i.e., each $\mathrm{ad}(H)-$eigenspace is one dimensional. Now, write $v_m \in V_m \setminus 0,\ m = n, n-2, \dots,$ then we have the following diagram for the $\mathfrak{sl}_2-$actions on V:



- (An useful fact). By the symmetry of the eigenvalues we see that if V is a representation of $\mathfrak{sl}_2$, then the number of irreducible components is $\dim(V_0) + \dim(V_1)$. In particular, if $\dim(V_0) + \dim(V_1) = 1$, then V is irreducible.

- For the previous calculation we see that each irreducible representation is completely characterized by n, the biggest $H-$eigenvalue.

- Let $\mu_k = k(n-1+k), k \geq 1,\ \mu_0 = 1$, then we renormalize $v_k = \frac{1}{\mu_k \cdots \mu_1 \mu_0} Y^k(v) \in V_{n-2k}$. With this normalization we get

$$Xv_k = \frac{\mu_k}{\mu_k \cdots \mu_1 \mu_0} Y^{k-1}(v) = v_{k-1},\ Yv_k = \mu_k v_{k+1}. \tag{1.128}$$

  Then in the basis $v_0 = v, v_1, \ldots, v_n$ we get matrix representations:

$$\rho_n(X) = \begin{pmatrix} 0 & 1 & & & \\ & 0 & \ddots & & \\ & & \ddots & \ddots & \\ & & & 0 & 1 \\ & & & & 0 \end{pmatrix},$$

$$\rho_n(Y) = \begin{pmatrix} 0 & & & & \\ \mu_1 & 0 & & & \\ & \ddots & \ddots & & \\ & & \ddots & 0 & \\ & & & \mu_r & 0 \end{pmatrix},\ \mu_k = k(n+1-k) \tag{1.129}$$

$$\rho_n(H) = \mathrm{Diag}(n, n-2, \ldots, -n+2, -n).$$

- We have an easy way to construct explictly nth irreducible representation of $\mathfrak{sl}_2$. First remember that if V and W are two $\mathfrak{g}-$modules, then $V \otimes W$ has a $\mathfrak{g}-$module structure given by:

$$X(v \otimes w) = Xv \otimes w + v \otimes Xw \text{ (like a derivation !)}. \tag{1.130}$$

  In particular, if V is a $\mathfrak{g}-$module, then $V^{\otimes k}$ is a $\mathfrak{g}-$module. Moreover, we have

$$X(v \otimes w - w \otimes v) = 0,\ \forall X \in \mathfrak{g}. \tag{1.131}$$

  and the $\mathfrak{g}$ action on $V^{\otimes k}$ pass to an action on $\mathrm{Sym}^k V$.

  Now, coming back to $\mathfrak{sl}_2-$modules. We have the $\mathbb{C}^2$ as the tautological representation of $\mathfrak{sl}_2$. Let $v = (1,0)^T, w = (0,1)^T \in \mathbb{C}^2$. Then

$$\mathrm{Sym}^k \mathbb{C}^2 = \langle v^k, v^{k-1}w, \ldots, vw^{k-1}, w^k \rangle \tag{1.132}$$

  is such that $H(v^{k-i}w^i) = (k-i-i)v^{k-i}w^i = (k-2i)v^{k-i}w^i$. So we have a complete string of $k, k-2, \ldots, -k+2, -k$ eigenvalues, each one spanning a one dimensional space, which must be the associated $H-$eigenspace by dimensional reasons. So $\mathrm{Sym}^k \mathbb{C}$ must be the kth irreducible representation of $\mathfrak{sl}_2$.

1.7 Root systems

Definition 1.7.1. Let V be a real vector space. A *root system* is a subset $\Delta \subset V \setminus 0$ satisfying the following axioms:

1. $\mathrm{Span}(\Delta) = V$
2. (String condition). For each $\alpha, \beta \in \Delta$ there exist $p, q \geq 0$ such that
$$[-p, q] \cap \mathbb{Z} = \{n \in \mathbb{Z} : \beta + n\alpha \in \Delta \cup \{0\}\},\ p - q = 2\frac{(\alpha, \beta)}{(\alpha, \alpha)}. \tag{1.133}$$
3. $\{n\alpha : n \in \mathbb{Z}\} \cap \Delta = \{\alpha, -\alpha\}$

Here $(\cdot, \cdot) \sim \mathrm{Diag}(1, \ldots, 1)$, an inner product on V (i.e., symmetric, nondegenerate and positive definite).

Reflections plays an important role in the context of root system. If $\alpha, \lambda \in V$, we define the reflection of λ around α, as the reflection of λ under "the" orthogonal line to α:
$$r_\alpha(\lambda) = \lambda - 2\mathrm{Proj}_\alpha(\lambda) = \lambda - 2\frac{(\alpha, \lambda)}{(\alpha, \alpha)}\alpha. \tag{1.134}$$

We have an important property: $r_\Delta(\Delta) \subset \Delta$. In fact, let $\alpha, \beta \in \Delta$, then by string condition there exist $p, q \geq 0$ such that
$$[-p, q] \cap \mathbb{Z} = \{n \in \mathbb{Z} : \beta + n\alpha \in \Delta \cup \{0\}\},\ p - q = 2\frac{(\alpha, \beta)}{(\alpha, \alpha)}. \tag{1.135}$$
So, in order to $r_\alpha(\beta) \in \Delta$, we need to show that $q - p \in [-p, q] \cap \mathbb{Z}$ and $r_\alpha(\beta) \neq 0$. But this is obvious, because $-p \leq q - p \leq q \iff 0 \leq q \leq q + p$ and $r_\alpha(\beta) = 0 \iff \beta = 0$ (we have the identity $r_\alpha^2 = \mathbb{1}_V$).

Now, comming back to semisimple Lie algebras.

Lemma 1.7.2. Let $\mathfrak{g}$ be a semisimple Lie algebra, $\mathfrak{h} \subset \mathfrak{g}$ a Cartan subalgebra and the related decomposition
$$\mathfrak{g} = \bigoplus_{\alpha \in \Delta} \mathfrak{g}_\alpha \tag{1.136}$$
into weightspaces $\mathfrak{g}_\alpha = V_\alpha^{\mathfrak{h}}$. Then we have:

1. $\dim(\mathfrak{g}_\alpha) = 1,\ \forall \alpha \in \Delta$
2. $[\mathfrak{g}_\alpha, \mathfrak{g}_\beta] \subset \mathfrak{g}_{\alpha+\beta}$ for $\alpha, \beta, \alpha + \beta \in \Delta$
3. Δ satisfy the string condition

Proof: For the first item we will use the representation theory of $\mathfrak{sl}_2$. In fact, we already know that there exist $(e,f) \in \mathfrak{g}_\alpha \times \mathfrak{g}_{-\alpha}$ such that $K(e,f) = 2/K(\alpha,\alpha)$. So

$$[e,f] = K(e,f)\nu^{-1}(\alpha) = \frac{2}{K(\alpha,\alpha)}\nu^{-1}(\alpha) = h. \tag{1.137}$$

In particular we see that see that $[h,e] = 2e$, $[h,f] = -2f$ and f,h,e is a $\mathfrak{sl}_2-$triple. Now, if $\dim(\mathfrak{g}_{-\alpha}) \geq 2$, then $K(e,\cdot) : \mathfrak{g}_{-\alpha} \to \mathbb{C}$ must have a nonzero kernel, i.e., there exist $v \in \mathfrak{g}_{-\alpha} \setminus 0$ such that $K(e,v) = 0$. So, in this case $[e,v] = 0$ and $[h,v] = -\alpha(h)v$. However

$$\alpha(h) = \nu\nu^{-1}(\alpha)(h) = \frac{2}{K(\alpha,\alpha)}K(\nu^{-1}\alpha,\nu^{-1}\alpha) = \frac{2}{K(\alpha,\alpha)}K(\alpha,\alpha) = 2. \tag{1.138}$$

So $V = \langle v, \mathrm{ad}(f)v, \dots \rangle$ is a $\mathfrak{sl}_2-$module with highest weight equal to -2, which is a contradiction.

For the third item let $\alpha, \beta \in \Delta$ and consider the $\mathfrak{sl}_2 \cong \mathbb{C}\langle f,h,e\rangle$ module given by

$$V = \bigoplus_{n\in\mathbb{Z}} \mathfrak{g}_{\beta+n\alpha}. \tag{1.139}$$

Let $p,q \geq 0$ be the biggest integers such that $\mathfrak{g}_{\beta-p\alpha}, \mathfrak{g}_{\beta+q\alpha} \neq 0$. Remember the following calculation:

$$\lambda(h) = \nu\nu^{-1}(\lambda)(h) = 2\frac{K(\lambda,\alpha)}{K(\alpha,\alpha)} \tag{1.140}$$

and therefore

$$(\beta+n\alpha)(h) = 2(\frac{K(\beta,\alpha)}{K(\alpha,\alpha)} + n) \in \mathbb{Z} \ (\Rightarrow 2\frac{K(\beta,\alpha)}{K(\alpha,\alpha)} \in \mathbb{Z}). \tag{1.141}$$

So, if $v \in \mathfrak{g}_{\beta+q\alpha} \setminus 0$, then $[h,v] = 2(\frac{K(\beta,\alpha)}{K(\alpha,\alpha)} + q)v$, $[e,v] = 0$ and by the symmetry of the $\mathrm{ad}(h)-$eigenvalues we must have $\mathrm{ad}(f)^N(v) \neq 0$ where $N = 2(\frac{K(\beta,\alpha)}{K(\alpha,\alpha)} + q)$. In particular

$$\mathbb{C}\mathrm{ad}(f)^N v = \mathfrak{g}_{\beta-(2\frac{K(\beta,\alpha)}{K(\alpha,\alpha)}+q)\alpha} \tag{1.142}$$

bacause of item (1). Therefore we see that

$$p \geq 2\frac{K(\beta,\alpha)}{K(\alpha,\alpha)} + q. \tag{1.143}$$

Now, changing $\alpha \to -\alpha$ and interchanging the roles of p and q we see that

$$q \geq -2\frac{K(\beta,\alpha)}{K(\alpha,\alpha)} + p \Rightarrow p - q = 2\frac{K(\beta,\alpha)}{K(\alpha,\alpha)}. \tag{1.144}$$

For the second item we just observe that since $\alpha + \beta \in \Delta$, then $q \geq 1$ and the map $\mathrm{ad}(e) : \mathfrak{g}_\alpha \to \mathfrak{g}_{\alpha+\beta}$ is surjective by the representation theory of $\mathfrak{sl}_2$. □

Remark 1.7.3. We know that $\dim(h) = \text{Rank}(\mathfrak{g})$ by the construction of h as the generalized eigenspace $E_0^{\text{ad}(X)}$ for some X regular. Moreover, we have $\text{Span}(\Delta) = \mathfrak{h}^*$. If not, by the exact sequence

$$0 \to \text{Ann Span}(\Delta) \hookrightarrow \mathfrak{h} \to \text{Span}(\Delta)^* \to 0 \tag{1.145}$$

we must have dim $\text{AnnSpan}(\Delta) > 0$, i.e., there exist $h \in \mathfrak{h}(\cong \mathfrak{h}^{**})$ such that $\alpha(h) = 0$, $\forall \alpha \in \Delta$. This means that $[h, \mathfrak{g}_\alpha] = 0$, $\forall \alpha \in \Delta$ and $h \in Z(\mathfrak{g}) = 0$, which is a contradiction.

In the proof of the next theorem we will show that

$$K(\alpha,\alpha) = \sum_{\beta\in\Delta} K(\alpha,\beta)^2 \iff \frac{1}{K(\alpha,\alpha)} = \sum_{\beta\in\Delta} (\frac{K(\alpha,\beta)}{K(\alpha,\alpha)})^2 \in \mathbb{Q}. \tag{1.146}$$

So we have $K(\alpha,\alpha) \in \mathbb{Q}$ and $K(\alpha,\beta) \in \mathbb{Q}$ for all $\alpha, \beta \in \mathbb{Q}$. Therefore if we define $V_\mathbb{Q} = \{\sum_{\alpha\in\Delta} c_\alpha \alpha : c_\alpha \in \mathbb{Q}\} \subset \mathfrak{h}^*$, then the Killing form restricts to a form $K : V_\mathbb{Q} \times V_\mathbb{Q} \to \mathbb{Q}$, which we will show to be nondegenerate. Accepting this for a while, we can define an Euclidean space $(V_\mathbb{R}, K)$, where $V_\mathbb{R} = \mathbb{R} \otimes_\mathbb{Q} V_\mathbb{Q}$. In this context, $(V_\mathbb{R}, \Delta)$ will be a root system. We claim that $\dim_\mathbb{Q}(V_\mathbb{Q}) = l$.

In fact, choose a basis $\{\alpha_1, \ldots, \alpha_l\}$ of $\mathfrak{h}^*$ in Δ and write $\{\beta_1, \ldots, \beta_l\}$ to the dual basis with respect to K. We have

$$\beta_i = \sum_j c_{ij}\alpha_j,\ c_{ij} \in \mathbb{Q}. \tag{1.147}$$

In fact, let $C = (c_{ij})$ which is an invertible matrix (maybe in $\mathbb{C}$). Then, the previous equation is equivalent to $\alpha_j = \sum_i (C^{-1})_{ji}\beta_i$ and $(C^{-1})_{ji} = K(\alpha_j, \alpha_i) = M_{ji} \in \mathbb{Q}$. So $C = M^{-1} \in GL_l(\mathbb{Q})$, where M is the Gram matrix of K. This proves that $c_{ij} \in \mathbb{Q}$. Now, if $\beta \in \Delta$, the previous equation says that

$$\beta = \sum_i K(\beta,\beta_i)\alpha_i,\ K(\beta,\beta_i) = \sum_j c_{ij}K(\beta,\alpha_j) \in \mathbb{Q}. \tag{1.148}$$

So $\Delta \subset \mathbb{Q}\langle\alpha_1, \ldots, \alpha_l\rangle$, which proves the claim, because $V_\mathbb{Q} = \text{Span}_\mathbb{Q}(\Delta)$ by definition.

Theorem 1.7.4. Let $\mathfrak{g}$ be a semisimple Lie algebra of rank l and Δ be its set of roots with respect to some choice of Cartan subalgebra. Define $V_\mathbb{R} = \mathbb{R} \otimes V_\mathbb{Q}$, where $V_\mathbb{Q} = \text{Span}_\mathbb{Q}(\Delta) \subset \mathfrak{h}^*$. Then $(V_\mathbb{R}, \Delta)$ is a rank l root system.

Proof: We still need to prove the 3. axiom of Δ. If $n\alpha \in \Delta$, then

$$p - q = 2\frac{(\alpha, n\alpha)}{(n\alpha, n\alpha)} \in \mathbb{Z} \Rightarrow \frac{2}{n} \in \mathbb{Z}. \tag{1.149}$$

Therefore $n = \pm 2, \pm 1$. However, since $\mathfrak{g}_{\pm 2\alpha} = [\mathfrak{g}_{\pm\alpha}, \mathfrak{g}_{\pm\alpha}] = 0$ ($\mathfrak{g}_\alpha$ is one dimension), we can not have $2\alpha \in \Delta$. So $n = \pm 1$.

It remains to show that $K(\alpha, \alpha) \geq 0$ (the form must be positive definite). However, since $\mathfrak{h}$ is spanned by semisimple elements, for any $X, Y \in \mathfrak{h}$ we have matrix representation of $\mathrm{ad}(X) \circ \mathrm{ad}(Y)$ given by $\mathrm{Diag}(\alpha(X)\alpha(Y) : \alpha \in \Delta, 0, \ldots, 0)$ with respect to the basis $\{e_\alpha : \alpha \in \Delta\} \cup \{h_1, \ldots, h_{\dim(\mathfrak{h})}\}$, where $\mathfrak{g}_\alpha = \mathbb{C}e_\alpha$. Therefore

$$K(X, Y) = \sum_{\alpha \in \Delta} \alpha(X)\alpha(Y) \tag{1.150}$$

and if $\lambda, \mu \in \mathfrak{h}^*$ we get

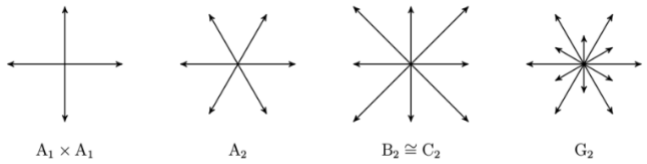
$$\begin{aligned} K(\lambda, \mu) &= K(\nu^{-1}(\lambda), \nu^{-1}(\mu)) \\ &= \sum_{\alpha \in \Delta} \alpha(\nu^{-1}(\lambda))\alpha(\nu^{-1}(\mu)) \\ &= \sum_{\alpha \in \Delta} K(\lambda, \alpha)K(\alpha, \mu). \end{aligned} \tag{1.151}$$

Finally $K(\alpha, \alpha) = \sum_{\beta \in \Delta} K(\beta, \alpha)^2 \geq 0$ and the equality occurs only if

$$K(\beta, \alpha) = 0, \ \forall \beta \in \Delta \Rightarrow \beta(\nu^{-1}(\alpha)) = 0 \ \forall \beta \in \Delta \tag{1.152}$$

which happens only if $\nu^{-1}(\alpha) = 0 \iff \alpha = 0$. □

Lemma 1.7.5. (Classification of rank 2 roots systems). There exist four rank 2 roots systems:



Rank 2 root systems.

Proof: We prove it in several steps:

- Claim: Every two adjacent roots have the same angle between then.

  Let $\alpha_0, \alpha_1 \in \Delta$ be two adjacent roots with minimum angle between then. We define

$$\alpha_{n+1} = -r_{\alpha_n}(\alpha_{n-1}) \in \Delta, \ n \geq 1 \tag{1.153}$$

and $\mathrm{Arg}(v) = \cos^{-1}(\frac{(v,\alpha_0)}{\|v\|\|\alpha_0\|}) \in [0, 2\pi)$. With this definition, we see that the angle between α_n and α_{n-1} for $n \geq 1$ still θ. In fact

$$\begin{aligned}\frac{(\alpha_n, \alpha_{n-1})}{\|\alpha_n\|\|\alpha_{n-1}\|} &= \frac{(r_{\alpha_{n-1}}(\alpha_{n-2}), \alpha_{n-1})}{\|r_{\alpha_{n-1}}(\alpha_{n-2})\|\|\alpha_{n-1}\|} \\ &= \frac{(\alpha_{n-2}, \alpha_{n-1})}{\|\alpha_{n-2}\|\|\alpha_{n-1}\|} \ (r_{\alpha_{n-1}} \text{ is an isometry}) \\ &\cdots \\ &= \frac{(\alpha_1, \alpha_0)}{\|\alpha_1\|\|\alpha_0\|}.\end{aligned} \tag{1.154}$$

In this condition we claim that we have $\mathrm{Arg}(\alpha_n) = n\theta \bmod 2\pi$, where $\theta = \mathrm{Arg}(\alpha_1)$. We argue by induction, the base case being just the definition. The induction step follows by the calculation

$$\begin{aligned}\cos \mathrm{Arg}(\alpha_{n+1}) &= \frac{(\alpha_{n+1}, \alpha_0)}{\|\alpha_{n+1}\|} \\ &= 2\frac{(\alpha_{n-1}, \alpha_n)}{\|\alpha_n\|\|\alpha_{n-1}\|} \cdot \frac{(\alpha_n, \alpha_0)}{\|\alpha_n\|} - \frac{(\alpha_{n-1}, \alpha_0)}{\|\alpha_{n-1}\|} \\ &= 2\cos(\theta)\cos(n\theta) - \cos((n-1)\theta) \\ &= \cos((n+1)\theta).\end{aligned} \tag{1.155}$$

Now let $n = \mathrm{Max}\{k \geq 1 : n\theta \leq 2\pi\}$. If $n\theta < 2\pi$, then $2\pi < (n+1)\theta < \theta + 2\pi$, which is a contradiction with the hipothesis that θ is the smallest angle between two adjacent roots. Therefore, we see that $n\theta = 2\pi$ and $\theta = 2\pi/n$. Since the only colinear roots are $\{\pm\alpha\}$, we see that we have obtained all roots by this process, because of the condition of minimality of θ.

- Remember that if $\alpha, \beta \in \Delta$, then

$$2\frac{(\alpha, \beta)}{(\alpha, \alpha)} \in \mathbb{Z}. \tag{1.156}$$

Therefore, if α, β are two adjacent roots we get

$$4\cos^2(\theta) = 4\frac{(\alpha, \beta)}{(\alpha, \alpha)(\beta, \beta)} \in \{0, 1, 2, 3, 4\} \text{ (apply Cauchy-Scharwz).} \tag{1.157}$$

So, we have $|\cos(\theta)| \in \{0, \frac{1}{2}, \frac{1}{\sqrt{2}}, \frac{\sqrt{3}}{2}, 1\}$ and $\theta \in \{\frac{\pi}{2}, \frac{\pi}{3}, \frac{\pi}{4}, \frac{\pi}{6}\}$. Here 0 is excluded because $\beta \neq \alpha$. The next step is to describe the roots system, i.e., find explictly all elements of Δ for each angle θ obtained. Before we proceed, we have a observation: Each θ is of the form π/n and we saw in the previous item that $\mathrm{Arg}(\Delta) = \{\frac{k\pi}{n} : k = 0, \ldots, 2n-1\}$, so $\#\Delta = 2n$.

- Case $n = 4$: In this case we have 8 roots. Lets find them in polar coordinates using complex number notation. Suppose that $\alpha, \beta \in \Delta \subset \mathbb{C}$ are two adjacent roots. We can assume $\text{Arg}(\alpha) = 0$ and $\text{Arg}(\beta) > 0$ (this condition will be used in all the cases). We have
$$4\cos^2(\theta) = 2\frac{(\alpha,\beta)}{(\alpha,\alpha)} \cdot 2\frac{(\alpha,\beta)}{(\beta,\beta)} = 2 \tag{1.158}$$
and without lost of generality we assume that
$$2\frac{(\alpha,\beta)}{(\alpha,\alpha)} = 2,\ 2\frac{(\alpha,\beta)}{(\beta,\beta)} = 1 \Rightarrow \frac{(\beta,\beta)}{(\alpha,\alpha)} = 2. \tag{1.159}$$
So we can assume $(\beta,\beta) = 2,\ (\alpha,\alpha) = 1$, which implies $\alpha = 1$ and $\beta = \sqrt{2}\exp(\frac{i\pi}{4})$. Now, since $2\frac{(\alpha,\beta)}{(\alpha,\alpha)} = 2$, we see that $\beta - 2\alpha \in \Delta$, i.e., $\sqrt{2}\exp(\frac{3i\pi}{4}) \in \Delta$. Furthermore $2\frac{(\alpha,\beta)}{(\beta,\beta)} = 1$ implies that $\alpha - \beta = -i \in \Delta$. Then the four roots in the upper plane are
$$1, \sqrt{2}\exp(\frac{i\pi}{4}), i \text{ and } \sqrt{2}\exp(\frac{3i\pi}{4}). \tag{1.160}$$
These roots determines all the 8 roots of Δ.

- Case $n = 3$: In this case we have 6 roots. We have
$$4\cos^2(\theta) = 2\frac{(\alpha,\beta)}{(\alpha,\alpha)} \cdot 2\frac{(\alpha,\beta)}{(\beta,\beta)} = 2 \tag{1.161}$$
and we can assume
$$2\frac{(\alpha,\beta)}{(\alpha,\alpha)} = 1,\ 2\frac{(\alpha,\beta)}{(\beta,\beta)} = 1. \tag{1.162}$$
So $(\alpha,\alpha) = (\beta,\beta) = 1$, which implies $\alpha = 1$ and $\beta = \exp(\frac{i\pi}{3})$. Now, since $2\frac{(\alpha,\beta)}{(\alpha,\alpha)} = 1$, we have $\beta - \alpha = \exp(\frac{2i\pi}{3}) \in \Delta$. From this we get all the 3 roots in the upper plane. This determines all the other roots in Δ:
$$\exp(\frac{ki\pi}{3}),\ k = 0, 1, 2, 3, 4, 5. \tag{1.163}$$

- Case $n = 2$: In this case we have 4 roots. We have
$$4\cos^2(\theta) = 2\frac{(\alpha,\beta)}{(\alpha,\alpha)} \cdot 2\frac{(\alpha,\beta)}{(\beta,\beta)} = 0 \tag{1.164}$$
and we can assume
$$2\frac{(\alpha,\beta)}{(\alpha,\alpha)} = 0 \Rightarrow (\alpha,\beta) = 0. \tag{1.165}$$
So $\alpha = 1$ and $\beta = mi, m \geq 0$ and it remains to find m.

- Case $n = 6$: In this case we have 12 roots. Lets find them. We have

$$4\cos^2(\theta) = 2\frac{(\alpha,\beta)}{(\alpha,\alpha)} \cdot 2\frac{(\alpha,\beta)}{(\beta,\beta)} = 3 \tag{1.166}$$

and we can assume

$$2\frac{(\alpha,\beta)}{(\alpha,\alpha)} = 1,\ 2\frac{(\alpha,\beta)}{(\beta,\beta)} = 3 \tag{1.167}$$

so we take $(\alpha,\alpha) = 3(\beta,\beta) = 3$. Then if $\alpha = \sqrt{3}$, we get $\beta = \exp(\frac{i\pi}{6})$. Now, condition $2\frac{(\alpha,\beta)}{(\alpha,\alpha)} = 1$ implies that $\beta - \alpha = \exp(\frac{5i\pi}{6}) \in \Delta$. Moreover, condition $2\frac{(\alpha,\beta)}{(\beta,\beta)} = 3$ implies that

$$\begin{aligned} &\alpha - 2\beta = -i \in \Delta \\ &\alpha - 3\beta = \sqrt{3}\exp(\frac{4i\pi}{3}) \in \Delta \end{aligned} \tag{1.168}$$

This determines all the 6 root in the upper half plane and therefore, all the other roots in Δ.

We could have used the previous algorithm to construct all roots in each case: For the initial adjacent roots α_0, α_1, all other roots are of the form $\alpha_{n+1} = -r_{\alpha_n}(\alpha_{n-1}),\ n \geq 1$. For example, in the case $n = 4$ we have $\alpha_0 = 1$ and $\alpha_1 = \sqrt{2}\exp(\frac{i\pi}{4})$. Then

$$\begin{aligned} \alpha_2 &= -1 + \mathrm{Re}(1 \cdot \overline{\sqrt{2}\exp(\frac{i\pi}{4})})\sqrt{2}\exp(\frac{i\pi}{4}) \\ &= -1 + 2\frac{1}{\sqrt{2}}\exp(\frac{i\pi}{4}) \\ &= i \end{aligned} \tag{1.169}$$

and

$$\begin{aligned} \alpha_3 &= -\sqrt{2}\exp(\frac{i\pi}{4}) + 2\mathrm{Re}[(-i)\sqrt{2}\exp(\frac{i\pi}{4})]i \\ &= -\sqrt{2}\exp(\frac{i\pi}{4}) - 2i \\ &= -1 - i \\ &= \sqrt{2}\exp(\frac{3i\pi}{4}). \end{aligned} \tag{1.170}$$

Example 1.7.6. The Lie algebra $\mathfrak{so}_8$ has root system given by:

$$\Delta = \{L_i - L_j : i \neq j\} \cup \{\pm(L_i + L_j) : i \neq j\} \tag{1.171}$$

for $i, j \in \{1, 2, 3, 4\}$. Let $f : \mathrm{Span}(L_i)_{1 \leq i \leq 4} \to \mathbb{R}$ be a linear map such that $f(L_1) > f(L_2) > f(L_3) > f(L_4) > 0$. Then, with respect to this linear functional we get

$$\Delta_+ = \{L_i - L_j : i > j\} \cup \{L_i + L_j : i \neq j\} \tag{1.172}$$

and $\Pi = \{L_1 - L_2, L_2 - L_3, L_3 - L_4, L_3 + L_4\}$.

It's a interesting exercise to decompose $\mathfrak{so}_8$ as a sum of $\mathfrak{sl}_2$-modules with respect to the adjoint action of the positive element X_α, $\alpha = L_1 - L_2$. To do this, we list the corresponding strings of $\mathfrak{sl}_2-$modules. The arrows "$\to$" corresponds to the action of $\mathrm{ad}(X_\alpha)$ on the respective root spaces.

$$\begin{aligned}
&-L_1 + L_2 \to H_{L_1 - L_2} \to L_1 - L_2 \to 0 \Rightarrow V_{(2)} \text{ (because } (\alpha, L_1 - L_2) = 2) \\
&L_2 - L_3 \to L_1 - L_3 \to 0 \Rightarrow V_{(1)} \text{ (because } (\alpha, L_1 - L_3) = 1) \\
&L_2 - L_4 \to L_1 - L_4 \to 0 \Rightarrow V_{(1)} \text{ (because } (\alpha, L_1 - L_4) = 1) \\
&L_3 - L_4 \to 0 \Rightarrow V_{(0)} \text{ (because } (\alpha, L_3 - L_4) = 0) \\
&L_1 + L_2 \to 0 \Rightarrow V_{(0)} \text{ (because } (\alpha, L_1 + L_2) = 0) \\
&L_2 + L_3 \to L_1 + L_3 \to 0 \Rightarrow V_{(1)} \text{ (because } (\alpha, L_1 + L_3) = 1) \\
&L_2 + L_4 \to L_1 + L_4 \to 0 \Rightarrow V_{(1)} \text{ (because } (\alpha, L_1 + L_4) = 1) \\
&L_3 + L_4 \to 0 \Rightarrow V_{(0)} \text{ (because } (\alpha, L_3 + L_4) = 0)
\end{aligned} \tag{1.173}$$

So the positive root spaces gives a contribution $V_{(2)} \oplus V_{(1)}^{\oplus 4} \oplus V_{(0)}^{\oplus 3}$ to the decomposition of $\mathfrak{so}_2$ into $\mathfrak{sl}_2-$modules. Also, there is only one element H_α of the Cartan subalgebra $\mathfrak{h}$ in this decomposition. So it remais to determine the decomposition of $\bigoplus_{\alpha \in \Delta_+} \mathfrak{g}_{-\alpha}$ and the rest of the Cartan subalgebra. However, we can see by the same reasoning that

$$\bigoplus_{\alpha \in \Delta_+} \mathfrak{g}_{-\alpha} \cong V_{(2)} \oplus V_{(1)}^{\oplus 4} \oplus V_{(0)}^{\oplus 3} \tag{1.174}$$

and since $\mathfrak{h} = \mathbb{C}H_\alpha \oplus \mathrm{Ker}(\alpha)$, we get the contribution of the remaining elements of the Cartan subalgebra:

$$\mathrm{Ker}(\alpha) \cong V_{(0)}^{\oplus 3} \text{ (because } \dim(\mathfrak{h}) = 4). \tag{1.175}$$

So, the final result is $\mathfrak{g} \cong V_{(2)} \oplus V_{(1)}^{\oplus 8} \oplus V_{(0)}^{\oplus 9}$. As a test, we sum the corresponding dimensions:

$$\dim(V_{(2)} \oplus V_{(1)}^{\oplus 8} \oplus V_{(0)}^{\oplus 9}) = 3 + 8 \cdot 2 + 9 \cdot 1 = 3 + 16 + 9 = 28. \tag{1.176}$$

Which confirms the calculation $\dim(\mathfrak{so}_8) = 2 \cdot 4^2 - 4 = 28$.

1.8 Simple Lie algebras

1.8.1 Irreducible root systems

Definition 1.8.1. A root system (V, Δ) is sai do to be decomposable if there exist a decomposition $V = V_1 \oplus V_2$ such that $\Delta = (\Delta \cap V_1) \cup (\Delta \cap V_2)$. An irreducible root system, is a root system that is not decomposable.

Lemma 1.8.2. If (V, Δ) is decomposable with $\Delta = \Delta_1 \cup \Delta_2,\ \Delta_i \subset V_i, V = V_1 \oplus V_2$, then $V_1 \perp V_2$.

Proof: Suppose that $\alpha \in \Delta_1, \beta \in \Delta_2$ and $(\alpha, \beta) \neq 0$. Since

$$p - q = 2\frac{(\alpha, \beta)}{(\alpha, \alpha)} \neq 0 \tag{1.177}$$

we have two cases:

- If $p > q$, then $p \geq 1$ and $\beta - \alpha \in \Delta$. But this can not happen, because if $\beta - \alpha \in \Delta_1 \overset{\alpha \in V_1}{\Rightarrow} \beta \in \Delta \cap V_1 = \Delta_1$. In a similar way we see that we can not have $\beta - \alpha \in \Delta_2$.
- If $q > p$, then $\beta + \alpha \in \Delta$ an the exact same reasoning apply to show that this let to a contradiction.

□

Remark 1.8.3. Since Δ contain a basis of V it can not be the case that Δ is contained in a proper subspace of V. So, if $V = V_1 \oplus V_2$ is a nontrivial decomposition, we must have $\Delta_i = \Delta \cap V_i \neq \varnothing$.

Remark 1.8.4. Maybe we can show that (V_i, Δ_i) are root systems, but we will not need it.

Theorem 1.8.5. Let $\mathfrak{g}$ be a Lie algebra with associated root system $(\mathfrak{h}^*, \Delta)$ where $\mathfrak{h}$ is a Cartan subalgebra. Then

$$\mathfrak{g} \text{ is simple } \iff (\mathfrak{h}^*, \Delta) \text{ is irreducible.} \tag{1.178}$$

Proof:($\Leftarrow$). Suppose that $\mathfrak{g} = \mathfrak{g}_1 \oplus \mathfrak{g}_2$ with $\mathfrak{g}_i$ semisimple with a choice of Cartan subalgebras $\mathfrak{h}_i$ $(i = 1, 2)$. Then, we see that $\mathfrak{h} = \mathfrak{h}_1 \oplus \mathfrak{h}_2 \subset \mathfrak{g}$ is Cartan. In fact, since

$[\mathfrak{h}_1, \mathfrak{h}_2] = 0$ we see that $\mathfrak{h}^k = \mathfrak{h}_1^k \oplus \mathfrak{h}_2^k$, so $\mathfrak{h}$ is nilpotentl. Moreover

$$\begin{aligned} &[(X,Y), \mathfrak{h}_1 \oplus \mathfrak{h}_2] \subset \mathfrak{h}_1 \oplus \mathfrak{h}_2 \iff \\ &([X, \mathfrak{h}_1], [(Y, \mathfrak{h}_2)]) \subset \mathfrak{h}_1 \oplus \mathfrak{h}_2 \iff \\ &[X, \mathfrak{h}_i] \subset \mathfrak{h}_i \ (i = 1, 2). \end{aligned} \tag{1.179}$$

Then $X \in N(\mathfrak{h}_1) = \mathfrak{h}_1$ and $Y \in N(\mathfrak{h}_2) = \mathfrak{h}_2$, so $(X, Y) \in \mathfrak{h}$. In particular this shows that

$$\mathrm{Rank}(\mathfrak{g}_1 \oplus \mathfrak{g}_2) = \mathrm{Rank}(\mathfrak{g}_1) + \mathrm{Rank}(\mathfrak{g}_2). \tag{1.180}$$

Now identify $\mathfrak{h}^* = \mathfrak{h}_1^* \oplus \mathfrak{h}_2^*$ with the rule $(\alpha, \beta)(v, w) = \alpha(v) + \beta(w)$. Then we claim that if $\alpha \in \Delta_1$, then $(\alpha, 0) \in \Delta$. In fact, for $v \in \mathfrak{g}_\alpha \setminus 0$, we have

$$\begin{aligned} [(H_1, H_2), (v, 0)] &= ([H_1, v], 0) \\ &= (\alpha(H_1)v, 0) \\ &= \alpha(H_1)(v, 0) \\ &= (\alpha, 0)(H_1, H_2) \cdot (v, 0). \end{aligned} \tag{1.181}$$

So $\mathfrak{g}_{(\alpha,0)} \neq 0$ and $(\alpha, 0) \in \Delta$. In a similar way we show that $(0, \beta) \in \Delta$ for $\beta \in \Delta_2$. Therefore

$$(\Delta_1 \times 0) \cup (0 \times \Delta_2) \subset \Delta. \tag{1.182}$$

Conversely, if $(\alpha, \beta) \in \Delta$, then there exist $(v, w) \in \mathfrak{g}_{(\alpha,\beta)} \setminus 0$. So

$$([H_1, v], [H_2, w]) = [(H_1, H_2), (v, w)] = (\alpha(H_1) + \beta(H_2)) \cdot (v, w), \ \forall (H_1, H_2) \in \mathfrak{h}_1 \oplus \mathfrak{h}_2. \tag{1.183}$$

If $v \neq 0$, then taking by taking $H_1 = 0$ on the previous equation we see that

$$(0, [H_2, w]) = \beta(H_2) \cdot (v, w) \tag{1.184}$$

and $\beta(H_2)v = 0 \Rightarrow \beta(H_2) = 0, \ \forall H_2 \in \mathfrak{h}_2$. Therefore $\beta = 0$ and by applying the same reasoning for $H_2 = 0$ we will conclude that $\alpha \in \Delta_1$ while $w = 0$. So this proves the other inclusion.

($\Rightarrow$). Suppose that $(\mathfrak{h}^*, \Delta)$ is decomposable, so there exist decomposition

$$\mathfrak{h}^* = \mathfrak{h}_1^* \oplus \mathfrak{h}_2^*, \ \Delta_i = \Delta \cap \mathfrak{h}_i^*, \Delta = \Delta_1 \cup \Delta_2. \tag{1.185}$$

In this condition, we claim that $\mathfrak{g}_1 = \bigoplus_{\alpha \in \Delta_1} \mathfrak{g}_\alpha$ is a proper ideal in $\mathfrak{g}$. In fact, we see that $\mathfrak{g}_\beta, \ \beta \in \Delta_2$ is not contained in $\mathfrak{g}_1$. Now, it's clear that $\mathfrak{g}_1$ is invariant by the adjoint action of $X_\alpha, \ \alpha \in \Delta_1$. It also happens that it is invariant by the adjoint action of $X_\beta, \ \beta \in \Delta_2$. In fact if $[X_\beta, X_\alpha] \neq 0$ for $\alpha \in \Delta_1$ and $\beta \in \Delta_2$, then

$$\alpha + \beta \in \Delta \tag{1.186}$$

which can not happen, as we saw before. □

Lemma 1.8.6. We have a nice criterion to find irreducible root systems (V, Δ). In fact, if for any $\alpha, \beta \in \Delta$ we have a sequence $\gamma_0, \gamma_1, \ldots, \gamma_{n+1}$ such that

- $\gamma_0 = \alpha$ and $\gamma_{n+1} = \beta$
- $\gamma_i + \gamma_{i+1} \in \Delta$

then Δ is irreducible.

Proof: To show this, assume that $V = V_1 \oplus V_2$ and $\Delta = \Delta_1 \cup \Delta_2,\ \Delta_i = V_i \cap \Delta$. Let $\alpha \in \Delta_1, \beta \in \Delta_2$ and $(\gamma_i)_0^{n+1}$ a sequence of roots satisfying the condition of the lemma. Then we see that $\alpha + \gamma_1 \in \Delta \overset{\alpha \in V_1}{\Rightarrow} \gamma_1 \in \Delta_1$. So, proceding by induction and using the condition $\gamma_i + \gamma_{i+1} \in \Delta$ we will conclude that $\gamma_i \in \Delta_1,\ \forall i$. In particular, this we have $\beta = \gamma_{n+1} \in \Delta_1$, which is a contradiction. □

1.8.2 Root systems of simple Lie algebras

An useful about invariant forms on simple Lie algebras.

Lemma 1.8.7. Any invariant bilinear form $(\cdot, \cdot)$ on a simple Lie algebra is symmetric.

Proof: Observe that for all $X, Y, Z \in \mathfrak{g}$ we have

$$\begin{aligned} ([X,Y],Z) &= (X,[Y,Z]) \\ &= -(X,[Z,Y]) \\ &= -([X,Z],Y) \\ &= ([Z,X],Y) \\ &= (Z,[X,Y]). \end{aligned} \tag{1.187}$$

Since a simple Lie algebra is perfect, i.e., $[\mathfrak{g}, \mathfrak{g}] = \mathfrak{g}$, we are done. □

Lemma 1.8.8. Let $\mathfrak{g} \subset \mathrm{End}(V)$ be a simple Lie algebra. Then any invariant form will be a multiple of the trace form:

$$(X, Y) = \mathrm{Tr}(XY). \tag{1.188}$$

Proof: We prove that the vector space of invariant bilienar forms is 1. Let $(\cdot, \cdot)_i,\ i = 1, 2$ two invariant forms and define $\nu : \mathfrak{g} \overset{\cong}{\to} \mathfrak{g}^* : X \mapsto (X, \cdot)_1$. Then, for any $X \in \mathfrak{g}$, we define $M(X) = \nu^{-1}(X, \cdot\,)_2$. It's clear that $M \in \mathrm{End}(\mathfrak{g})$. We have more: M is lie algebra morphism

$\mathfrak{g} \to \mathfrak{g}$. In fact for any $X, Y, Z \in \mathfrak{g}$ we have

$$\begin{aligned}(M[X,Y],Z)_1 &= ([X,Y],Z)_2 \text{ (definition of } M)\\ &= (X,[Y,Z])_2 \text{ (invariance of } (\cdot,\cdot)_2)\\ &= (Mx,[Y,Z])_1 \text{ (definition of } M)\\ &= ([Mx,Y],Z)_1 \text{ (invariance of } (\cdot,\cdot)_1).\end{aligned} \tag{1.189}$$

So, by nondegeneracy of $(\cdot,\cdot)_1$ we have $M[X,Y] = [Mx,Y]$, i.e., $M \circ \mathrm{ad}(X) = \mathrm{ad} \circ M(X)$ and M is a intertwining operator $\mathfrak{g} \to \mathfrak{g}$ relative to the adjoint action of $\mathfrak{g}$ on itself. Therefeore $M = c\mathbb{1}_{\mathfrak{g}}$ by the Schur's Lemma. This means that $c(\cdot,\cdot)_1 = (\cdot,\cdot)_2$.

Finally, since the trace form is an invariant bilinear form, we get the result. □

Next, we will compute the dimension, rank and the Cartan matrix of some simple classical Lie algebras.

Family A_n:

This family corresponds to the Lie algebras $\mathfrak{sl}_n,\ n \geq 1$. First, we prove that $\mathfrak{sl}_n$ is a semisimple Lie algebra. We will use lemma (1.3.17). Since $I_V \notin \mathfrak{sl}_n$ we need to show that its tautological representation is irreducible. Let $W \subset V = \mathbb{R}^n$ be a nonzero invariant subspace and let $w \in W \setminus 0$. Write $w = (a_1, \ldots, a_n)$ and let j be such that $a_j \neq 0$. Then $E_{1j}(w) = a_j e_1 \in W \Rightarrow e_1 \in W$. We continue in this way by applying $E_{21}, E_{32}, \ldots, E_{n,n-1}$ to conclude that $e_1, \ldots, e_n \in W$, so $W = V$. Now, we apply lemma (1.4.13) to conclude that $\mathfrak{h} = \{\mathrm{Diag}(\lambda_1, \ldots, \lambda_n) : \sum_j \lambda_j = 0\}$ is a Cartan subalgebra of $\mathfrak{sl}_n$. From this we see that $E_{ij} \subset \mathfrak{g}_{L_i - L_j},\ i \neq j$ and since the weight spaces are one dimensional we get $\mathfrak{g}_{L_i - L_j} = \mathbb{C}E_{ij}$. So we get the weightspace decomposition of $\mathfrak{sl}_n$:

$$\mathfrak{sl}_n = \mathfrak{h} \oplus \bigoplus_{i \neq j} \mathfrak{g}_{L_i - L_j},\ \mathfrak{g}_{L_i - L_j} = \mathbb{C}E_{ij},\ \mathfrak{h} = \{\mathrm{Diag}(\lambda_1, \ldots, \lambda_n) : \sum_j \lambda_j = 0\}. \tag{1.190}$$

So, the root system associated to $\mathfrak{sl}_n$ is:

$$\Delta = \{L_i - L_j : i \neq j,\ 1 \leq i, j \leq n\} \subset \mathrm{Span}(L_i)_{i=1}^n \cong \mathbb{R}^n. \tag{1.191}$$

Note that Δ lies in the subspace $\{a_1 L_1 + \cdots + a_n L_n : \sum_j a_j = 0\}$. Now, the choice of linear functional $f : L_i \mapsto n - i + 1$ yields positive roots:

$$\Delta_+ = \{L_i - L_j : i < j,\ 1 \leq i, j \leq n\}. \tag{1.192}$$

The simple roots are $\Pi = \{\alpha_1, \ldots, \alpha_{n-1}\},\ \alpha_i = L_i - L_{i+1}$. So far we got:

$$\dim(\mathfrak{sl}_n) = n^2 - 1,\ \mathrm{Rank}(\mathfrak{sl}_n) = n - 1,\ \#\Delta = n(n-1). \tag{1.193}$$

Now we need to compute the Cartan matrix of $\mathfrak{sl}_n$. Recall that if $\alpha, \beta \in \mathfrak{h}^*$, then

$$\beta(H_\alpha) = \frac{2}{(\alpha,\alpha)}\beta(\nu^{-1}(\alpha)) = 2\frac{(\alpha,\beta)}{(\alpha,\alpha)}. \tag{1.194}$$

So the Cartan matrix can be expressed by $(\alpha_j(H_{\alpha_i}))_{i,j=1}^{n-1}$. Furthermore, we know that if $(Y,X) \in \mathfrak{g}_{-\alpha} \times \mathfrak{g}_\alpha$, then $[X,Y] = K(X,Y)\nu^{-1}(\alpha)$. Now, define $H = [X,Y]$ and suppose that $[H,X] = 2X$. Then $\alpha(H) = K(X,Y)(\alpha,\alpha) = 2$, i.e., $H = [X,Y] = H_\alpha$. This shows that $H_{i,j} = [E_{ij}, E_{ji}] = \mathrm{Diag}(0,\dots,0,\hat{1},\dots,\hat{-1},0,\dots,0)$, ($\hat{} = i\text{th}, \hat{} = j\text{th}$ position) is the coroot relative to $L_i - L_j$, i.e., $H_{i,j} = H_{L_i - L_j}$, $i < j$.

Now define the form $(L_i, L_j) = \delta_{ij}$. Then we see that, with respect to this form we have

$$2\frac{(\alpha_i,\alpha_j)}{(\alpha_i,\alpha_i)} = \begin{cases} -1 & \text{if } |i-j| = 1 \\ 0 & \text{if } |i-j| > 1 \\ 2 & \text{if } i = j. \end{cases} \tag{1.195}$$

This agrees with the values $\alpha_j(H_{\alpha_i})$, so this orthonormal form must be a multiple of the Killing form transposed to $\mathfrak{h}^*$.

Finally we show that $\mathfrak{sl}_n$ is in fact a simple Lie algebra. To do this we use lemma (1.8.6): observe that if $\alpha = L_i - L_j, \beta = L_k - L_l$ ($i < j < k < l$), then we take

$$\alpha = \gamma_0 = L_i - L_j,\ \gamma_1 = L_j - L_k,\ \beta = \gamma_2 = L_k - L_l. \tag{1.196}$$

Family C_n:

This family corresponds to the simplectic Lie algebra $\mathfrak{sp}_{2n}$ defined by

$$\mathfrak{sp}_{2n} = \{X \in M_{2n}(\mathbb{C}) : X^T\Omega + \Omega X = 0\},\ \Omega = \begin{pmatrix} 0 & I_n \\ -I_n & 0 \end{pmatrix} \in M_{2n}(\mathbb{C}). \tag{1.197}$$

More explictly: $\mathfrak{sp}_{2n} = \left\{ \begin{pmatrix} A & B \\ C & -A^T \end{pmatrix} : B = B^T, C = C^T \right\}$.

- We claim that $\mathfrak{sp}_{2n}$ is semisimple. To show this we use lemma (1.3.17). Since $I_{2n} \notin \mathfrak{sp}_{2n}$ we only need to show that the tautological representation is irreducible. Observe that $E_{j,n+j}, E_{n+j,k}, E_{ij} - E_{n+j,n+i} \in \mathfrak{sp}_{2n}$, $(1 \le i,j \le n)$. If $W \subset V$ is a nonzero invariant subspace, then there exist $w = (a_1,\dots,a_{2n})^T \in W \setminus 0$. Without lost of generalization we can assume that $a_k \neq 0$ for some k, $(1 \le k \le n)$. If this is not the case, then there exist $a_{n+k} \neq 0$ for some k, $(1 \le k \le n)$ and we apply $E_{k,n+k}w = a_{n+k}e_k \in W$. Now, with the assumption that $a_k \neq 0$ we apply $E_{k,n+k}E_{n+k,k}w = a_k e_k \in W$ to conclude that $e_k \in W$. Finally, we can use the elements $E_{ik} - E_{n+k,n+i}$, $i = 1,\dots,n$ to conclude that $e_1,\dots,e_n \in W$ and again we use $E_{n+i,i}$ to conclude that $e_{n+1},\dots,e_{2n} \in W$. So $W = V$ and $\mathfrak{sp}_{2n}$ is semisimple.

- Since we know that $\mathfrak{sp}_{2n}$ is semisimple, we use the fact that

$$\text{Diag}(\lambda_1, \ldots, \lambda_n, -\lambda_1, \ldots, -\lambda_n) \in \mathfrak{sp}_{2n} \text{ for } \lambda_i \neq \lambda_j \tag{1.198}$$

to conclude that its diagonal elements forms a Cartan subalgebra

$$\mathfrak{h} = \{\text{Diag}(\lambda_1, \ldots, \lambda_n, -\lambda_1, \ldots, -\lambda_n) : \lambda_i \in \mathbb{C},\ i = 1, \ldots, n\}\,. \tag{1.199}$$

Remark 1.8.9. In the following, we will identify $\mathfrak{h} \cong \{\text{Diag}(\lambda_1, \ldots, \lambda_n) : \lambda_i \in \mathbb{C}\}$. So we will construct its weight space decomposition using linear functionals $L_i,\ i = 1, \ldots, n$. In fact we have the identification

$$L_{n+i} + L_i = 0 \tag{1.200}$$

on $\mathfrak{h}$.

The corresponding weight space decomposition is the following:

$$\begin{aligned} Y_{ij} &= E_{i,n+j} + E_{j,n+i}\ (1 \leq i \leq j \leq n) \\ X_{ij} &= E_{ij} - E_{n+j,n+i}\ (1 \leq i, j \leq n, i \neq j) \\ Z_{ij} &= E_{n+i,j} + E_{n+j,i}\ (1 \leq i \leq j \leq n). \end{aligned} \tag{1.201}$$

To find the corresponding weights we compute

$$\begin{aligned} [H, Y_{ij}] &= (\lambda_i + \lambda_j)E_{i,n+j} + (\lambda_j + \lambda_i)E_{j,n+i} = (L_i + L_j)(H)Y_{ij} \\ [H, X_{ij}] &= (\lambda_i - \lambda_j)E_{ij} - (-\lambda_j + \lambda_i)E_{n+j,n+i} = (L_i - L_j)(H)X_{ij} \\ [H, Z_{ij}] &= (-\lambda_i - \lambda_j)E_{n+i,j} + (-\lambda_j - \lambda_i)E_{n+j,i} = -(L_i + L_j)(H)Z_{ij}. \end{aligned} \tag{1.202}$$

So

$$\Delta = \{L_i - L_j : i \neq j\} \cup \{\pm(L_i + L_j) : i \leq j\},\ i, j \in \{1, \ldots, n\}. \tag{1.203}$$

The choice of linear functional $f : \mathfrak{h}^* \to \mathbb{R}$ such that $f(L_1) > \cdots > f(L_n) > 0$ yields positive roots given by

$$\Delta_+ = \{L_i - L_j : i < j\} \cup \{L_i + L_j : i \leq j\}. \tag{1.204}$$

By excluding roots which are um of other positive roots we get the set of simple roots

$$\Pi = \{L_1 - L_2, \ldots, L_{n-1} - L_n, 2L_n\}. \tag{1.205}$$

We will write $\alpha_i = L_i - L_{i+1}, i = 1, \ldots, n-1$ and $\alpha_n = 2L_n$ to the simple roots. So far we got

$$\dim(\mathfrak{sp}_{2n}) = 2n^2 + n,\ \text{Rank}(\mathfrak{sp}_{2n}) = n,\ \#\Delta = 2n^2. \tag{1.206}$$

- Now we compute the Cartan matrix of $\mathfrak{sp}_{2n}$. To do this we use the same approach made in the case A_n. Remember that $\mathbb{C}Y_{ij} = \mathfrak{g}_{L_i+L_j}, \mathbb{C}X_{ij} = \mathfrak{g}_{L_i-L_j}$ and $\mathbb{C}Z_{ij} = \mathfrak{g}_{-(L_i+L_j)}$. So we find the coroots:

$$\begin{aligned} &[Y_{ij}, Z_{ij}] = H_{i,n+j} + H_{j,n+i} = H_{L_i+L_j},\ i \neq j \\ &[Y_{ii}/2, Z_{ii}/2] = H_{i,n+i} = H_{2L_i} \\ &[X_{ij}, X_{ji}] = H_{i,j} + H_{n+j,n+i} = H_{L_i-L_j},\ i < j. \end{aligned} \tag{1.207}$$

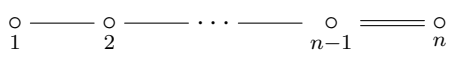
  Now we get

$$\alpha_j(H_{\alpha_i}) = \begin{cases} 2 & \text{if } i = j \\ 0 & \text{if } |i-j| > 1 \\ -1 & \text{if } |i-j| = 1 \end{cases} \qquad (1 \leq i, j \leq n) \tag{1.208}$$

  and

$$\alpha_n(H_{\alpha_i}) = \begin{cases} 0 & \text{if } i < n-1 \\ -2 & \text{if } i = n-1 \end{cases}, \ \alpha_i(H_{\alpha_n}) = \begin{cases} 0 & \text{if } i < n-1 \\ -1 & \text{if } i = n-1 \end{cases} \quad (1 \leq i \leq n) \tag{1.209}$$

  So, the Cartan matrix of $\mathfrak{sp}_{2n}$ is compatible with the form $(L_i, L_j) = \delta_{ij}$. Furthermore, its associated Dynkin diagram is:



  which is connected, so $\mathfrak{sp}_{2n}$ is simple.

Family D_n:

This family corresponds to the Lie algebra $\mathfrak{so}_{2n}$. It's usual presentation is as the set of matrices X satisfying $X^T + X = 0$ (skew symmetric matrices). However, this presentation is not useful (to us), because we can not use the lemma (1.4.13) to decribe its Cartan subalgebras.

In general, let (q, V) be a quadratic space, i.e., V is a vector space and $q : V \to \mathbb{F}$ is a quadratic form. So, by definition there exist a positive definite, symmetric, bilinear form $B : V \otimes V \to \mathbb{F}$ such that $q(v) = B(v, v), \forall v \in V$. In this setup, we define the group $O_{V,B} = \{g \in \mathrm{End}(V) : q(gv) = q(v),\ \forall v \in V\}$.

We say that two quadratic spaces (q_i, V_i) $(i = 1, 2)$ are isomorphic if there exist a isomorphism $T : V_1 \to V_2$ such that $q_2 \circ T = q_1$. So, with this definition we see that we have an isomorphism

$$O_{V_1,B_1} \xrightarrow{\cong} O_{V_2,B_2} : A \mapsto TAT^{-1}. \tag{1.210}$$

In fact

$$\begin{aligned} q_2(TgT^{-1}v) &= q_1(gT^{-1}v) \\ &= q_1(T^{-1}v) \\ &= q_2(TT^{-1}v) \\ &= q_2(v). \end{aligned} \tag{1.211}$$

We can prove that $O_{V,B} \subset GL(V)$ is a (embedded) Lie group. With this in mind we can calculate its Lie algebra: If $X \in \mathfrak{o}_{V,B}$, then $\exp(tX) \in O_{V,B}$ and

$$0 = \frac{d}{dt}|_{t=0} B(\exp(tX)v, \exp(tX)v) = B(Xv, v) + B(v, Xv). \tag{1.212}$$

For a choice of basis, we identify $B(u, v) = u^T Av$, so the previous equation yields $X^T A + AX = 0$, that is $\mathfrak{so}_{V,B} = \{X : X^T A + AX = 0\}$, where A is the matrix representation of B.

Now, we will prove that over $\mathbb{C}$ we have $\mathfrak{so}_n \cong \{X :\in M_n(\mathbb{C}) : X^T F + FX = 0\}$, where $F = \sum_{i=1}^n E_{n-i+1,i}$. The general case is similar to the following calculation for $n = 2$. At each step we multiply the first matrix by $L_P \circ R_{P^T}$ where P is an elementary matrix. So we do an operation on its lines and repeat the same operation on its collumns. The second matrix register only the matrix P, that is we only execute the corresponding row operations on the second matrix.

$$\begin{aligned}
\left(\begin{array}{cc|cc} 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 \end{array}\right) &\sim \left(\begin{array}{cc|cc} 2 & 1 & 1 & 1 \\ 1 & 0 & 0 & 1 \end{array}\right) \\
&\sim \left(\begin{array}{cc|cc} 2 & 0 & 1 & 1 \\ 0 & -\frac{1}{2} & -\frac{1}{2} & \frac{1}{2} \end{array}\right) \\
&\sim \left(\begin{array}{cc|cc} 1 & 0 & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ 0 & -\frac{1}{2} & -\frac{1}{2} & \frac{1}{2} \end{array}\right) \\
&\sim \left(\begin{array}{cc|cc} 1 & 0 & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ 0 & 1 & -\frac{i}{\sqrt{2}} & \frac{i}{\sqrt{2}} \end{array}\right)
\end{aligned} \tag{1.213}$$

The last line has a crucial step where we *need* to introduce complex numbers. In fact, over $\mathbb{R}$, following the same procedure we can show that each symmetric matrix is similar to a matrix

$$\mathrm{Diag}(\underbrace{1, \ldots, 1}_{p \text{ times}}, \underbrace{-1, \ldots, -1}_{n \text{ times}}) = \mathrm{Diag}(1^p, (-1)^n). \tag{1.214}$$

The number $p - n$ is called the signature of the matrix and is an invariant: Suppose that $A \in M_m(\mathbb{R})$ is a symmetric matrix and $v_1, \ldots, v_n,\ w_1, \ldots, w_n$ are two basis of $\mathbb{R}^m$ such that the corresponding matrix representation of A is $\mathrm{Diag}(1^p, (-1)^n),\ \mathrm{Diag}(1^{p'}, (-1)^{n'})$ (resp.). Let $V = \mathrm{Span}(v_1, \ldots, v_p)$ and $W = \mathrm{Span}(w_{p+1}, \ldots w_n)$. Now, if $v \in V \cap W$, then

$$v^T Av \geq 0 \ \text{ and } \ v^T Av \leq 0 \Rightarrow v = 0 \tag{1.215}$$

because A is positive definite by assumption. So $V \cap W = 0$ and

$$\begin{aligned}
&\dim(V + W) = \dim(V) + \dim(W) = p + n' \leq \dim(\mathbb{R}^m) = m \\
&\quad \Rightarrow p + (m - p') \leq m \\
&\quad \Rightarrow p \leq p'.
\end{aligned} \tag{1.216}$$

By symmetry we conclude that $p = p'$.

Now, lets compute the root system of $\mathfrak{so}_{2n}$. By the previous discussion, we see that over $\mathbb{C}$ all symmetric matrices are similar. So we use the realization $F = \begin{pmatrix} 0 & I_n \\ I_n & 0 \end{pmatrix}$ for the bilinear form. With this realization we get

$$\mathfrak{so}_{2n} \cong \{X \in M_{2n}(\mathbb{C}) : X^T F + FX = 0\} = \left\{ \begin{pmatrix} A & B \\ C & -A^T \end{pmatrix} : C = -C^T, B = -B^T \right\}. \quad (1.217)$$

- Claim: $\mathfrak{so}_{2n}$ is semisimple. Define

$$\begin{aligned} X_{ij} &= E_{ij} - E_{n+j,n+i} \\ Y_{ij} &= E_{i,n+j} - E_{j,n+i} \\ Z_{ij} &= E_{n+i,j} - E_{n+j,i} \end{aligned} \quad (1.218)$$

  It's clear that $I_{2n} \notin \mathfrak{so}_{2n}$. Now we show that its tautological representation is irreducible. Let $W \subset \mathbb{C}^{2n}$ be a nonzero invariant subspace and $w = (a_1, \ldots, a_{2n})^T \in W \backslash 0$. If $\exists\ a_k \neq 0$ $(1 \leq k \leq n)$, then $X_{1k} w = a_k e_1 \in W$ and by applying $X_{i1},\ i = 1, \ldots, n$ we see that $e_1, \ldots, e_n \in W$. Furthermore, $Z_{ij} e_j = e_{n+j} \in W\ i \neq j$ and we see that $W = \mathbb{C}^{2n}$. However, if $a_k = 0, \forall k = 1, \ldots, n$ then $a_{n+k} \neq 0$ for some $1 \leq k \leq n$ because $w \neq 0$. So we apply $X_{1i} Y_{ik} w = a_{n+k} e_1,\ i \neq k$ to conclude that $e_1 \in W$ and repeat all the process. By lemma (1.3.17), we see that $\mathfrak{so}_{2n}$ is semisimple.

- Now, since

$$\mathrm{Diag}(\lambda_1, \ldots, \lambda_n - \lambda_1, \ldots, -\lambda_n) \in \mathfrak{so}_{2n}\ \lambda_i \neq \lambda_j \quad (1.219)$$

  we see that $\mathfrak{h} = \{\mathrm{Diag}(\lambda_1, \ldots, \lambda_n - \lambda_1, \ldots, -\lambda_n) :\ \lambda_i \in \mathbb{C}\}$ is a Cartan subalgebra by lemma (1.4.13).

- We show that

$$\begin{aligned} X_{ij} &= E_{ij} - E_{n+j,n+i} \\ Y_{ij} &= E_{i,n+j} - E_{j,n+i} \\ Z_{ij} &= E_{n+i,j} - E_{n+j,i} \end{aligned} \quad (1.220)$$

  give a decomposition of $\mathfrak{so}_{2n}$ into weight spaces. In fact, if $H = \mathrm{Diag}(\lambda_1, \ldots, \lambda_n - \lambda_1, \ldots, -\lambda_n)$, then

$$\begin{aligned} &[H, X_{ij}] = (\lambda_i - \lambda_j) E_{ij} - (-\lambda_j + \lambda_i) E_{n+j,n+i} = (L_i - L_j)(H) X_{ij}\ (i \neq j) \\ &[H, Y_{ij}] = (\lambda_i + \lambda_j) E_{i,n+j} - (\lambda_j + \lambda_i) E_{j,n+i} = (L_i + L_j)(H) Y_{ij}\ (i \neq j) \\ &[H, Z_{ij}] = (-\lambda_i - \lambda_j) E_{n+i,j} - (-\lambda_j - \lambda_i) E_{n+j,i} = -(L_i + L_j)(H) Z_{ij}\ (i \neq j) \end{aligned} \quad (1.221)$$

  Then we deduce the root system of $\mathfrak{so}_{2n}$:

$$\Delta = \{L_i - L_j : i \neq j\} \cup \{\pm(L_i + L_j) : i < j\},\ i, j \in \{1, \ldots, 2n\}. \quad (1.222)$$

Choose linear functional $f(L_1) > \cdots > f(L_n) > 0$, then we get

$$\Delta_+ = \{L_i - L_j : i < j\} \cup \{L_i + L_j : i < j\} = \{L_i \pm L_j : i < j\} \tag{1.223}$$

and simple roots $\Pi = \{\alpha_1, \ldots, \alpha_n\}$ where $\alpha_i = L_i - L_{i+1}$ $(i < n)$ and $\alpha_n = L_{n-1} + L_n$. Then we already got

$$\dim(\mathfrak{so}_{2n}) = 2n^2 - n.\ \mathrm{Rank}(\mathfrak{so}_{2n}) = n,\ \#\Delta = 2(n^2 - n). \tag{1.224}$$

Remark 1.8.10. Since $L_{n+i} + L_i = 0$ on $\mathfrak{h}$, we identify $\mathfrak{h}^* \cong \mathrm{Span}(L_i)_{i=1}^n$.

- Next we compute its Cartan matrix. To do this remember that for $i \neq j$ we have

$$\mathbb{C}X_{ij} = \mathfrak{g}_{L_i - L_j},\ \mathbb{C}Y_{ij} = \mathfrak{g}_{L_i + L_j},\ \mathbb{C}Z_{ij} = \mathfrak{g}_{-(L_i + L_j)} \tag{1.225}$$

and

$$[Y_{ij}, Z_{ij}] = H_{L_i + L_j},\ [X_{ij}, X_{ji}] = H_{L_i - L_j}. \tag{1.226}$$

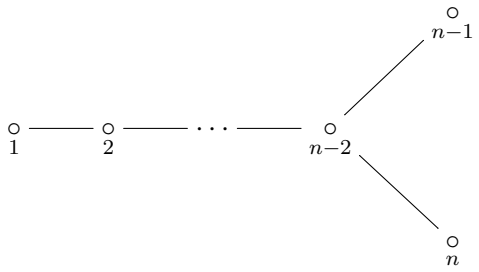
Now we get

$$\alpha_j(H_{\alpha_i}) = \begin{cases} 2 & \text{if } i = j \\ 0 & \text{if } |i - j| > 1 \\ -1 & \text{if } |i - j| = 1 \end{cases} \tag{1.227}$$

and

$$\alpha_n(\alpha_i) = \begin{cases} -1 & \text{if } i = n-2 \\ 0 & \text{if } i = n-1 \\ 0 & \text{if } i < n-2 \end{cases},\ \alpha_i(H_{\alpha_n}) = \begin{cases} -1 & \text{if } i = n-2 \\ 0 & \text{if } i = n-1 \\ 0 & \text{if } i < n-2 \end{cases} \tag{1.228}$$

Its Dynkin diagram is



which is connected, so $\mathfrak{so}_{2n}$ is simple.

1.8.3 Cartan matrices and Dynkin diagrams

The idea here is to encode a semisimple Lie algebra by a graph, called Dynkinn diagram. We need to remember a series of facts.

- (Chevalley). Any two Cartan subalgebras $\mathfrak{h}, \mathfrak{h}'$ in a semisimple Lie algebra $\mathfrak{g}$ are conjugate. That is, there exist $\phi \in \mathrm{Ad}(G) \subset \mathrm{Aut}(\mathfrak{g})$ such that $\phi(\mathfrak{h}) = \mathfrak{h}'$. So

  The root system of a semisimple Lie algebra does not deppend on the choice of Cartan subalgebra.

  In fact, if Δ' is the root system associated to the Cartan subaglebra $\mathfrak{h}'$, then for each $\beta \in \Delta'$ we have $\mathfrak{g}_\beta = \mathfrak{g}_{\phi^*\beta}$. So $\phi^* : \Delta' \to \Delta$ and since $K(\phi(X), \phi(Y)) = K(X, Y)$, $\forall X, Y \in \mathfrak{g}$ we also get that $(\phi^*\alpha, \phi^*\beta) = (\alpha, \beta)$. Therefore the linear map $\phi^* : (\mathfrak{h}')^* \to \mathfrak{h}^*$ induces an isomorphism of root systems $\Delta' \cong \Delta$.

- Now, let (V, Δ) be a root system. Now let $f, f' : V \to \mathbb{C}$ be two linear functionals that induces positive roots Δ_+, Δ'_+. Then, there exist an element of the Weyl group W of Δ such that $w(\Delta_+) = \Delta'_+$. In particular we have $w(\Pi) = \Pi'$, this follows from the fact that each positive root lies in the orbit of a simple root with respect to the action of the Weyl group. Since $W \subset O(V)$, we have a matrix that is independent of the choice of positive roots:

$$A = (A_{ij}),\ A_{ij} = 2\frac{(\alpha_i, \alpha_j)}{(\alpha_i, \alpha_i)} \tag{1.229}$$

  where $\Pi = \{\alpha_1, \ldots, \alpha_l\}$ is some listing of roots. Of course, this matrix may deppend of the listing of the roots, but this is not important.

Lemma 1.8.11. Two root systems with the same Cartan matrix are isomorphic.

Proof: Let $(V, \Delta), (V', \Delta')$ be two root systems with the same Cartan matrix, that is

$$\frac{(\alpha_i, \alpha_j)}{(\alpha_j, \alpha_j)} = \frac{(\beta_i, \beta_j)}{(\beta_j, \beta_j)} \tag{1.230}$$

where $\Pi = \{\alpha_1, \ldots, \alpha_l\}, \Pi' = \{\beta_1, \ldots, \beta_l\}$. Then

$$\frac{(\alpha_i, \alpha_i)}{(\alpha_j, \alpha_j)} = \frac{(\beta_i, \beta_i)}{(\beta_j, \beta_j)},\ \forall i, j. \tag{1.231}$$

Fixing $j = 1$ we see that $(\alpha_i, \alpha_i) = k(\beta_i, \beta_i)$ for $k > 0$. Therefore we see that $(\alpha_i, \alpha_j) = k(\beta_i, \beta_j)$, $\forall i, j$. Now, the map $\phi : \alpha_i \mapsto \sqrt{k}\beta_i$ defines a map $V \to V'$, which is an isometry by the previous equations. Moreover, since homotheties does not change root systems, we will assume that $k = 1$.

Now it remains to show that $\phi(\Delta) = \Delta'$. We know that $\phi(\Pi) = \Pi'$, so $\phi(\Delta_+) \subset \mathbb{Z}_{\geq 0}\Pi'$. However, if $\beta \in \Delta'_+ \setminus \Pi$, then there is a set of simple reflections $r_{\beta_{i_1}}, \ldots, r_{\beta_{i_k}}$ such that $r_{\beta_{i_1}} \cdots r_{\beta_{i_k}}(\beta) \in \Pi'$. But since $r_{\phi(\alpha)} = \phi \circ r_\alpha \circ \phi^{-1}$, this means that $\phi \circ r_{\alpha_{i_1}} \cdots r_{\alpha_{i_k}}(\phi^{-1}\beta) \in \Pi'$. Therefore

$$\begin{aligned} & r_{\alpha_{i_1}} \cdots r_{\alpha_{i_k}}(\phi^{-1}\beta) \in \phi^{-1}(\Pi') = \Pi \\ & \Rightarrow r_{\alpha_{i_1}} \cdots r_{\alpha_{i_k}}(\phi^{-1}\beta) = \alpha_i, \text{ for some } i \\ & \Rightarrow \phi^{-1}\beta = r_{\alpha_{i_k}} \cdots r_{\alpha_{i_1}}(\alpha_i) \in \Delta_+ \text{ (because } (\alpha_i, \alpha_j) \leq 0) \\ & \Rightarrow \Delta'_+ \subset \phi(\Delta_+) \\ & \Rightarrow \Delta'_+ = \phi(\Delta_+) \text{ (by symmetry).} \end{aligned} \tag{1.232}$$

□

Remark 1.8.12. We also can show that any root system can be recovered from its Cartan matrix.

- Now we know that the Cartan matrix is an invariant of the root system, so it's an invariant of the semisimple Lie algebra associated to this root system. Then we will try to classify Cartan matrices.

  Remember that

$$4\cos^2(\theta) = 4\frac{(\alpha,\beta)^2}{\|\alpha\|^2\|\beta\|^2} = 2\frac{(\alpha,\beta)}{(\alpha,\alpha)} \cdot 2\frac{(\beta,\alpha)}{(\beta,\beta)} \leq 4 \tag{1.233}$$

  while $2\frac{(\alpha,\beta)}{(\alpha,\alpha)}, 2\frac{(\beta,\alpha)}{(\beta,\beta)} \in \mathbb{Z}$. So, if θ is the angle between two different roots α_i and α_j, we get $n_{ij} = A_{ij}A_{ji} \in \{0,1,2,3\}$. It can not be 4, because this would implies $\theta \in \{0, \pi\}$, which is not the case. Now, we immediately get tha main properties of Cartan matrices, $A = (A_{ij})$ must satisfy the following conditions:

  1. $A_{ii} = 2$, $\forall i$
  2. If $A_{ij} = -2, -3$ then $A_{ji} = -1$
  3. $A_{ij} = 0 \iff A_{ji} = 0$

For example, the Rank 1 and 2 Cartan matrices are:

$$\begin{gathered}(2)\\ \begin{pmatrix}2 & 0\\ 0 & 2\end{pmatrix}\\ \begin{pmatrix}2 & -1\\ -1 & 2\end{pmatrix}\\ \begin{pmatrix}2 & -1\\ -2 & 2\end{pmatrix}\ \begin{pmatrix}2 & -2\\ -1 & 2\end{pmatrix}\\ \begin{pmatrix}2 & -1\\ -3 & 2\end{pmatrix}\ \begin{pmatrix}2 & -3\\ -1 & 2\end{pmatrix}\end{gathered} \tag{1.234}$$

- To classify Cartan matrices of higher ranks, we will need to introduce a graph called Dynkin diagram. The Dynkin diagram associated to a Cartan matrix $A = (A_{ij})_{1\leq i,j\leq l}$ is a graph with l vertex $V_1, \ldots, V_l$, where each pair of vertex (V_i, V_j) is connected by $n_{ij} = A_{ij}A_{ji}$ edges.

  Remark 1.8.13. The rank 2 Dynkin diagrams are:

- The Dynkin diagram of a simple Lie algebra must be connected. To see this we need the following fact: If (V, Δ) is an irreducible root system with some choice of simple roots Π, then we can not have decomposition $\Pi = \Pi_1 \cup \Pi_2,\ \Pi_1 \perp \Pi_2$. In fact, if such decomposition exists, let $V_i = \mathrm{Span}(\Pi_i), i = 1, 2$. Then we claim that $\Delta = \Delta_1 \cup \Delta_2,\ \Delta_i = V_i \cap \Delta$. First we see that the Weyl group respect such decomposition: Let $r_\alpha,\ \alpha \in \Pi_1$ be a simple reflection, then

$$\begin{aligned} &r_\alpha(\lambda) = \lambda - 2\frac{(\alpha,\lambda)}{(\alpha,\alpha)}\alpha \in V_1 \ \text{ if } \lambda \in V_1\\ &r_\alpha(\lambda) = \lambda \in V_2 \ \text{ if } \lambda \in V_2 \quad (\text{because } \alpha \perp V_2)\end{aligned} \tag{1.235}$$

  So, if $\beta \in \Delta_+$ then there exist $w \in W = \langle r_\alpha : \alpha \in \Pi\rangle$ such that $w(\beta) \in \Pi$. Assume that $w(\beta) \in \Pi_1$ and let $\beta_i,\ i = 1, 2$ be the components of β in $V_i,\ i = 1, 2$. Then

$$w(\beta) = w(\beta_1) + w(\beta_2) \in \Pi_1 \Rightarrow w(\beta_2) = 0. \tag{1.236}$$

  So $w(\beta - \beta_1) = 0 \Rightarrow \beta - \beta_1 = 0$, i.e. $\beta \in V_1$.

  From this result, we see that the Dynkin diagram of a simple Lie algebra must be connected, otherwise any disjoint connected components will create a orthogonal decomposition in the set of simple roots.

- The last observation is that for any Dynkin diagram with l vertex we can associate a quadratic form by the formula:

$$Q(X_1, \ldots, X_l) = 2\sum_i X_i^2 - \sum_{i \neq j} \sqrt{n_{ij}} X_i X_j. \tag{1.237}$$

Of course, the associated matrix of this form is just the Cartan matrix associated to the Dynkin diagram. We can think that there is an ambiguity here, because this form deppends on the way we label a Dynkin diagram:

Dynkin diagram	Cartan matrix
	$\begin{pmatrix} 2 & -1 & 0 \\ -1 & 2 & -2 \\ 0 & -1 & 2 \end{pmatrix}$
	$\begin{pmatrix} 2 & 0 & -1 \\ 0 & 2 & -2 \\ -1 & -1 & 2 \end{pmatrix}$

However, the resulting matrices are equivalent, in the sense that they differ by $L_P \circ R_{P^T}$, where P is some elementary matrix. In the example given above P is the matrix that interchange lines 2 and 3. So, the resulting quadratic forms are equal up to a choice of basis.

Lemma 1.8.14. The form $Q(X_1, \ldots, X_l)$ is positive definite if it comes from a root system of a semisimple Lie algebra.

Proof: Since $n_{ij} = A_{ij}A_{ji} = \frac{(\alpha_i,\alpha_j)^2}{(\alpha_i,\alpha_i)(\alpha_j,\alpha_j)}$, we see that $\sqrt{n_{ij}} = -\frac{(\alpha_i,\alpha_j)}{\|\alpha_i\|\|\alpha_j\|}$ and

$$\begin{aligned} Q(X_1, \ldots, X_l) &= \sum_{i,j} \frac{(\alpha_i, \alpha_j)}{\|\alpha_i\|\|\alpha_j\|} X_i X_j \\ &= (\sum_i \frac{X_i \alpha_i}{\|\alpha_i\|}, \sum_j \frac{X_j \alpha_j}{\|\alpha_j\|}) \geq 0. \end{aligned} \tag{1.238}$$

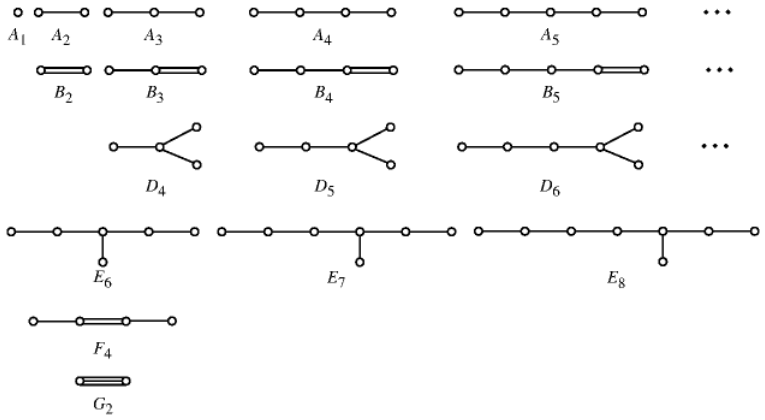
and the equality happens only if $\sum_i \frac{\alpha_i}{\|\alpha_i\|} X_i = 0$, i.e., $X_i = 0 \ \forall i$ because α_i are all linearly independent.

Lemma 1.8.15. (Sylvester criterion) A symmetric matrix $A \in M_n(\mathbb{R})$ is positive definite if $xAx^T > 0, \forall x \neq 0$. The Sylvester criterion says that such is the case if and only if the leading minors

$$\begin{pmatrix} a_{11} & \cdots & a_{1i} \\ \cdots & \cdots & \cdots \\ a_{i1} & \cdots & a_{ii} \end{pmatrix}, \ 1 \leq i \leq n \tag{1.239}$$

have positive determinant.

- So to classify Dynkin diagrams, we will restrict our attention to graphs that are connected, have at most 3 edges connecting each pair of vertex and with positive definite quadratic form. We claim that each of the following graphs satisfies these properties:



They are cleary connected and with at most 3 edges connecting each vertex. Now, we need to show that the associated quadratic form is positive definite. To do this, we look first at rank 2 Dynkin diagrams and observe that the associated Cartan matrices satisfies the Sylvester criterion. Now, for the remaning familes $A_l, B_l, D_l,\ l \geq 3$ we apply an induction. To do this we observe that each such graph Γ_l has a vertex with only one simple edge. We number this vertex by l and the other vertex connected to it we number by $l-1$. Therefore

$$\begin{aligned}\det(\Gamma_l) &= \det \begin{pmatrix} & & & & 0 \\ & & & & \vdots \\ & & & & 0 \\ & & & 2 & -1 \\ 0 & \cdots & 0 & -1 & 2 \end{pmatrix} \\ &= 2\det(\Gamma_{l-1}) - (-1)\det \begin{pmatrix} \Gamma_{l-2} & \begin{matrix} 0 \\ -1 \end{matrix} \end{pmatrix} \\ &= 2\det(\Gamma_{l-1}) - \det(\Gamma_{l-2}).\end{aligned} \tag{1.240}$$

where $\Gamma_{l-1}, \Gamma_{l-2}$ are the Dynkin diagrams obteined by removing the vertex l and $l, l-1$ respectively. By taking the first cases we see that $\det(\Gamma_l) > 0, \forall l \geq 1$ for each family A_l, B_l, D_l. To the exceptional families we can do just an inspection.

- Now we will construct another set of graphs called extended Dynkin diagrams. To do this, let (V, Δ) be a root system with simple roots $\Pi = \{\alpha_1, \ldots, \alpha_l\}$. We define the

extended Cartan matrix by

$$\widetilde{A}_{ij} = 2\frac{(\alpha_i, \alpha_j)}{(\alpha_i, \alpha_i)},\ 1 \leq i, j \leq l+1 \tag{1.241}$$

where $\alpha_{l+1} = -\theta$ (θ being the highest root). We define the associated extended Dynkin diagram in the same way.

The extended Cartan matrix have similar properties of standard Cartan matrices. That is $\widetilde{A}_{ii} = 2\ \forall i, \widetilde{A}_{ij} \leq 0\ (i \neq j)$ and $0 \leq \widetilde{A}_{ij}\widetilde{A}_{ji} \leq 3\ (i \neq j)$. To see this, we observe that if θ is the highest root, then $(\theta, \alpha_i) \geq 0\ \forall i$. By contradiction, suppose that this is not the case, i.e., there exist $i\ (1 \leq i \leq l)$ such that $(\theta, \alpha_i) < 0$. Since θ is maximal, we have $\theta + \alpha_i \notin \Delta_+$, so $q = 0$. Therefore

$$0 \leq p = 2\frac{(\theta, \alpha_i)}{(\alpha_i, \alpha_i)} < 0 \tag{1.242}$$

which is a contradiction. Furthermore, the inequality $0 \leq 2\frac{(\theta,\alpha_i)}{(\alpha_i,\alpha_i)} \cdot 2\frac{(\alpha_i,\theta)}{(\theta,\theta)} \leq 3$ follows from the fact that $\alpha_i, \theta \in \Delta$, so we have the string condition satisfied and $2\frac{(\theta,\alpha_i)}{(\alpha_i,\alpha_i)}, 2\frac{(\alpha_i,\theta)}{(\theta,\theta)} \in \mathbb{Z}$. Finally, this number is just $4\cos^2(\gamma)$, where γ is the angle between θ and α_i. Of course, since $\theta \notin \Pi$ ($\#\Pi > 1$) we can not have $\gamma = 0$.

Remark 1.8.16. Let $A \in M_n(\mathbb{F})$ be a $n \times n$ matrix over a field $\mathbb{F}$ and $1 \leq i_1 \leq \cdots \leq i_r \leq n$. We define the corresponding principal minor of A with respec to $I = \{i_1, \ldots, i_r\}$ by

$$\det(A_{i_k i_l})_{1 \leq k,l \leq r} \tag{1.243}$$

Finally, we claim that all the principal minors of $\widetilde{A}$ are positive. First write

$$\widetilde{A} = \mathrm{Diag}(\tfrac{2}{(\alpha_i,\alpha_i)})_{i=1}^{l+1} \cdot Q,\ Q = ((\alpha_i, \alpha_j))_{i,j=1}^{l+1}. \tag{1.244}$$

Then for any subset $I \subset \{1, \ldots, l+1\}$ of l elemetns, the corresponding minor of Q is the associated gram matrix of $(\cdot, \cdot)$ is some basis. So, by Silvester criterion, all of its leading minors are positive. Then we see that all principal minors of $\widetilde{A}$ are positive.

Example 1.8.17. Compute the extended Dynkin diagram for the Lie algebra $\mathfrak{sl}_4$.

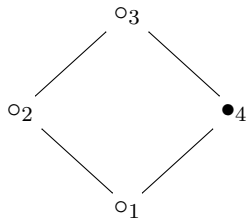
The roots are $\Delta = \{L_i - L_j : i \neq j\} \subset \bigoplus_{i=1}^4 \mathbb{C}L_i$. The functional $f(L_i) = n + 1 - i$ yields the positive roots $\Delta_+ = \{L_i - L_j : i < j\}$. So, in this case the simple roots are

$$\alpha_i = L_i - L_{i+1}, i = 1, 2, 3 \tag{1.245}$$

while the maximal root is $\theta = \sum_i \alpha_i = L_1 - L_4$. The extended Cartan matrix is

$$\begin{pmatrix} 2 & -1 & 0 & -1 \\ -1 & 2 & -1 & 0 \\ 0 & -1 & 2 & -1 \\ -1 & 0 & -1 & 2 \end{pmatrix} \tag{1.246}$$

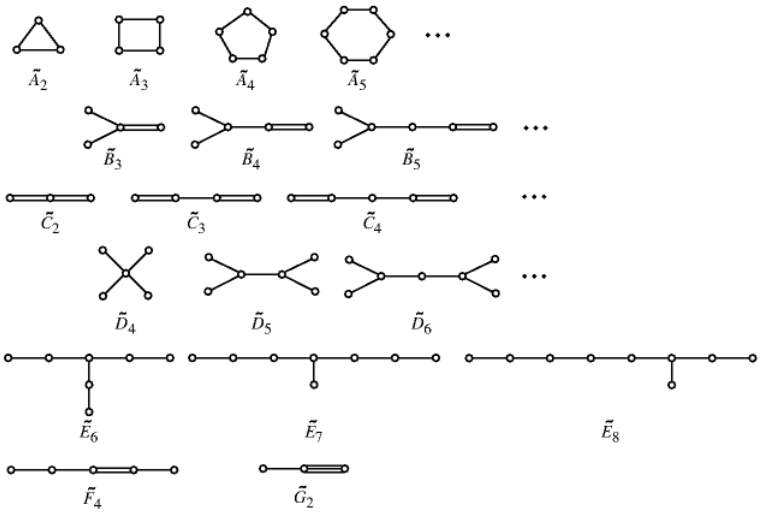
with associated extended Dynkin diagram given by:



Now, to see that $\det(\widetilde{A}) = 0$, we just observe that the last column is a linear combination of the previous collumns:

$$\begin{aligned} C_{l+1} &= \begin{pmatrix} 2\frac{(\alpha_1,\alpha_{l+1})}{(\alpha_1,\alpha_1)} \\ 2\frac{(\alpha_2,\alpha_{l+1})}{(\alpha_2,\alpha_2)} \\ \vdots \\ 2 \end{pmatrix} \\ &= -\sum_{j=1}^{l} m_j \begin{pmatrix} 2\frac{(\alpha_1,j)}{(\alpha_1,\alpha_1)} \\ 2\frac{(\alpha_2,\alpha_j)}{(\alpha_2,\alpha_2)} \\ \vdots \\ 2\frac{(\alpha_{l+1},\alpha_j)}{(\alpha_{l+1},\alpha_{l+1})} \end{pmatrix} \\ &= -\sum_j m_j C_j. \end{aligned} \tag{1.247}$$

Next, follows the list all the extended Dynkind diagrams obtained using the above procedure for extension.



- By the properties of extended Dynkin diagrams discussed before we see that any connected subgraph obtained from an extended Dynkin diagram by excluding vertices or decreasing the number of edges, satisfies the above properties of Dynkin diagrams: It's connected, with at most 3 edges and the associated quadratic form is positive definite (recall the discussion of principal minors of an extended Cartan matrix).

Theorem 1.8.18. The Dynkin diagram of a simple Lie algebra is one of the ones:

$$A_l,\ l \geq 1;\ B_l,\ l \geq 2;\ D_l,\ l \geq 4;\ E_6;\ E_7;\ E_8;\ F_4;\ G_2. \tag{1.248}$$

1.9 Cohomology

Aplication: To show that finite dimensional $\mathfrak{g}-$modules, $\mathfrak{g}$ semisimple, are completely reducible: If $A \subset B$ is a submodule, then there exist $C \subset B$ submodule such that $B \cong A \oplus C$.

- (**Main motivation**). Our initial step is to observe that if $A \subset B$ is a submodule, then we have a complementary subspace C. That is, at the level of vector spaces we have the splitting sequence:

$$0 \longrightarrow A \xrightarrow{i} B \underset{s}{\overset{p}{\rightleftarrows}} C \longrightarrow 0,\ p \circ s = \mathbb{1}_C.$$

  The point is: we do not know if s is a map of $\mathfrak{g}-$modules. In fact, we do not know even if B is a $\mathfrak{g}-$module.

- (**Extensions**). So we try the inverse path: Start with $\mathfrak{g}-$modules A, C and define the category of extensions of C by A as the set of exact sequences of $\mathfrak{g}-$modules

$$0 \longrightarrow A \xrightarrow{i} B \xrightarrow{p} C \longrightarrow 0$$

Where morphisms between two sequences are defined by a $\mathfrak{g}-$module morphism f such that the following diagram commutes:

$$\begin{array}{ccccccccc} 0 & \longrightarrow & A & \xrightarrow{i} & B & \xrightarrow{p} & C & \longrightarrow & 0 \\ \vdots & & \downarrow{\scriptstyle \mathbb{1}_A} & & \downarrow{\scriptstyle f} & & \downarrow{\scriptstyle \mathbb{1}_C} & & \vdots \\ 0 & \longrightarrow & A & \xrightarrow{i} & B' & \xrightarrow{p} & C & \longrightarrow & 0 \end{array}$$

By the five lemma, f is an isomorphism of vector spaces.

Definition 1.9.1. We define $\mathrm{Ext}(C, A)$ as the set of isomorphisms classes of extensions of C by A.

- (**Sections**). We know that if the exact sequence

$$0 \longrightarrow A \xrightarrow{i} B \xrightarrow{p} C \longrightarrow 0$$

splitts, then we have the $\mathfrak{g}-$module morphism $B \cong A \oplus C$. By linear algebra, we know that there exist a linear map $s \in \mathrm{Hom}(C, B)$ such that $p \circ s = \mathbb{1}_C$. So, we try to understand when the map

$$\xi_B : \mathfrak{g} \to \mathrm{Hom}(C, A) : x \mapsto [c \mapsto xs(c) - sx(c)] \tag{1.249}$$

vanish. Observe that $\mathrm{Im}(\xi_B)$ really lies in A, because

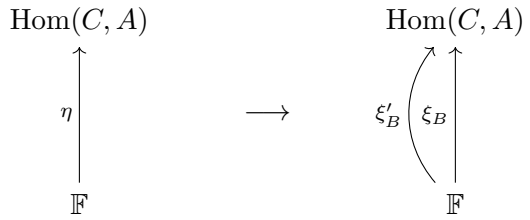
$$\begin{aligned} p\xi_B(x) &= pxs(c) - psx(c) \\ &= xps(c) - psx(c) \\ &= x(c) - x(c) \quad (\text{because } p \circ s = \mathbb{1}_C) \\ &= 0. \end{aligned} \tag{1.250}$$

So $\xi_B(x) \in \mathrm{Ker}(p) = \mathrm{Im}(i) \cong A$. Observe that the morphism ξ_B deppends on the choice of extension and the choice of splitting. Fix an extension and let ξ_B, ξ'_B be associated to two sections s, s'. We have

$$(\xi_B - \xi'_B)(x) : c \mapsto x(s - s')c - (s - s')xc. \tag{1.251}$$

Then we see that $\xi_B - \xi'_B$ comes from a map $\eta = s - s' \in \mathrm{Hom}(C, A)$, because $p(s - s') = 0$, that is $\mathrm{Im}(s - s') \in \mathrm{Ker}(p) = \mathrm{Im}(i) \cong A$.

- (**Sistematization**). Lets try to sistematizate what we have done. Previously we constructed a kind of a "map":



In the general framekork, we have:

$$\begin{aligned} d : \mathrm{Hom}_{\mathbb{F}}(\mathbb{F}, M) &\longrightarrow \mathrm{Hom}_{\mathbb{F}}(\mathfrak{g}, M) \\ m &\longmapsto [x \mapsto x \cdot m] \end{aligned} \tag{1.252}$$

Moreover, we have the following observation: If A, C are two $\mathfrak{g}-$modules, $\mathrm{Hom}(C, A)$ also has a $\mathfrak{g}-$module structure given by:

$$(x \cdot f)(c) = xf(c) - f(xc). \tag{1.253}$$

So, by the previous discussion we have $\xi_B - \xi'_B = d(\eta),\ \eta = s - s' \in \mathrm{Hom}(C, A)$. Now we arrive at the main definition of this section.

Definition 1.9.2. (Chevalley-Eilenberg complex). Let M be a $\mathfrak{g}-$module and $C^k(\mathfrak{g}, M) = \mathrm{Hom}_{\mathbb{F}}(\Lambda^k \mathfrak{g}, M)$. Define the map

$$\begin{aligned} &d : C^k(\mathfrak{g}, M) \to C^{k+1}(\mathfrak{g}, M) \\ &df(x_1, \ldots, x_{k+1}) = \sum_{i=1}^{k} (-1)^{i+1} x_i \cdot f(x_1 \wedge \cdots \wedge \widehat{x_i} \wedge \cdots \wedge x_{k+1}) \\ &\qquad + \sum_{i<j} (-1)^{i+j} f([x_i, x_j] \wedge x_1 \wedge \cdots \wedge \widehat{x_i} \wedge \cdots \wedge \widehat{x_j} \wedge \cdots \wedge x_{k+1}) \end{aligned} \tag{1.254}$$

The Chevalley-Eilenberg complex is the complex $(C^k(\mathfrak{g}, M), d^k)$.

In particular, we have

$$\begin{aligned} &d^0 : C^0(\mathfrak{g}, M) \to C^1(\mathfrak{g}, M) \\ &d^0(m)[x_1] = x_1 \cdot m,\ m \in M \cong \mathrm{Hom}(\mathbb{F}, M) \end{aligned} \tag{1.255}$$

and

$$\begin{aligned} &d^1 : C^1(\mathfrak{g}, M) \to C^2(\mathfrak{g}, M) \\ &d^1(f)[x_1 \wedge x_2] = x_1 f(x_2) - x_2 f(x_1) - f([x_1, x_2]) \end{aligned} \tag{1.256}$$

Definition 1.9.3. We write $Z^n = \mathrm{Ker}(d^n)$, $B^n = \mathrm{Im}(d^{n-1})$ and $H^n(\mathfrak{g}, M) = Z^n/B^n$.

- Now, coming back to extension of modules. Given an extension

$$0 \to A \xrightarrow{i} B \xrightarrow{p} C \to 0 \tag{1.257}$$

and the map $\xi_B \in \mathrm{Hom}(\mathfrak{g}, \mathrm{Hom}(C, A))$ constructed previously, we see that

$$d^1(\xi_B)(x \wedge y) = x \cdot \xi_B(y) - y \cdot \xi_B(x) - \xi_B([x, y]) \tag{1.258}$$

maps c to

$$\begin{aligned} & x \cdot \xi_B(y)c - y \cdot \xi_B(x)c - \xi_B([x, y])c \\ & \quad = x\xi_B(y)c - \xi_B(y)(xc) - y\xi_B(x)c + \xi_B(x)(yc) - \xi_B([x, y])c \\ & \quad = xys(c) - xsy(c) - ysx(c) + syx(c) \\ & \qquad - yxs(c) + ysx(c) + xsy(c) - sxy(c) - \xi_B([x, y])c \\ & \quad = [x, y]s(x) - s([x, y]c) - \xi_B([x, y])c \\ & \quad = 0. \end{aligned} \tag{1.259}$$

Remark 1.9.4. we must be careful here, we are using $\cdot$ to denote the action of $\mathfrak{g}$ on $\mathrm{Hom}(C, A)$!

Therefore $\xi_B \in Z^1(\mathfrak{g}, \mathrm{Hom}(C, A))$. In particular, we see that the class

$$[\xi_B] \in H^1(\mathfrak{g}, \mathrm{Hom}(C, A)) \tag{1.260}$$

is independent of the choice of the section s, in fact $\xi_B - \xi'_B = d(s - s')$. So this class deppends only on the extension of C by A.

Now we have all the tools to state the main theorems of this section.

Theorem 1.9.5. The map $(0 \to A \to B \to C \to 0) \mapsto [\xi_B]$ defines a bijection

$$\psi : \mathrm{Ext}(C, A) \xrightarrow{\cong} H^1(\mathfrak{g}, \mathrm{Hom}(C, A)) \tag{1.261}$$

Proof: First we show that this map is well defined. Suppose that we have an isomorphism $f : B \to B'$ of exact sequences. That is f is a $\mathfrak{g}-$module isomorphism that makes the following diagram commutes:

$$\begin{array}{ccccccccc} 0 & \longrightarrow & A & \xrightarrow{i} & B & \xrightarrow{p} & C & \longrightarrow & 0 \\ \vdots & & \downarrow{\scriptstyle \mathbb{1}_A} & & \downarrow{\scriptstyle f} & & \downarrow{\scriptstyle \mathbb{1}_C} & & \vdots \\ 0 & \longrightarrow & A & \xrightarrow{i'} & B' & \xrightarrow{p'} & C & \longrightarrow & 0 \end{array}$$

Choose sections $s \in \mathrm{Hom}(C, B), s' \in \mathrm{Hom}(C, B')$ such that $ps = p's' = \mathbb{1}_C$. Here we need to be more formal with definitions. We have $\xi_B = i^{-1} \circ (xs - sx)$. Now we observe that

$$i' \circ \xi_B(x) = fi \circ \xi_B(x) = x \circ fs - fs \circ x. \tag{1.262}$$

Here we have used the hypothesis that f is a $\mathfrak{g}-$module morphism. Moreover, we have

$$i' \circ (\xi_B(x) - \xi_{B'}(x)) = x \circ (fs - s') - (fs - s') \circ x. \tag{1.263}$$

However

$$\begin{aligned} p'(fs - s') &= p'fs - p's' \\ &= ps - p's' \\ &= 0. \end{aligned} \tag{1.264}$$

Therefore $\mathrm{Im}(fs - s') \subset \mathrm{Ker}(p') = \mathrm{Im}(i')$. Since $i' : A \to \mathrm{Im}(i')$ is a $\mathfrak{g}-$module isomorphism and $\mathrm{Im}(fs - s') \subset \mathrm{Im}(i')$, we have

$$\xi_B(x) - \xi_{B'}(x) = d((i')^{-1} \circ (fs - s')) \Rightarrow [\xi_B(x)] = [\xi_{B'}(x)] \in H^1. \tag{1.265}$$

Now we construct an inverse ϕ for this map. Let $\xi \in Z^1(\mathfrak{g}, \mathrm{Hom}(C, A))$, then we define a module structure on $A \oplus C$ given by $x(a, c) = (xa + \xi(x)c, xc)$. To show that this in fact defines a $\mathfrak{g}-$module we use the fact that $d(\xi) = 0$, that is

$$0 = d(\xi)(x \wedge y) = x \cdot \xi(y) - y \cdot \xi(x) - \xi([x, y]). \tag{1.266}$$

Then

$$\begin{aligned} [x, y](a, c) &= ([x, y]a + \xi([x, y])c, [x, y]c) \\ &= ([x, y]a + x \cdot \xi(y)c - y \cdot \xi(x)c, [x, y]c) \\ &= ([x, y]a + x\xi(y)c - \xi(y)xc - y\xi(x)c + \xi(x)yc, [x, y]c). \end{aligned} \tag{1.267}$$

But

$$\begin{aligned} xy(a, c) &= x(ya + \xi(y)c, yc) \\ &= (xya + x\xi(y)c + \xi(x)yc, xyc). \end{aligned} \tag{1.268}$$

So we see that $[x, y](a, c) = (xy - yx)(a, c)$. We claim that $Z^1 \to \mathrm{Ext}(C, A)$ descends to a map $H^1 \to \mathrm{Ext}(C, A)$. To see this we need to show that if $[\xi] = [\xi']$, then the associated modules $A \oplus C$ induced by ξ and ξ' are isomorphic. First, we know that $\xi - \xi' = d\lambda,\ \lambda \in \mathrm{Hom}(C, A)$. The linear isomorphism $\varphi_\lambda \in \mathrm{End}(C, A)$ which sends $(a, c) \mapsto (a + \varphi_\lambda(c), c)$ makes the diagram

$$\begin{array}{ccccccccc} 0 & \longrightarrow & A & \xrightarrow{i} & A\oplus C & \xrightarrow{p} & C & \longrightarrow & 0 \\ \vdots & & \downarrow{\scriptstyle \mathbb{1}_A} & & \downarrow{\scriptstyle \varphi_\lambda} & & \downarrow{\scriptstyle \mathbb{1}_C} & & \vdots \\ 0 & \longrightarrow & A & \xrightarrow{i} & A\oplus C & \xrightarrow{p} & C & \longrightarrow & 0 \end{array}$$

commutes. Furthermore, we have $x\cdot\varphi_\lambda = \varphi_\lambda\cdot x$ if and only if $\xi-\xi' = d\lambda$. So φ_λ is a $\mathfrak{g}-$module isomorphism. Therefore, we have a well defined map $\phi: H^1\to \mathrm{Ext}(C,A)$.

Finally, we show that $\psi\phi = \mathbb{1}_{H^1}$ and $\phi\psi = \mathbb{1}_{\mathrm{Ext}(C,A)}$. For the first one, we observe that if we associate the corresponding module structure on $A\oplus C$ using $\xi_0\in Z^1$, then for any $c\in C$ we have

$$i\circ\xi(x)c = x\circ s(c) - s\circ x(c) = x(0,c)-(0,xc) = (\xi_0(x)c, xc)-(0,xc) = (\xi_0(x)c,0) \quad (1.269)$$

and $\xi(x) = \xi_0(x)$, $\forall x\in\mathfrak{g}$. Therefore $\psi\phi([\xi_0]) = [\xi_0]$. Now, for the second identity, let $[\xi] = \psi(0\to A\xrightarrow{i} B\xrightarrow{p} C\to 0)$, where $\xi = i^{-1}(x\circ s - s\circ x)$. To show that $\phi([\xi]) = [0\to A\xrightarrow{i} B\xrightarrow{p} C\to 0]$, we must find a $\mathfrak{g}-$module morphism f making the diagram

$$\begin{array}{ccccccccc} 0 & \longrightarrow & A & \longrightarrow & A\oplus C & \longrightarrow & C & \longrightarrow & 0 \\ \vdots & & \downarrow{\scriptstyle \mathbb{1}_A} & & \downarrow{\scriptstyle f} & & \downarrow{\scriptstyle \mathbb{1}_C} & & \vdots \\ 0 & \longrightarrow & A & \xrightarrow{i} & B & \xrightarrow{p} & C & \longrightarrow & 0 \end{array}$$

commutes. There is no much of choice left: We try $f: A\oplus C\to B$, $f(a,c) = i(a)+\alpha s(c)$. So we see that, in order to the prevoius diagram commute, we need: $pf(a,c) = c\Rightarrow \alpha = 1$. Furthermore $f(a,0) = i(a)$, so the previous diagram commutes and f is an isomorphism by the 5 lemma. We need to show that f is a $\mathfrak{g}-$module morphism. We compute:

$$xf(a,c) - f(xa+\xi(x)c, xc) = xi(a)+xs(c)-i(xa)-i(\xi(x)c)-s(xc). \quad (1.270)$$

But, i is a $\mathfrak{g}-$module morphism, so $i(xa) = xi(a)$, while $i(\xi(x)c) = xs(c)-sx(c)$ by definition. □

Theorem 1.9.6. (Whitehead lemma). If $\mathfrak{g}$ is a semisimple Lie algebra and V is a finite dimensional $\mathfrak{g}-$module, then $H^1(\mathfrak{g},V) = 0$.

Proof: The proof will be divided in several steps.

1. Let $\mathfrak{g}_0 = \{X\in\mathfrak{g} : Xv = 0,\ \forall v\in V\}$, then $\mathfrak{g}_0$ is an ideal. We claim that both Lie algebras $\mathfrak{g}_0, \overline{\mathfrak{g}} = \mathfrak{g}/\mathfrak{g}_0$ are semisimple. To prove this we choose a basis $\{X_1,\dots,X_k\}$ of $\mathfrak{g}_0$ and extend it to a basis of $\mathfrak{g}$ through the set $\{X_{k+1},\dots,X_n\}$. It's a general property of semisimple Lie algebras that its ideals are also semisimple, so $\mathfrak{g}_0$ is semisimple (to see this, just define $\mathfrak{a} = \mathfrak{g}_0\cap(\mathfrak{g}_0)^\perp$ and apply CC(sol) to see that it's a solvable ideal). So we just need to prove that $\overline{\mathfrak{g}}$ is semisimple.

For $X, Y \in \mathfrak{g}$ compute

$$\begin{aligned} K^{\mathfrak{g}}(X,Y) &= \mathrm{Tr}\begin{pmatrix} X_{11} & X_{12} \\ 0 & X_{22} \end{pmatrix}\begin{pmatrix} Y_{11} & Y_{12} \\ 0 & Y_{22} \end{pmatrix} \\ &= \mathrm{Tr}(X_{11}Y_{11}) + \mathrm{Tr}(X_{22}Y_{22}) \\ &= \mathrm{Tr}(X_{11}Y_{11}) + K^{\overline{\mathfrak{g}}}(\overline{X},\overline{Y}). \end{aligned} \tag{1.271}$$

The last line follows from the property that if $\mathrm{ad}(X)Z = Z_0 + \sum_{i=k+1}^{n} a_k X_k$, then $\mathrm{ad}(\overline{X})\overline{Z} = \sum_{i=k+1}^{n} a_k \overline{X}_k$. Now, suppose that $K^{\overline{\mathfrak{g}}}(\overline{X},\overline{Y}) = 0,\ \forall Y \in \mathfrak{g}$ and let X' be the component of X in $\mathfrak{g}_0$. Then

$$K^{\mathfrak{g}}(X',Y) = \mathrm{Tr}(X_{11}Y_{11}) \Rightarrow K^{\mathfrak{g}}(X - X', Y) = 0, \forall Y \in \mathfrak{g}. \tag{1.272}$$

Therefore, by CC(ss) we see that $X - X' = 0$, that is $X \in \mathfrak{g}_0$.

Now we observe that $\overline{\mathfrak{g}} \hookrightarrow \mathrm{End}(V)$, so we can think $\overline{\mathfrak{g}} \subset \mathrm{End}(V)$.

2. Next we show that $H^1(\mathfrak{g}, V) = 0 \iff H^1(\overline{\mathfrak{g}}, V) = 0$. This shows that we can assume $\mathfrak{g} \subset \mathrm{End}(V)$ without lost of generality. This follows from the following observation, if $f \in Z^1(\mathfrak{g}, V)$, then $f|_{\mathfrak{g}_0} = 0$. In fact, since $\mathfrak{g}_0$ is semisimple, we have $[\mathfrak{g}_0, \mathfrak{g}_0] = \mathfrak{g}_0$. So, for any $Z \in \mathfrak{g}_0$ we can write $Z = [X, Y],\ X, Y \in \mathfrak{g}_0$. However, since $f \in Z^1$, we have

$$f(Z) = f([X,Y]) = Xf(Y) - Yf(X) = 0 \tag{1.273}$$

because $X, Y \in \mathfrak{g}_0$. So f descends to a map $\overline{f} : \overline{g} \to V$ and

$$\begin{aligned} f = d(\eta) &\iff f(X) = X \cdot \eta,\ \forall X \in \mathfrak{g} \\ &\iff \overline{f}(\overline{X}) = \overline{X} \cdot \eta,\ \forall X \in \mathfrak{g} \\ &\iff \overline{f} = d(\eta). \end{aligned} \tag{1.274}$$

3. From the previous observations we see that we can assume $\mathfrak{g} \subset \mathrm{End}(V)$. The advantage of this hypothesis is that we can apply CC(sol) using the trace form:

$$(\mathfrak{a}, [\mathfrak{a}, \mathfrak{a}])_V = 0 \iff \mathfrak{a} \text{ is solvable} \tag{1.275}$$

so the trace form is also nondegenerate. Then we can use it to construct dual basis and define the Casimir element.

Now, choose a dual basis $\{a_1, \ldots, a_n\}$ and $\{b_1, \ldots, b_n\}$ with respect to $(\ ,\ )_V$. That is $(a_i, b_j) = \delta_{ij}$. We have the Casimir element

$$\Omega = \sum_{i=1}^{n} a_i b_i \in Z(\mathcal{U}(\mathfrak{g})). \tag{1.276}$$

Decompose V into generalized eigenspaces

$$V = \bigoplus_{\gamma \in \mathbb{C}} V_\gamma,\ V_\gamma = E_\gamma^\Omega. \tag{1.277}$$

Since $\Omega \in Z(\mathcal{U}(\mathfrak{g}))$, we see that each V_γ is a submodule of V. Now we observe that $V_0 \neq V$. In fact, if $V = V_0$, then $0 = \mathrm{Tr}_V(\Omega) = \sum_i (a_i, b_i) = \dim(\mathfrak{g})$, which is not the case. So, by defining $V' = \bigoplus_{\gamma \in \mathbb{C}^*} V_\gamma$, we see that $V = V' \oplus V_0$ with $\dim(V_0), \dim(V') < \dim(V)$. The result will follow by induction on dimension of V provided we show that

$$\begin{aligned} &(a)\ H^1(\mathfrak{g}, V') = 0 \\ &(b)\ \text{By (a), we have } H^1(\mathfrak{g}, V) \cong H^1(\mathfrak{g}, V_0). \end{aligned} \tag{1.278}$$

We will prove (a) on the next item. Assume it for now and lets show that its implies the second item. We have a natural homomorphism

$$\psi : H^1(\mathfrak{g}, V_0) \to H^1(\mathfrak{g}, V). \tag{1.279}$$

We claim that under the condition $H^1(\mathfrak{g}, V') = 0$, this is an isomorphism. First we show injectivity: if $\psi([\xi]) = 0$, then there exist $\eta \in V$ such that $\xi = d(\eta)$. To show that $[\xi] = 0$ in $H^1(\mathfrak{g}, V_0)$, we need to show that we can take $\eta \in V_0$. To see this, write $\eta = \sum_{\gamma \in S} \eta_\gamma,\ \eta_\gamma \in V_\gamma$. Then, since each V_γ is a submodule, we have $x \cdot \eta = \xi(x) \in V_0 \iff x \cdot \eta = x \cdot \eta_0$, so $\xi = d(\eta_0)$ and $[\xi] = 0$ in $H^1(\mathfrak{g}, V_0)$. Now we prove that ψ is surjective, let $[\xi] \in H^1(\mathfrak{g}, V)$, so $\xi \in Z^1(\mathfrak{g}, V)$. The projection $p : V \to V'$ is a $\mathfrak{g}-$module homomorphism. Moreover, we claim that $p\xi \in Z^1(\mathfrak{g}, V')$. This follows from the calculation

$$\begin{aligned} d(p\xi)[X \wedge Y] &= Xp\xi(Y) - Yp\xi(X) - p\xi([X,Y]) \\ &= pX\xi(Y) - pY\xi(X) - p\xi([X,Y]) \quad (p \text{ is a module morphism}) \\ &= p(X\xi(Y) - Y\xi(X) - \xi([X,Y])) \\ &= pd(\xi)[X \wedge Y] \\ &= 0. \end{aligned} \tag{1.280}$$

Now we use the hypothesis that $H^1(\mathfrak{g}, V') = 0$ to see that this implies that $p\xi = d(\eta),\ \eta \in V'$. Also, observe that $pd(\eta)(x) = p(x \cdot \eta) = x \cdot \eta$ because V' is invariant by the $\mathfrak{g}$ action. So we see that $p \circ (\xi - d(\eta)) = 0 \Rightarrow \mathrm{Im}(\xi - d(\eta)) \subset V_0$. So we can write the class $[\xi - d(\eta)] \in H^1(\mathfrak{g}, V_0)$ which is clearly mapped to $[\xi]$ through ψ. This proves the claim.

4. Now we prove that $H^1(\mathfrak{g}, V') = 0$. To show this, we need a lemma.

Lemma 1.9.7. We have the following identity:

$$X \cdot (\sum_i a_i \xi(b_i)) = \Omega \circ \xi(X). \tag{1.281}$$

Proof: This follows form the calculation

$$\begin{aligned} X \cdot (\sum_i a_i \xi(b_i)) &= \sum_i X a_i \xi(b_i) \\ &= \sum_i ([X, a_i]\xi(b_i) + a_i X \xi(b_i)) \\ &= \sum_i ([X, a_i]\xi(b_i) + a_i b_i \xi(X) + a_i \xi([X, b_i])) \quad \text{(because } \xi \in Z^1\text{)} \\ &= \sum_i ([X, a_i]\xi(b_i) + a_i \xi([X, b_i])) + \Omega \circ \xi(X) \quad \text{(definition of } \Omega\text{)} \\ &= \sum_{i,j} (([X, a_i], b_j) a_j \xi(b_i) + ([X, b_i], a_j) a_i \xi(b_j)) + \Omega \circ \xi(X) \\ &= \sum_{i,j} (([X, a_i], b_j) a_j \xi(b_i) + ([X, b_j], a_i) a_j \xi(b_i)) + \Omega \circ \xi(X) \\ &= \sum_{i,j} (([X, a_i], b_j) + ([X, b_j], a_i)) a_j \xi(b_i) + \Omega \circ \xi(X) \\ &= \Omega \circ \xi(X). \end{aligned} \tag{1.282}$$

The last line follows by invariance of $(\ ,\ )_V$. Now, the proof of the claim becomes easy. Let $\xi \in Z^1(\mathfrak{g}, V')$, then $\Omega \circ \xi(X) = X \cdot v_0$ where $v_0 = \sum_i a_i \xi(b_i) \in V_0$. However, $\Omega|_{V'}$ is invertible, so we can write $\xi(X) = \Omega^{-1}(X \cdot v_0) = X \cdot (\Omega^{-1}(v_0))$. Therefore

$$\xi = d(\Omega^{-1}(v_0)) \tag{1.283}$$

and $[\xi] = 0$.

□

Corollary 1.9.8. Each finite dimensional $\mathfrak{g}-$module is completely decomposable.

Proof: Let $A \subset B$ an invariant subspace. So we have an exact sequence $(0 \to A \to B \to C \to 0) \in \mathrm{Ext}(C, A)$, where $C = B/A$. Since $H^1(\mathfrak{g}, \mathrm{Hom}(C, A)) = 0$, we get an isomorphism $B \cong A \oplus C$, where the module structure of $A \oplus C$ comes from the zero class $[0] \in H^1$, that is $x(a, c) = (xa, xc)$. □

Chapter 2

Nilpotent orbits

2.1 Adjoint and coadjoint orbits

2.1.1 Review on Lie groups and infinitesimal actions

- Let $F : M \to N$ be a diffeomorphism and $X \in \mathfrak{X}(M)$. We define the pushforward $F_*X \in \mathfrak{X}(N)$ to be vector field defined by the equation $F_*X|_{F(p)} = F_*X_p$. It's a general fact that if $X, Y \in \mathfrak{X}(M)$, then $F_*[X, Y] = [F_*X, F_*Y]$.

  Moreover, if $X \in \mathfrak{X}(M)$ and $Y \in \mathfrak{X}(N)$, we say that X and Y are $F-$related (and we will write $X \sim_F Y$) if $F_*X = Y$.

- Let G be a Lie group. Define $\mathfrak{X}_{\text{inv}}(G) = \{X \in \mathfrak{X}(G) : \ell_{g*}X = X\}$ to be the set of left invariant vector fields of G. Then, it's clear that $[\mathfrak{X}_{\text{inv}}(G), \mathfrak{X}_{\text{inv}}(G)] \subset \mathfrak{X}_{\text{inv}}(G)$, so $\mathfrak{X}_{\text{inv}}(G)$ is a Lie algebra. We have an isomorphism

$$\mathfrak{g} \xrightarrow{\cong} \mathfrak{X}_{\text{inv}}(G) = T_eG : X \mapsto [g \mapsto \ell_{g,*e}X] \tag{2.1}$$

  As a matter of notation, we will write u^L for the left invariant vector field generated by $u \in \mathfrak{g}$, i.e., $u^L(g) = \ell_{g,*}u$.

- A G- manifold M is a triple (M, G, ψ) where G is a Lie group and $\psi : G \times M \to M$ is a smooth map such that $\psi_g = \psi(g, \cdot)$ defines a homomorphism $G \to \text{Diff}(M) : g \mapsto \psi_g$. An infinitesimal generator of ψ is a vector field defined by

$$u_M(x) = \left.\frac{d}{dt}\right|_{t=0} \psi_{\exp(tu)}(X). \tag{2.2}$$

  This defines a map $\mathfrak{g} \to \mathfrak{X}(M) : u \mapsto u_M$ which we will show to be a Lie algebra anti-homomorphism.

  Example 2.1.1. (infinitesimal generator of Ad). Consider the adjoint action $G \times \mathfrak{g} \to$

$\mathfrak{g}$. We have

$$\begin{aligned} u_{\mathfrak{g}}(v) &= \left.\frac{d}{dt}\right|_{t=0} \mathrm{Ad}_{\exp(tu)}(v) \\ &= \left.\frac{d}{dt}\right|_{t=0} (r_{\exp(-tu)} \circ \ell_{\exp(tu)})_{*,e}(v) \\ &= \left.\frac{d}{dt}\right|_{t=0} d_{\exp(tu)} r_{\exp(-tu)}(v^L(\exp(tu))) \\ &= \left.\frac{d}{dt}\right|_{t=0} r_{\exp(tu)*}(v^L)\big|_e \\ &= \mathcal{L}_{u^L}(v^L)\big|_e \\ &= [u^L, v^L]_e \\ &= [u, v]. \end{aligned} \tag{2.3}$$

Therefore $\mathrm{Ad}_{*,e} = \mathrm{ad}$.

- We have three important results that we will need:

  1. Let $\varphi : M_1 \to M_2$ be a G equivariant map, i.e., $\varphi(gx) = g\varphi(x)$, $\forall x \in M_1$, then for any $u \in \mathfrak{g}$, we have $u_{M_1} \sim_\varphi u_{M_2}$.
  2. Let M be a G-manifold, then $\mathfrak{g} \to \mathfrak{X}(M) : u \mapsto u_M$ is a Lie algebra anti-homomorphism. In fact, the result is true for $(G, \psi(g,\cdot) = r_{g^{-1}})$. This follows frmo the equation

$$u_G(g) = \left.\frac{d}{dt}\right|_{t=0} r_{\exp(-tu)}(g) = -u^L(g). \tag{2.4}$$

Remember that $r_{\exp(tu)}$ defines the flux of the left invariant vector field u^L. Now, we observe that the result is also true for $\overline{M} = G \times M$ with action $g \cdot (a, x) = (r_{g^{-1}}(a), x)$. In fact

$$\begin{aligned} \pi_{G*(a,x)} u_{\overline{M}}(a, x) &= \left.\frac{d}{dt}\right|_{t=0} \pi_G \circ (r_{\exp(-tu)}(a), x) \\ &= \left.\frac{d}{dt}\right|_{t=0} r_{\exp(-tu)}(a) \\ &= -u^L(a). \end{aligned} \tag{2.5}$$

Therefore

$$[u_{\overline{M}}, v_{\overline{M}}] = -([u^L, v^L], 0) = -([u^L, v^L], 0) = -[u, v]_{\overline{M}}. \tag{2.6}$$

Now define the G-equivariant map $F : \overline{M} \to M : (a, x) \mapsto a^{-1}x$. By applying F_* to the previous equation we get $[u_M, v_M] = -[u, v]_M$.

3. We have $g_*(u_M) = \mathrm{Ad}_g(u)_M$. This is true in $\overline{M} = G \times M$. In fact, we have

$$\begin{aligned}
\pi_G * (a,x) g_{*,(a,x)} u_{\overline{M}}(a,x) &= (\pi_G \circ \psi_g)_{*,(a,x)} u_{\overline{M}}(a,x) \\
&= \frac{d}{dt}\Big|_{t=0} \pi_G \circ \psi_g \circ (r_{\exp(-tu)}(a), x) \\
&= \frac{d}{dt}\Big|_{t=0} \pi_G(r_{g^{-1}} r_{\exp(-tu)}(a), x) \\
&= \frac{d}{dt}\Big|_{t=0} a \exp(-tu) g^{-1} \\
&= \frac{d}{dt}\Big|_{t=0} a g^{-1} c_g(\exp(-tu)) \\
&= \frac{d}{dt}\Big|_{t=0} a g^{-1} \exp(-\mathrm{Ad}_g(u)) \\
&= \frac{d}{dt}\Big|_{t=0} r_{\exp(-\mathrm{Ad}_g(u))}(a g^{-1}) \\
&= -(\mathrm{Ad}_g(u))^L (a g^{-1}).
\end{aligned} \tag{2.7}$$

Finally

$$\begin{aligned}
g_* u_{\overline{M}}\big|_{g\cdot(a,x)} &= -(\mathrm{Ad}_g(u)^L (a g^{-1}), 0) \\
&= -(\mathrm{Ad}_g(u)^L, 0)\big|_{g\cdot(a,x)}.
\end{aligned} \tag{2.8}$$

Now we have

$$g_* u_M = g_* F_* u_{\overline{M}} = F_* g_* u_{\overline{M}} = F_*(\mathrm{Ad}_g(u))_{\overline{M}} = (\mathrm{Ad}_g(u))_M. \tag{2.9}$$

- Let (M, G, ψ) be a G-manifold. For $x \in M$, we define the isotropy subgroup $G_x = \{g \in G : gx = x\}$. It's clear that $G_x \subset G$ is closed, so it's a Lie subgroup of G. Define $\mathfrak{g}_x = \{u \in \mathfrak{g} : u_M(x) = 0\}$, we claim that

$$\mathrm{Lie}(G_x) = \mathfrak{g}_x. \tag{2.10}$$

Let $\exp(tu) \in G_x,\ \forall t \in \mathbb{R}$, that is, assume that $u \in \mathrm{Lie}(G_x)$. Then, by definition, we have $\exp(tu)x = x$. So, by differentiating this equation we get $u_M(x) = 0$ and $u \in \mathfrak{g}_x$. Conversely, assume that $u \in \mathfrak{g}_x$. Let $g = \exp(ts),\ s \in \mathbb{R}$, then

$$\frac{d}{dt}\Big|_{t=s} \exp(tu)x = g_{*,x}\left(\frac{d}{dt}\Big|_{t=0} \exp(tu)x\right) = 0 \tag{2.11}$$

and $\exp(tu) \cdot x = x, \forall t \in \mathbb{R}$. Therefore $u \in \mathrm{Lie}(G_x)$.

- Now, if $\mathcal{O}_x = Gx$, it's a fact that if we identify $G/G_x \cong \mathcal{O}_x$ by the orbit stabilizer theoerem, then $\mathcal{O}_x$ inherits a differentiable structure such that the inclusion $i : \mathcal{O}_x \hookrightarrow M$ is an imersion. In this context we claim that

$$i_{*,x} T_x \mathcal{O}_x = \{u_M(x) : u \in \mathfrak{g}\}. \tag{2.12}$$

Lets prove it. We know that $i : G/G_x \hookrightarrow M : [g] \mapsto gx$ is an imersion. First observe that

$$\begin{aligned} \left.\frac{d}{dt}\right|_{t=0} i \circ \pi \circ r_{\exp(tu)}(g) &= \left.\frac{d}{dt}\right|_{t=0} g \exp(tu) \cdot x \\ &= g_{*,x} u_M(x) \\ &= (\mathrm{Ad}_g(u))_M(gx). \end{aligned} \tag{2.13}$$

Since $\pi : G \to G/G_x$ is a submersion, we conclude that $i_{*,[g]} T_{[g]}(G/G_x) \subset \{u_M(g \cdot x) : u \in \mathfrak{g}\}$. Now define the map

$$\mathfrak{g} \to \{u_M(g \cdot x) : u \in \mathfrak{g}\} \tag{2.14}$$

which has kernel equal to $\mathrm{Lie}(G_{g \cdot x})$. Therefore we have an isomorphism

$$\mathfrak{g}/\mathrm{Lie}(G_{g \cdot x}) \xrightarrow{\cong} \{u_M(g \cdot x) : u \in \mathfrak{g}\}. \tag{2.15}$$

Therefore the dimension of the last subspace is equal to

$$\begin{aligned} \dim(G) - \dim(G_{g \cdot x}) &= \dim(G) - \dim(g G_x g^{-1}) \\ &= \dim(G) - \dim(G_x) \\ &= \dim(G/G_x). \end{aligned} \tag{2.16}$$

Therefore, by dimensional reasons we have $i_{*,[g]} T_{[g]}(G/G_x) = \{u_M(g \cdot x) : u \in \mathfrak{g}\}$ and since i_* is injective we can identify $T_x \mathcal{O}_x = \{u_M(x) : u \in \mathfrak{g}\}$.

2.1.2 Co-adjoint orbits and KKS form

Let G be a Lie group. We will show that the orbits on $\mathfrak{g}^*$ under the co-adjoint action have a natural structure of a symplectic manifold.

KKS: Let $\xi \in \mathcal{O}$, we define $\omega \in \bigwedge^2 T^*_\xi \mathcal{O}$ by the formula:

$$\omega_\xi(u_{\mathfrak{g}^*}(\xi), v_{\mathfrak{g}^*}(\xi)) = \langle \xi, [u, v] \rangle. \tag{2.17}$$

We need to verify a few things:

- ω_ξ is well defined. Assume that $u_{\mathfrak{g}^*}(\xi) = u'_{\mathfrak{g}^*}(\xi)$ and $v_{\mathfrak{g}^*}(\xi) = v'_{\mathfrak{g}^*}(\xi)$, then

$$\begin{aligned}\omega_\xi(u'_{\mathfrak{g}^*}(\xi), v'_{\mathfrak{g}^*}(\xi)) &= -u'_{\mathfrak{g}^*}(\xi)(v') \\ &= -u_{\mathfrak{g}^*}(\xi)(\xi)(v') \\ &= \langle \xi, [u, v'] \rangle \\ &= v'_{\mathfrak{g}^*}(\xi)(u) \\ &= v_{\mathfrak{g}^*}(\xi)(u) \\ &= \langle \xi, [u, v] \rangle.\end{aligned} \tag{2.18}$$

- It's skew symmetric and nondegenerate (clear): $\langle \xi, [u, v] \rangle = 0 \; \forall v \iff u_{\mathfrak{g}^*}(\xi) = 0$.
- It's $G-$invariant, so it's smooth. This can be verified directly by using the relation $g_*(u_{\mathfrak{g}^*}) = (\mathrm{Ad}_g(u))_{\mathfrak{g}^*}$.
- By the previous item we see that $\psi^*_{e^{tu}}\omega = \omega$. Therefore

$$\mathcal{L}_{\mathfrak{g}^*}\omega = 0 \iff (d\iota_{u_{\mathfrak{g}^*}} + \iota_{u_{\mathfrak{g}^*}} d)\omega = 0. \tag{2.19}$$

  Claim: We have $\iota_{u_{\mathfrak{g}^*}}\omega = du$ where we identify $u \in \mathfrak{g} \cong C^\infty_{\mathrm{Lin}}(\mathfrak{g}^*)$.

  This can be seen directly by the equation:

$$\begin{aligned}\iota_{u_{\mathfrak{g}^*}}\omega\big|_\xi(v_{\mathfrak{g}^*}(\xi)) &= \langle \xi, [u, v] \rangle \\ &= \langle \xi, [u, v] \rangle \\ &= v_{\mathfrak{g}^*}(\xi)(u) \\ &= du\big|_\xi v_{\mathfrak{g}^*}(\xi) \text{ (by the identification } \mathfrak{g}^{**} = \mathfrak{g})\end{aligned} \tag{2.20}$$

- For $G = SO(3)$ the co-adjoint and adjoint orbits are the same. We have an isomorphism of Lie algebras $\mathbb{R}^3 \cong \mathfrak{so}(3)$ which maps $e_i \in \mathbb{R}^3$ to the associated matrix representation of the map $(e_i)_\times : v \mapsto e_i \times v$. Moreover, we have the identity

$$A(u \times v) = Au \times Av, \; \forall A \in SO(3). \tag{2.21}$$

  With this equation we get $(Au)_\times v = Au \times v = Au_\times A^{-1}(v) = \mathrm{Ad}_A(u_\times)v$. So this isomorphism is $SO(3)-$equivariant, with the adjoint orbits being spheres in $\mathbb{R}^3$.

Remark 2.1.2. If $\nu : \mathfrak{g} \to \mathfrak{g}^*$ is an isomorphism that intetwine the adjoint and co-adjoint actions of G. Then we can induce a symplectic form on adjoint orbits using KKS. The formula is the one

$$\omega_\xi(u_{\mathfrak{g}}(\xi), v_{\mathfrak{g}}(\xi)) = \langle \nu(\xi), [u, v] \rangle. \tag{2.22}$$

Going back to definitions: If $\xi \in (\mathbb{R}^3)^*$, we have

$$\begin{aligned}\omega_\xi(u_\mathfrak{g}(\xi), v_\mathfrak{g}(\xi)) &= \langle \xi, [u, v] \rangle \\ &= \langle \xi, u \times v \rangle \text{ ($\mathbb{R}^3$ is a Lie algebra with the croos product)} \\ &= \xi \cdot (u \times v) \text{ (By the isomorphism } \mathbb{R}^3 \cong (\mathbb{R}^3)^*)\end{aligned} \tag{2.23}$$

Remark 2.1.3. Let $S \subset M$ a hypersurface, M oriented with orientation form ω. Let $N \in \mathfrak{X}(M)$ be a vector field nowhere tangent to S, then $\iota_N \omega|_S$ is an orientation form for S.

By the previous remark we can use the vector field $N = x\frac{\partial}{\partial x} + y\frac{\partial}{\partial y} + z\frac{\partial}{\partial z}$ to induce an orientation form

$$\sigma = xdy \wedge dz - ydx \wedge dz + zdx \wedge dy\big|_{\mathbb{S}^2} \tag{2.24}$$

on the spheres. In fact, if $f : (x, y, z) \mapsto x^2 + y^2 + z^2 - 1$, then $df(N) = 2(x^2 + y^2 + z^2)$ is never zero on $\mathbb{S}^2$, so N is nowhere tangent to $\mathbb{S}^2$.

Remark 2.1.4. If $f : \mathbb{R}^n \to \mathbb{R}$ is smooth with regular value c, then $S = f^{-1}(0)$ is a regular submanifold with tangent spaces

$$i_{*p} T_p S \cong \nabla f(p)^\perp \tag{2.25}$$

In our case we have $\nabla f(\xi) = \xi$, so $T_\xi \mathbb{S}^2 = \xi^\perp = \{\xi \times u : u \in \mathbb{R}^3\}$. If we use the area form

$$\begin{aligned}\sigma_\xi(\xi \times u, \xi \times v) &= \xi \cdot ((\xi \times u) \times (\xi \times v)) \text{ (use the area form)} \\ &= -(\xi \times u) \cdot (\xi \times (\xi \times v)) \\ &= -(\xi \times u) \cdot (\xi(\xi \cdot v) - v(\xi \cdot \xi)) \\ &= \|\xi\|(\xi \times u) \cdot v \\ &= \xi \cdot (u \times v)\|\xi\|.\end{aligned} \tag{2.26}$$

In conclusion, we see the relation between the KKS and the area form:

$$\omega_{\text{kks}} = \frac{1}{r}\sigma \tag{2.27}$$

where $r = \|\xi\|$ is the radius of the sphere.

2.1.3 Hamiltonian actions

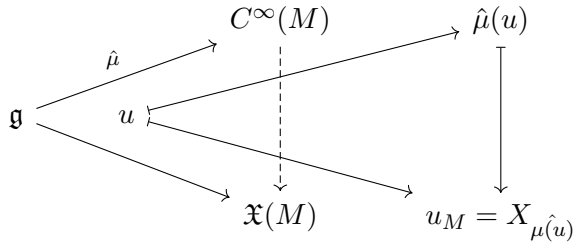
Let $X \in T_0\mathbb{R}^n \cong \mathbb{R}^n$, then $\widetilde{X}(g) = (g, X)$. In fact

$$\begin{aligned} \ell_{g*,0}(\frac{\partial}{\partial x_i}\Big|_g)x_j &= \frac{\partial}{\partial x_i}\Big|_g x_j \circ \ell_g \\ &= \frac{\partial}{\partial x_i}\Big|_g (x_j + x_j(g)) = \delta_{ij}. \end{aligned} \tag{2.28}$$

In general if $\widetilde{X} \in \mathfrak{X}_{\text{inv}}(G)$, then we set $\exp(X) = \gamma_1^{\widetilde{X}}(e)$. If $G = \mathbb{R}^n$, then $\gamma_t^{\widetilde{X}}(0) = tX$ and $e^X = X$.

Let (M, ω) be a symplectic manifold. Then

- $G-$action is symplectic if $\mathcal{L}_X\omega = 0$. This condition is equivalent to $\psi_g^*\omega = \omega \ \forall g \in G$ if G is connected.
- $G-$action is Hamiltonian if there exist a linear map $\hat{\mu} : \mathfrak{g} \to C^\infty(M)$ such that the following diagram commutes:



Remark 2.1.5. We have a bijection

$$\{\hat{\mu} : \mathfrak{g} \to C^\infty(M)\} \leftrightarrow \{\mu : M \to \mathfrak{g}^*\} \tag{2.29}$$

- The action is weakly Hamiltonian if

$$u_M = X_{\hat{\mu}(u)} \iff \iota_{u_M}\omega = d\hat{\mu} = d\langle \mu, u\rangle. \tag{2.30}$$

- The action is Hamiltonian if, in addition, we have

$$\mu \circ \psi_g = \text{Ad}_g^* \circ \mu \overset{G \text{ connected}}{\iff} \hat{\mu}([u, v]) = \{\hat{\mu}(u), \hat{\mu}(v)\}. \tag{2.31}$$

  The relation between the two definitions is $\hat{\mu}(u)\big|_x = \langle \mu(x), u\rangle$.

Now, look at the following equation for $x \in M, v \in \mathfrak{g}$:

$$\langle \mu \circ \psi_g(x), v\rangle = \langle \text{Ad}_g^* \circ \mu(x), v\rangle = \langle \mu(x), \text{Ad}_{g^{-1}}(v)\rangle. \tag{2.32}$$

This equation means that $\hat{\mu}(v) \circ \psi_g(x) = \hat{\mu}(\text{Ad}_{g^{-1}}(v))\big|_x$. If we put $g = e^{tu}, u \in \mathfrak{g}$ and differentiate this equation, we get $\hat{\mu}([u, v]) = \{\hat{\mu}(u), \hat{\mu}(v)\}$. For the converse we need the hypothesis of G being connected.

Remark 2.1.6. Remember that $\mathfrak{X}_{\text{Ham}}(M) \subset \mathfrak{X}_{\text{Sym}}(M)$.

In our case we have $\mathcal{L}_{u_M}\omega = 0$. Since u_M has flux given by $\psi_{e^{tu}}$, we get $\psi_{e^{tu}}^*\omega = \omega$. Therefore $\psi_g^*\omega = \omega,\ \forall g \in G^\circ$.

Theorem 2.1.7. (Noether principle). Let (M, ω, μ) be a Hamiltonian G-space and $H \in C^\infty(M)$. Then H is G-invariant if and only if μ is preserved by the Hamiltonian flux of H.

Proof: Remember that H is G invariant if $\psi_g^* H = H \iff \mathcal{L}_{u_M} H = 0$ if G is connected. Also, μ is preserved by the Hamiltonian flux of H if $\varphi_t^*\mu = \mu$ where $\varphi_t = \varphi_t^{X_H}$ is the flux of X_H. This equation is true if

$$\begin{aligned} \mathcal{L}_{X_H}\langle \mu, u\rangle &= \left.\frac{d}{dt}\right|_{t=0} \varphi_t^*\langle \mu, u\rangle \\ &= \left.\frac{d}{dt}\right|_{t=0} \langle \varphi_t^*\mu, u\rangle = 0\ \forall u \in \mathfrak{g} \\ &\iff \varphi_t^*\mu = \mu,\ \forall t \in \mathbb{R}. \end{aligned} \tag{2.33}$$

But we see that

$$\begin{aligned} \mathcal{L}_{X_H}\langle \mu, u\rangle &= (d\iota_{X_H} + \iota_{X_H} d)\langle \mu, u\rangle \\ &= \iota_{X_H} d\langle \mu, u\rangle \\ &= \iota_{X_H}\iota_{u_M}\omega \\ &= -dH(u_M) \\ &= -\mathcal{L}_{u_M} H. \end{aligned} \tag{2.34}$$

□

Example 2.1.8. A general situation: (M, ω) is symplectic with $\omega = -d\theta$ exact. Suppose that we have a G on M such that $\psi_g^*\omega = \omega$. In this case we define

$$\iota_{u_M}\theta = \langle \mu, u\rangle. \tag{2.35}$$

It's easy to see that this defines a Hamiltonian action in M.

Example 2.1.9. (Cotangent lifting). Take a G-action ψ_g on M, then we can lift it to a G-action $\widehat{\psi}_g$ on T^*M. So we have $\widehat{\psi}_g^*\alpha = \alpha$ and thus we get a Hamiltonian map given by $\langle \mu, u\rangle = \iota_{u_M}\alpha$.

Example 2.1.10. Let $G = \mathbb{R}^n$ acting on $M = \mathbb{R}^n$ by translation $g \cdot q = q + g$. With the identification $T^*M \cong \mathbb{R}^n \times \mathbb{R}^n$ we have $g \cdot (q, p) = (g + q, p)$. We can explictly compute the

moment map:

$$\begin{aligned} u = (a,b) &\mapsto u_M = \sum_i a_i \frac{\partial}{\partial q_i} + \sum_i b_i \frac{\partial}{\partial p_i} \\ \iota_{u_M} \sum_i p_i dq_i &= \sum_i p_i dq_i(u_M) = \sum_i p_i a_i \\ \mu(q,p) &: \mathbb{R}^n \to \mathbb{R}^n : e_i \mapsto p_i \end{aligned} \tag{2.36}$$

So $\mu(q,p) = p$ in the choosen basis.

Example 2.1.11. Let $G = SO(3)$ acting in $\mathbb{R}^3$ in the usual way a lift it to an action on $T^*\mathbb{R}^3 = \mathbb{R}^3 \times \mathbb{R}^3$. Now observe that

$$\begin{aligned} \psi_{g*,v}(\frac{\partial}{\partial x_i}\Big|_v) x_j &= \frac{\partial}{\partial x_i}\Big|_v x_j \circ \psi_g \\ &= d(x_j \circ \psi_g)_v(e_i) \\ &= x_j \circ \psi_g(e_i) \\ &= g_{ji} \end{aligned} \tag{2.37}$$

This proves that $\psi_{g*,v}(\sum_i a_i \frac{\partial}{\partial x_i}\Big|_v) = \sum_i (ga)_i \frac{\partial}{\partial x_i}\Big|_{gv}$. Therefore, the corresponding action on $T\mathbb{R}^3$ is

$$g \cdot (q,p) = (gq, gp). \tag{2.38}$$

Remember that the inverse of the dualization preserve base points and inverse the dual map, so the action on $T^*\mathbb{R}^3$ is

$$g \cdot (q,p) = (gq, (g^{-1})^T p) = (gq, gp),\ g \in SO(3). \tag{2.39}$$

Finnaly we see that

$$\begin{aligned} u_M(q,p) &= \frac{d}{dt}\Big|_{t=0} (e^{tu}q, e^{tu}p) = (uq, up) \\ u_M &= \sum_i (uq)_i \frac{\partial}{\partial q_i} + \sum_i (up)_i \frac{\partial}{\partial p_i} \\ \iota_{u_M} \sum_i p_i dq_i &= \sum_i p_i u_{ij} q_j \\ \mu(q,p) &: \mathfrak{so}(3) \to \mathbb{R} : \ell_1 \mapsto p_3 q_2 - p_2 q_3, \cdots \Rightarrow \mu(q,p) = q \times p \end{aligned} \tag{2.40}$$

2.1.4 Marsden, Weinstein, Meyer theorem

2.1.5 Lazy and sophisticated approach for coadjoint orbits

Remark 2.1.12. Let ψ be a nondegenerate, symmetric bilinear form on a vector space E. If $W \subset E$ is a subspace then $E = W \oplus W^{\perp}$.
In fact, if $\nu : E \to E^*$ is the linear map induced by ψ, then we see that $\nu(W) = \mathrm{Ann}(W)$. Moreover, the exact sequence

$$0 \to \mathrm{Ann}(W) \to E^* \to W^* \to 0 \tag{2.41}$$

yields $\dim(\mathrm{Ann}(W)) + \dim(W) = \dim(E)$. The dimension formula yields $\dim(W) + \dim(W^{\perp}) = \dim(E)$, so $W \cap W^{\perp} = (0)$. Putting together all these results, we get an isomorphism $W \oplus W^{\perp} \to E : (w, w') \mapsto w + w'$.

Lemma 2.1.13. Let $\mathfrak{g} = \mathfrak{c} \oplus [\mathfrak{g}, \mathfrak{g}]$ be a reductive Lie algebra and ψ a symmetric, nondegenerate, invariant bilinear form on $\mathfrak{g}$. Then, for any two simple components $\mathfrak{s}_1, \mathfrak{s}_2 \subset [\mathfrak{g}, \mathfrak{g}]$ we have $\psi(\mathfrak{s}_1, \mathfrak{s}_2) = 0$, $\psi(\mathfrak{s}_1, \mathfrak{c}) = 0$.

Proof: We just observe that $[\mathfrak{s}_1, \mathfrak{s}_2] = 0$, while $[\mathfrak{s}, \mathfrak{s}] = \mathfrak{s}$ for $\mathfrak{s}$ simple. Therefore

$$\psi(\mathfrak{s}_1, \mathfrak{s}_2) = \psi([\mathfrak{s}_1, \mathfrak{s}_1], \mathfrak{s}_2) = \psi(\mathfrak{s}_1, [\mathfrak{s}_1, \mathfrak{s}_2]) = 0. \tag{2.42}$$

In the same way we see that $\psi(\mathfrak{s}, \mathfrak{c}) = 0$. □

Lemma 2.1.14. (1.3.9 - Lazy and sophisticated approach): Let $\mathfrak{g}$ be a reductive Lie algebra with nondegenerate, symmetric, invariant bilinear form ψ. If $f = \psi(X, \cdot) \in \mathfrak{g}^*$, then

$$f|_{\mathfrak{g}^f} = 0 \iff X \text{ is nilpotent.} \tag{2.43}$$

Proof: First assume $f = \psi(X, \cdot)$ with X nilpotent, i.e., $X \in [\mathfrak{g}, \mathfrak{g}]$ and $\mathrm{ad}(X)$ is a nilpotent endomorphism of $\mathfrak{g}$. Now, let $[\mathfrak{g}, \mathfrak{g}] = \mathfrak{s}_1 \oplus \cdots \oplus \mathfrak{s}_r$ and $X = X_1 + \cdots + X_r, Z = Z_0 + Z_1 + \cdots + Z_r$ for $Z_0 \in \mathfrak{c}$, $X_i, Z_i \in \mathfrak{s}_i$.

Then

$$\begin{aligned} f(Z) = \psi(X, Z) &= \sum_{i,j} \psi(X_i, Z_j) \\ &= \sum_i \psi(X_i, Z_i) \\ &= \sum_i \alpha_i K^{\mathfrak{s}_i}(X_i, Z_i) \\ &= \sum_{i,j} \alpha_j K(X_i, Z_j) \\ &= K(X, Z'),\ Z' = \sum_j \alpha_j Z_j \\ &= \mathrm{Tr}(\mathrm{ad}(X) \circ \mathrm{ad}(Z')),\ Z' = \sum_j \alpha_j Z_j. \end{aligned} \tag{2.44}$$

However, if $Z \in \mathfrak{g}^f = \mathfrak{g}^X$, then $[Z, X] = 0 \Rightarrow [Z_i, X_i] = 0\ \forall i$ and $Z' \in \mathfrak{g}^X$ as well. In fact, we have $Z \in \mathfrak{g}^X \iff Z' \in \mathfrak{g}^X$ because $\alpha_i \neq 0,\ \forall i$. Then $[\mathrm{ad}(X), \mathrm{ad}(Z')] = 0$ and since $\mathrm{ad}(X)$ is nilpotent, we see that $\mathrm{ad}(X) \circ \mathrm{ad}(Z')$ is nilpotent and thus must be of trace zero.

Now assume that $f|_{\mathfrak{g}^f} = 0$. We need to show two things: That $X \in [\mathfrak{g}, \mathfrak{g}]$ and $\mathrm{ad}(X)$ is nilpotent. For the first one we have $f|_{\mathfrak{g}^f} = \psi(X, \cdot)|_{\mathfrak{g}^X} = 0$, so $X \in (\mathfrak{g}^X)^\perp$. Moreover, it's clear that $\psi([X, \mathfrak{g}], \mathfrak{g}^X) = 0$ by invariance. So $[X, \mathfrak{g}] \subset (\mathfrak{g}^X)^\perp$, and since the two subspaces have the same dimension, we get $[X, \mathfrak{g}] = (\mathfrak{g}^X)^\perp$. So we see that $X \in (\mathfrak{g}^X)^\perp = [X, \mathfrak{g}]$.

Now we need to show that $\mathrm{ad}(X)$ is nilpotent. To see this, we use the fact that $X = [H, X]$ for some $H \in \mathfrak{g}$. Now, decompose $\mathfrak{g}$ into generalized eigenspaces:

$$\mathfrak{g} = \bigoplus_{s \in \mathbb{C}} E_s^{\mathrm{ad}(H)} = \bigoplus_{s \in \mathbb{C}} \mathfrak{g}_s. \tag{2.45}$$

We claim that if $W \in \mathfrak{g}_s$, then $\mathrm{ad}(X)W \in \mathfrak{g}_{s+1}$. To see this, apply

$$\begin{aligned} (\mathrm{ad}(H) - s - 1)^N \mathrm{ad}(X)W &= \sum_{k=0}^{s} \binom{N}{k} \mathrm{ad}(H)^k \mathrm{ad}(X)W(-s-1)^{N-k} \\ &= \sum_{k=0}^{N} \binom{N}{k} \mathrm{ad}(X)(\mathrm{ad}(H) + 1)^k W(-s-1)^{N-k} \\ &= \mathrm{ad}(X)(\mathrm{ad}(H) + 1 - s - 1)^N W \\ &= \mathrm{ad}(X)(\mathrm{ad}(H) + 1)^N W = 0. \end{aligned} \tag{2.46}$$

□

Remark 2.1.15. Since ψ is nondegenerate, we have the decomposition at the level of vector spaces:

$$\mathfrak{g} = \mathfrak{g}^X \oplus (\mathfrak{g}^X)^{\perp}. \tag{2.47}$$

Now, by dimension formula we see that $\dim(\mathfrak{g}^X)^{\perp} = \dim(\mathfrak{g}) - \dim(\mathfrak{g}^X) = \dim([X, \mathfrak{g}])$. Since $K([X, \mathfrak{g}], \mathfrak{g}^X) = 0$ by invariance, see that $[X, \mathfrak{g}] \subset (\mathfrak{g}^X)^{\perp}$ and the equality follows by dimensional reasons.

2.1.6 Standard triples

For each $r \geq 1$ we have an irreducible representation of $\mathfrak{sl}_2$ on $\mathbb{C}^{r+1}$ given by:

$$\begin{aligned}
\rho_r(X) &= \begin{pmatrix} 0 & 1 & & & \\ & 0 & \ddots & & \\ & & \ddots & \ddots & \\ & & & 0 & 1 \\ & & & & 0 \end{pmatrix}, \\
\rho_r(Y) &= \begin{pmatrix} 0 & & & & \\ \mu_1 & 0 & & & \\ & \ddots & \ddots & & \\ & & \ddots & 0 & \\ & & & \mu_r & 0 \end{pmatrix}, \ \mu_i = i(r+1-i) \\
\rho_r(H) &= \mathrm{Diag}(r, r-2, \ldots, -r+2, -r).
\end{aligned} \tag{2.48}$$

In particular this defines a homomorphism $\mathfrak{sl}_2 \to \mathfrak{sl}_{r+1}$. So, to each nilpotent orbit $\mathcal{O} = \mathcal{O}_{d_1,\ldots,d_r}$ in $\mathfrak{sl}_n$ ($n = d_1 + \cdots + d_r$), we can find a nilpositive element $X \in \mathcal{O}$. In fact, if $k = l([d_1, \ldots, d_r])$ we put $\rho(X) = \bigoplus_{i=1}^{k} \rho_{d_i - 1}(X) = X_{d_1,\ldots,d_r} \in \mathfrak{sl}_n$ and $\{\rho(X), \rho(H), \rho(Y)\}$ is a $\mathfrak{sl}_2-$triple.

2.2 Jacobson-Morozov theorem

Theorem 2.2.1. Let $X \in \mathfrak{g}$ be a nilpotent element of a complex semisimple Lie algebra. Then X is the nilpositive element of some $\mathfrak{sl}_2-$triple in $\mathfrak{g}$.

Proof: The proof will be divided in four steps:

- Induction on dimension: The base case if $\dim(\mathfrak{g}) = 3$, i.e., $\mathfrak{g} = \mathfrak{sl}_2$. If $X \in \mathfrak{g}$ is nilpotent, then we see that there exists $g \in GL_n$ such that $\mathrm{Ad}(g)X = \left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right)$. Putting

$g \to g/\sqrt{\det(g)}$, we see that we have in fact an element $g \in G = SL_2$ such that $\mathrm{Ad}(g)X = \left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right)$. So we have an automorphism $\phi \in \mathrm{Aut}(\mathfrak{g})$ such that $\phi(X)$ is the nilpositive element of the standard triple in $\mathfrak{sl}_2$. Now, we will proceed the proof by assuming $\dim(\mathfrak{g}) > 3$ and that X doesn't lie in any proper semisimple Lie algebra of $\mathfrak{g}$.

- By the proof of theorem (2.1.14), we see that $K(X, \mathfrak{g}^X) = 0$ (here, since $\mathfrak{g}$ is semisimple, we can just take $\psi = K$). Therefore, $X \in (\mathfrak{g}^X)^\perp$ which is equal to $[X, \mathfrak{g}]$ and there exists $H \in \mathfrak{g}$ such that $[H, X] = 2X$.

  In this condition, we can assume H to be semisimple: Since $\mathrm{ad}(H)$ restricts to $\mathbb{C}X$ and $\mathrm{ad}(H)_{s,n}$ are polynomials in $\mathrm{ad}(H)$, then its semisimple and nilpotent parts also restricts to $\mathbb{C}X$. In particular $\mathrm{ad}(H_n)|_{\mathbb{C}X}^{\dim(\mathbb{C}X)} = 0$, i.e., $\mathrm{ad}(H_n)(X) = 0$. Therefore $[H, X] = [H_s, X]$ and we can pass notation to use H as H_s.

- In this part of the proof we will use the induction step to show that $K(H, \mathfrak{g}^X) = 0$, i.e., $H \in (\mathfrak{g}^X)^\perp = [X, \mathfrak{g}]$.

  Suppose that $H \notin (\mathfrak{g}^X)^\perp$, so $K(H, \mathfrak{g}^X) \neq 0$. We know that $\mathfrak{g}^X$ is $\mathrm{ad}(H)-$invariant:

  $$\mathrm{ad}(X)\mathrm{ad}(H)Y = (\mathrm{ad}(H)\mathrm{ad}(X) + \mathrm{ad}([X, H]))Y = -2\mathrm{ad}(X)Y = 0. \tag{2.49}$$

  So, since $\mathrm{ad}(H)$ is semisimple, its restriction to $\mathfrak{g}^X$ is also semisimple and we have the eigenspace decomposition

  $$\mathfrak{g}^X = \bigoplus_{i=1}^{k} E_{\lambda_i}^{\mathrm{ad}(H)|_{\mathfrak{g}^X}} = \bigoplus_{i=1}^{k} \mathfrak{g}_{\lambda_i}^X. \tag{2.50}$$

  In this case we see that if $\lambda_i \neq 0$, then

  $$K(H, \mathfrak{g}_{\lambda_i}^X) = \frac{1}{\lambda_i} K(H, [H, \mathfrak{g}_{\lambda_i}^X]) = 0. \tag{2.51}$$

  So there exist $Z \in \mathfrak{g}_0^X = (\mathfrak{g}^X)^H$ such that $K(H, Z) \neq 0$.

  If Z is nilpotent, then since $[Z, H] = 0$, we see that $K(Z, H) = \mathrm{Trad}(Z) \circ \mathrm{ad}(H) = 0$ because $\mathrm{ad}(Z) \circ \mathrm{ad}(H)$ is nilpotent. Therefore, we will assume that $Z_s \neq 0$. Here we observe that $Z_s \in (\mathfrak{g}_X)^H$. In fact, whenever $\mathrm{ad}([W, Z]) = 0$, we have $\mathrm{ad}([W, Z_s]) = 0$ and $[W, Z_s] = 0$.

  Finally, we see that since $Z_s \neq 0$, then $\mathfrak{g}^{Z_s} \overset{\neq}{\subset} \mathfrak{g}$ and $[\mathfrak{g}^{Z_s}, \mathfrak{g}^{Z_s}]$ is a proper semisimple subalgebra of $\mathfrak{g}$ (we are using the fact that centralizers of single elements $\mathfrak{g}^W$ are reductive algebras). However, since $Z_s \in (\mathfrak{g}_X)^H$, we have $H, X \in \mathfrak{g}^{Z_s}$. But $2X = [H, X] \in [\mathfrak{g}^{Z_s}, \mathfrak{g}^{Z_s}]$, which contradicts the Induction hypothesis.

- Now we know that there exist $H \in \mathfrak{g}$ semisimple such that $[H, X] = 2X$. Moreover, we know that $H = [X, Y]$ for some $Y \in \mathfrak{g}$. Now, decompose $\mathfrak{g}$ into $\mathrm{ad}(H)-$eigenspaces:

  $$\mathfrak{g} = \bigoplus_{i=1}^{k} E_{\lambda_i}^{\mathrm{ad}(H)} = \bigoplus_{i=1}^{k} \mathfrak{g}_{\lambda_i}. \tag{2.52}$$

We see that $\mathrm{ad}(X)\mathfrak{g}_{\lambda_i} \subset \mathfrak{g}_{\lambda_i+2}$. Moreover

$$H = [X, Y] = \sum_{i=1}^{k} [X, Y_i],\ Y_i \in \mathfrak{g}_{\lambda_i}. \tag{2.53}$$

Therefore, since $H \in \mathfrak{g}_0$, we see that $H = [X, Y_{-2}]$, where Y_{-2} is the eigenvector relative to the eigenvalue -2 (we are abusing notation here !). Changing $Y \to Y_{-2}$ we get the standard triple $\{Y, H, X\}$ with X being the nilpositive element fo the standard triple.

□

Remark 2.2.2. In the previous proof we saw that we can take the neutral element H as a semisimple element.

2.3 Conjugation theorems

Some important results to remember:

- Let $\{Y, H, X\}, \{Y', H, X\}$ two standar triples with $H = H_s$ being the neutral element and $X = X_n$ the nilpositive element. Then $Y = Y'$. This follows at once by the condition $[Y - Y', X] = H - H = 0$, so $Y - Y' \in \mathfrak{g}^X \cap \mathfrak{g}_{-2} = 0$. □

- Let $X = X_n \neq 0$, then $\mathfrak{g}^X \cap [X, \mathfrak{g}] = \bigoplus_{i>0} \mathfrak{g}_i^X$. In fact, if $W \in \mathfrak{g}_i^X \setminus 0,\ i > 0$, then W is a highest weight for the action of $\mathfrak{sl}_2 \cong \mathbb{C}\langle Y, H, X\rangle$ (remember the Jacobson Morozov theorem). Since W has weight greater than zero, we necessarily have $Z = [Y, W] \neq 0$ because of the representation theory of $\mathfrak{sl}_2$. Therefore $W = \frac{1}{i}[X, Z] \in [X, \mathfrak{g}]$. Conversely, if $W = \sum_{i\geq 0} W_i,\ W_i \in \mathfrak{g}_i^X$ is also in $[X, \mathfrak{g}]$, then $W_0 \in [X, \mathfrak{g}]$, because $W_i \in [X, \mathfrak{g}]$ for $i > 0$, as we saw before. However, if $W_0 = [X, Z]$, then we can assume $Z \in \mathfrak{g}_{-2}$. So if $Z \neq 0$, we have $\mathrm{ad}(X)^2 Z = [X, W_0] \neq 0$ by the representation theory of $\mathfrak{sl}_2$, which is a contradiciton. Then $W_0 = 0$ and $W \in \bigoplus_{i>0} \mathfrak{g}_i^X$. □

  Notation 2.3.1. We will write $\mathfrak{u}^X = \mathfrak{g}^X \cap [X, \mathfrak{g}] = \bigoplus_{i>0} \mathfrak{g}_i^X$. It's clearly a nilpotent ideal in $\mathfrak{g}^X = \bigoplus_{i\geq 0} \mathfrak{g}_i^X$.

- Fact: We can integrate $\mathfrak{u}^X$ to a simply connected subgroup $U^X \subset G^X_{\mathrm{ad}}$ (Lie third theorem). Moreover, since $\mathfrak{u}^X$ is nilpotent, the exponential map $\exp : \mathfrak{u}^X \to U^X$ is a *diffeomorphism*. Recall the following calculation:

$$\begin{aligned} \mathrm{Ad}_{\mathrm{ad}(Z)} H &= \exp(\mathrm{ad}(Z)) H \\ &= \sum_{i\geq 0} \frac{1}{i!} \mathrm{ad}(Z)^i H \end{aligned} \tag{2.54}$$

which is a finite sum, because $\mathfrak{u}^X$ is nilpotent. Moreover $\mathfrak{u}^X = \bigoplus_{i>0} \mathfrak{g}_i^X$ is $\mathrm{ad}(H)-$invariant and $\mathrm{ad}(Z)^i H \in \mathfrak{g}_i^X,\ i > 0$. Then $\mathrm{Ad}_{\exp(Z)} H \in H + \mathfrak{u}^X$. We will write

$$\mathrm{Ad}(U^X)H \subset H + \mathfrak{u}^X. \tag{2.55}$$

- (Kostant). Previously we saw that $\mathrm{Ad}(U^X)H \subset H + \mathfrak{u}^X$. Indeed, we have more: The two sets are equal $\mathrm{Ad}(U^X)H = H + \mathfrak{u}^X$ and for any $W \in \mathfrak{u}^X$ there exist an unique $Z \in \mathfrak{u}^X$ such that $\mathrm{Ad}_{\exp(Z)}(H) = H + W$.

  Let $\mathfrak{u}^X = \bigoplus_{i=1}^m \mathfrak{g}_i^X$. We construct Z inductively as follows. Let $Z_0 = 0$ and for $j \geq 1$ define

$$\begin{aligned} Z_{j+1} &= Z_j + \frac{1}{j+1} p_{j+1}(\mathrm{Ad}_{\exp(Z_j)}(H) - H - W), \\ &(p_j = \mathfrak{u}^X \to \mathfrak{g}_j^X \ \text{ is the projection}). \end{aligned} \tag{2.56}$$

  We claim that this sequence satisfies two conditions:

$$\begin{aligned} &(1)\ Z_j \in \bigoplus_{i=1}^{j} \mathfrak{g}_i^X \\ &(2)\ \mathrm{Ad}_{\exp(Z_j)}(H) - H - W \in \bigoplus_{i=j+1}^{m} \mathfrak{g}_i^X \end{aligned} \tag{2.57}$$

  for all $j = 1, \ldots, m$. If these two conditions are satisfied, then it'z clear that $Z = Z_m$ is the solution of the problem. Then we proceed by induction. For $j = 1$, we have $Z_1 = -W_1 \in \mathfrak{g}_1^X$ by definition of the projection. Moreover

$$\mathrm{Ad}_{\exp(Z_1)}(H) - H - W \in (\mathrm{ad}(Z_1)H - W) + \sum_{i \geq 2} \mathfrak{g}_i^X. \tag{2.58}$$

  This follows from the fact that $\mathrm{ad}(Z_1)^j : \mathfrak{g}_0^X \to \mathfrak{g}_j^X$. Moreover, by definition

$$\mathrm{ad}(Z_1)H - W = -\mathrm{ad}(W_1)H - W = W_1 - W \in \bigoplus_{j \geq 2} \mathfrak{g}_j^X. \tag{2.59}$$

  Now, assume the result is true for some $j \geq 1$. It's clear from the definition that $Z_{j+1} \in \bigoplus_1^{j+1} \mathfrak{g}_i^X + \mathfrak{g}_{j+2}^X = \bigoplus_1^{j+2} \mathfrak{g}_i^X$. Now, by Baker-Campbel-Hausdorff formula we have the sequence of equalities modulo $\bigoplus_{i \geq j+2} \mathfrak{g}_i^X$:

$$\begin{aligned} &\mathrm{Ad}_{\exp(Z_{j+1})} H - H - W \\ &= \mathrm{Ad}_{\exp(Z_j)} \mathrm{Ad}_{\exp(Z'_{j+1})/(j+1)} H - H - W \\ &= \mathrm{Ad}_{\exp(Z_j)} \left(\mathrm{Ad}_{\exp(Z'_{j+1})/(j+1)} H - H \right) + \mathrm{Ad}_{\exp(Z_j)} H - H - W \end{aligned} \tag{2.60}$$

Now, since $\bigoplus_{\geq j+2}\mathfrak{g}_i^X \subset \mathrm{Ker}(p_{j+1})$, we get

$$\begin{aligned} & p_{j+1}\mathrm{Ad}_{\exp(Z_j)}\left(Ad_{\exp(Z'_{j+1})/(j+1)}H - H\right) \\ & = p_{j+1}\mathrm{Ad}_{\exp(Z_j)}(\frac{[Z'_{j+1},H]}{j+1}) \quad ([H,Z'_{j+1}] = (j+1)Z'_{j+1}) \\ & = -(j+1)\frac{Z'_{j+1}}{j+1} \quad (Z'_{j+1}\in\mathfrak{g}_{j+1}^X) \\ & = -Z'_{j+1} \end{aligned} \tag{2.61}$$

Therefore, we have

$$\mathrm{Ad}_{\exp(Z_{j+1})}H - H - W = -Z'_{j+1} + \mathrm{Ad}_{\exp(Z_j)}H - H - W \mod \bigoplus_{\geq j+2}\mathfrak{g}_i^X \tag{2.62}$$

which implies that $p_{j+1}(\mathrm{Ad}_{\exp(Z_{j+1})}H - H - W) = 0$. The induction hypothesis yields $\mathrm{Ad}_{\exp(Z_j)}H - H - W \in \bigoplus_{i=j+1}^m \mathfrak{g}_i^X$. From this, we conclude the induction step.

- (Kostant). If $\{Y,H,X\},\{Y',H',X\}$ are two standard triples with the same nilpositive element. Then there exist $x\in U^X$ such that $\mathrm{Ad}_x(Y) = Y', \mathrm{Ad}_x(H) = H'$ and $\mathrm{Ad}_x(X) = X$.

  We have $[Y-Y',X] = H - H' \in [\mathfrak{g},X]$ while $[H-H',X] = 2X - 2X = 0$. So

  $$H - H' \in \mathfrak{g}^X \cap [X,\mathfrak{g}] = \mathfrak{u}^X. \tag{2.63}$$

  Therefore, by the previous theorem from Kostant we have $x\in U^X$ such that $\mathrm{Ad}_x(H) = H + (H'-H) = H'$. In particular $x = \exp(Z),\ Z\in\mathfrak{g}^X$, so $\mathrm{Ad}_x(X) = \exp(\mathrm{ad}(Z))X = X$ and $\{\mathrm{Ad}_x(Y),H',X\}$ is a standard triple. Since $\{Y',H',X\}$ is a standard triple with the same neutral and nilpositive element, we must have $\mathrm{Ad}_x(Y) = Y'$.

- (Mal'cev). If $\{Y,H,X\},\{Y',H,X'\}$ are two standard triples with the same neutral element. Then, these two triples are conjugate.

  Let $\mathfrak{g} = \bigoplus_{i\in\mathbb{Z}}\mathfrak{g}_i$ be the decomposition of $\mathfrak{g}$ into $\mathrm{ad}(H)-$eigenspaces. Also, define $\mathcal{P} = \{Z\in\mathfrak{g}_2 : \mathfrak{g}^Z \cap \mathfrak{g}_{-2}\}$ which contains the candidates for nilpositive elements for $\mathfrak{sl}_2-$triples containing H as its neutral element. In this case, the group $(G_{\mathrm{ad}}^H)^\circ$ acts in $\mathcal{P}$.

  Remark 2.3.2. The subgroup $G_{\mathrm{ad}}^H \subset G_{\mathrm{ad}}$ is defined as the group integrating $\mathfrak{g}^H$ and $(G_{\mathrm{ad}}^H)^\circ$ is it connected component containing the identity. The advantage to take the connected component of the identity is that

  $$(G_{\mathrm{ad}}^H)^\circ = \exp(\mathrm{ad}(\mathfrak{g}^H)). \tag{2.64}$$

In fact, we have for $x \in (G_{\text{ad}}^H)^\circ$

$$\begin{aligned} Z \in \mathcal{P} &\iff \mathfrak{g}^Z \cap \mathfrak{g}_{-2} = 0 \\ &\iff x(\mathfrak{g}^Z \cap \mathfrak{g}_{-2}) = 0 \quad (x \text{ is an isomorphism}) \\ &\iff \mathfrak{g}^{x\cdot Z} \cap x(\mathfrak{g}_{-2}) = 0 \quad (x \text{ is an isomorphism}) \\ &\iff \mathfrak{g}^{x\cdot Z} \cap \mathfrak{g}_{-2} = 0 \\ &\quad (\mathfrak{g}^H = \mathfrak{g}_0, \text{ and } \exp(\text{ad}(\mathfrak{g}_0))\mathfrak{g}_{-2} = \mathfrak{g}_{-2}) \\ &\iff x \cdot Z \in \mathcal{P}. \end{aligned} \tag{2.65}$$

The result will follow if we show that $\mathcal{P}$ is connected, while each orbit is clopen. For the first claim, we will prove more: $\mathcal{P}$ is path connected. Define the map $T : \mathfrak{g}_2 \to \text{Hom}(\mathfrak{g}_{-2}, \mathfrak{g}_0)$ given by $T(Z) = \text{ad}(Z)|_{\mathfrak{g}_{-2}}$. Now we see that

$$\begin{aligned} Z \in \mathcal{P} &\iff \mathfrak{g}^Z \cap \mathfrak{g}_{-2} = 0 \\ &\iff \text{Ker}T(Z) = 0 \\ &\iff \dim(\text{Im}T(Z)) = \dim([Z, \mathfrak{g}_{-2}]) = \dim(\mathfrak{g}_{-2}) \\ &\iff T(Z) \text{ has full rank} \end{aligned} \tag{2.66}$$

However, $T(Z)$ is a $\dim(\mathfrak{g}_{-2}) \times \dim(\mathfrak{g}_0)-$matrix and the condition of maximal rank is complementary to the condition that each of its $\dim(\mathfrak{g}_{-2}) \times \dim(\mathfrak{g}_{-2})-$minor is singular. Moreover, since each entry of $T(Z)$ is linear in Z, this last condition is equivalent to the vanishing of a set of polynomials in $\mathfrak{g}_2$. So we see that

$$\mathcal{P} = \mathfrak{g}_2 \setminus \{\det(M(Z)) = 0 : M(Z) \in M_{\dim(\mathfrak{g}_{-2})\times\dim(\mathfrak{g}_{-2})}(\mathbb{C}) \text{ a minor of } T(Z)\}. \tag{2.67}$$

So $\mathcal{P}$ is Zariski open and path connected. Let $v, w \in \mathcal{P}$, then for each minor $M(Z)$ of $T(Z)$ there exist a finite number of complex numbers such that $\det(M(\lambda v + (1-\lambda)w)) = 0$. Then, there exist a finite collection of complex numbers $\{\lambda_1, \ldots, \lambda_k\}$ such that $\det(M(\lambda v + (1-\lambda)w)) = 0$ for each $\dim(\mathfrak{g}_{-2}) \times \dim(\mathfrak{g}_{-2})-$minor $M(Z)$ of $T(Z)$. So, we get a continuous path $\gamma : [0,1] \to \mathbb{C} \setminus \{\lambda_1, \ldots, \lambda_k\}$, that composed with $\lambda \mapsto \lambda v + (1-\lambda)w$ yields a path connecting v and w in $\mathcal{P}$.

To see that each orbit is open, we will compute dimension of each orbit. We have

$$\begin{aligned} \dim((G_{\text{ad}}^H)^\circ \cdot Z) &= \dim(G_{\text{ad}}^H \cdot Z) \\ &= \dim(G^H / G_Z^H) \\ &= \dim(\mathfrak{g}^H) - \dim(\mathfrak{g}^H \cap \mathfrak{g}^Z) \\ &= \dim(\mathfrak{g}_0) - \dim(\mathfrak{g}^Z \cap \mathfrak{g}_0). \end{aligned} \tag{2.68}$$

Moreover, by the nondegeneracy and invariance of the Killing form, we get

$$0 = K([\mathfrak{g}^Z \cap \mathfrak{g}_0, Z], \mathfrak{g}_{-2}) = K(\mathfrak{g}^Z \cap \mathfrak{g}_0, [Z, \mathfrak{g}_{-2}]). \tag{2.69}$$

So $[Z, \mathfrak{g}_{-2}] \subset (\mathfrak{g}^Z \cap \mathfrak{g}_0)^\perp$ and since $K|_{\mathfrak{g}^H \times \mathfrak{g}^H}$ is nondegenerate, we get

$$\dim(\mathfrak{g}^Z \cap \mathfrak{g}_0)^\perp + \dim(\mathfrak{g}^Z \cap \mathfrak{g}_0) = \dim(\mathfrak{g}_0) \tag{2.70}$$

Therefore

$$\dim((G_{\mathrm{ad}}^H)^\circ \cdot Z) \geq \dim(\mathfrak{g}^Z \cap \mathfrak{g}_0)^\perp \geq \dim([Z, \mathfrak{g}_{-2}]) = \dim(\mathfrak{g}_{-2}) \tag{2.71}$$

and $(G_{\mathrm{ad}}^H)^\circ$ is of maximal dimensional in $\mathcal{P}$, so its open in $\mathcal{P}$. Now, since $\mathcal{P}$ is a disjoint union of orbits, we see that each orbit is also closed. □

2.4 Weighted Dynkin diagrams

There is some difference between mine and Collingwood notation:

	Mine	Collingwood
Roots	Δ	Φ
Simple roots	Π	Δ
Fundamental domain	D_Π	D_Δ

The idea here is to classify nilpotent orbits by distinguished semisimple orbits. Here we will fix a Cartan subalgebra $\mathfrak{h} \subset \mathfrak{g}$ with associated decomposition $\mathfrak{g} = \bigoplus_{\alpha \in \Delta} \mathfrak{g}_\alpha$.

$$\mathcal{O}_X \overset{\text{Jacobson-Morozov}}{\longrightarrow} \{Y, H, X\} \to \mathcal{O}_H. \tag{2.72}$$

So we will classify semisimple orbits that cames from neutral elements belonging to a $\mathfrak{sl}_2-$triple.

The idea is the following:

- Let $\{Y, H, X\}$ be a $\mathfrak{sl}_2-$triple with H being the neutral semisimple element. Since H is semisimple, it belongs to some Cartan algebra $\mathfrak{h}_X$ which must be conjugate to $\mathfrak{h}$. So we can take $H \in \mathfrak{h}$ because the adjoint action does not change orbits. Now, we can conjugate H again to take it in the "fundamental" domain D_Π (why ?).

- Let

$$\begin{aligned} \mathfrak{n} &= \bigoplus_{\alpha \in \Delta_+} \mathfrak{g}_\alpha, \\ \overline{\mathfrak{n}} &= \bigoplus_{\alpha \in \Delta_-} \mathfrak{g}_\alpha. \end{aligned} \tag{2.73}$$

  Then we have the Borel subalgebra $\mathfrak{b} = \mathfrak{h} \oplus \mathfrak{n}$ and the negative Borel subalgebra $\overline{\mathfrak{b}} = \mathfrak{h} \oplus \overline{\mathfrak{n}}$. Recall the definition of the *fundamental domain*:

$$D_\Pi = \{H \in \mathfrak{h} : \mathrm{Re}(\alpha(H)) \geq 0,\ \mathrm{Im}(\alpha(H)) = 0,\ \forall \alpha \in \Pi\}. \tag{2.74}$$

- We claim that if $H \in D_\Pi$, then it can be completely determined by a map in

$$\{0,1,2\}^\Pi \tag{2.75}$$

called *weighted Dynkin diagram*. First, remark that $\mathfrak{h}^* = \text{Span}(\Delta) = \text{Span}(\Pi)$. So $\alpha(H) = 0 \ \forall \alpha \in \Pi$ if and only if $H = 0$. Now, we have the $\text{ad}(H)-$eigenspace decomposition:

$$\mathfrak{g} = \bigoplus_{i \in \mathbb{Z}} \mathfrak{g}_i. \tag{2.76}$$

By the representation theory of $\mathfrak{sl}_2$ we see that Ker $\text{ad}(X)$ is spanned only by highest weight spaces:

$$\mathfrak{g}^X = \bigoplus_{i \geq 0} \mathfrak{g}_i^X, \ \mathfrak{g}_i^X = \mathfrak{g}^X \cap \mathfrak{g}_i \tag{2.77}$$

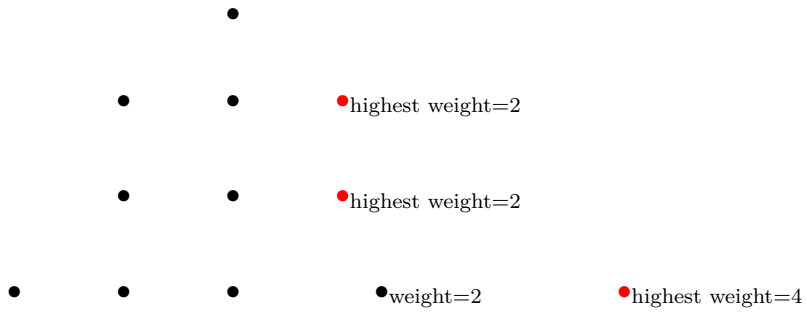
and in a similar way: $\mathfrak{g}^Y = \bigoplus_{i \leq 0} \mathfrak{g}_i^Y$.

Remark 2.4.1. We have $H \in \mathfrak{h}$, for some $\mathfrak{h}$ Cartan sublgebra. Then, the triangular decomposition $\mathfrak{g} = \bigoplus_{\alpha \in \Delta} \mathfrak{g}_\alpha$ provides a diagonalization of H. Now, if we decompose $\mathfrak{g} = \bigoplus_{i \in \mathbb{Z}} \mathfrak{g}_i$ as a direct sum of $\mathfrak{sl}_2 \cong \mathbb{C}\langle X, H, Y \rangle-$modules, then

$$\mathfrak{g}_i = \bigoplus_{\alpha} \mathfrak{g}_\alpha \cap \mathfrak{g}_i \tag{2.78}$$

because $\mathfrak{g}_\alpha$ is a weight space for $\mathfrak{sl}_2$. Therefore, for each i such that $\mathfrak{g}_i \neq 0$, there exist $\alpha \in \Delta$ such that $\mathfrak{g}_\alpha \cap \mathfrak{g}_i \neq 0$ and $i = \alpha(H)$.

Suppose that $\mathfrak{g}$ has the following decomposition as a $\mathfrak{sl}_2-$module:



Here $\mathfrak{g}_1^X$ has dimension 2, while $\mathfrak{g}_1$ has dimension 3. Moreover $\mathfrak{g}_3 = \mathfrak{g}_3^X$.

Remark 2.4.2. $\mathfrak{g}_i$, i highest weight for the action of $\mathbb{C}\langle Y, H, X \rangle$ is just $\mathfrak{g}_i^X$.

Remark 2.4.3. We have $\alpha(H) \in \mathbb{Z}_{\geq 0}$, by the representation theory of $\mathfrak{sl}_2$ and the fact that $H \in D_\Pi$.

Now, let $\alpha \in \Pi, \beta \in \Delta_-$, then $\alpha + \beta \in \Delta$ only if if $\alpha + \beta \in \Delta_-$. Otherwise, if $\alpha + \beta \in \Delta_+$, then $\alpha = (\alpha + \beta) - \beta$ will not be simple.

With this observation we see that

$$[X_\alpha, Y] \in \overline{\mathfrak{n}}, \ \forall \alpha \in \Pi, \tag{2.79}$$

because $Y \in \overline{\mathfrak{n}}$, since $\mathrm{ad}(H)Y = -2Y$ and $H \in D_\Pi$. Then

$$\begin{aligned} &[X_\alpha, Y] = 0 \Rightarrow X_\alpha \in \mathfrak{g}^Y \Rightarrow \alpha(H) \leq 0, \text{ so } \alpha(H) = 0 \\ &[X_\alpha, Y] \neq 0 \Rightarrow \alpha(H) - 2 \leq 0 \text{ (by the above observation), so } \alpha(H) \leq 2 \end{aligned} \tag{2.80}$$

- Now, for two nilpotent orbits $\mathcal{O}, \mathcal{O}'$ we have associated semisimple elements $H, H' \in D_\Pi$ and the corresponding Dynkin diagrams

$$D(H), D(H') \in \{0, 1, 2\}^\Pi. \tag{2.81}$$

If $D(H) = D(H') \neq 0$, then $H = H' \neq 0$ because of the observation made before: $\alpha(H - H') = 0, \ \forall \alpha \in \Pi \Rightarrow H - H' = 0$.

□

Lemma 2.4.4. We have $\#\Pi = \dim(\mathfrak{h}^*) = \mathrm{Rank}(\mathfrak{g})$.

Proof: We know that $\mathrm{Span}(\Pi) = \mathrm{Span}(\Delta)$, by definition of simple roots. So the result yields provided we show that the set Π is linearly independent. We prove it in two steps:

- Claim: If $\alpha, \beta \in \Pi$, then $(\alpha, \beta) \leq 0$. In fact, if $(\alpha, \beta) > 0$, then

$$p - q = 2\frac{(\alpha, \beta)}{(\alpha, \alpha)} > 0. \tag{2.82}$$

This means that $p > q \geq 0$, so $p \geq 1$ and $\beta - \alpha \in \Delta \Rightarrow \beta - \alpha \in \Delta_-$ and $\alpha - \beta \in \Delta_+$, which is a contradiction with the fact that $\alpha = (\alpha - \beta) + \beta$ is simple.

- Now suppose that $\sum_{\alpha \in \Pi} a_\alpha \alpha = 0$. Then if at least one of the sets $I = \{\alpha : a_\alpha > 0\}$ and $J = \{\alpha : a_\alpha < 0\}$ is nonempty, then we can write this linear dependence as

$$\gamma = \sum_{\alpha \in I} a_\alpha \alpha = \sum_{\alpha \in J} (-a_\alpha)\alpha \in \Delta_+ \cup \{0\}. \tag{2.83}$$

Therefore

$$(\gamma, \gamma) = \sum_{\alpha \in I} \sum_{\beta \in J} (-a_\alpha a_\beta)(\alpha, \beta) \leq 0. \tag{2.84}$$

So $(\gamma,\gamma)=0$ and $\gamma=0$. However, from this we deduce that for $\beta\in\Pi$

$$0=(\gamma,\beta)=\sum_{\alpha\in I}a_\alpha(\alpha,\beta),\ (\alpha,\beta)\leq 0, a_\alpha\geq 0. \tag{2.85}$$

This forces $(\alpha,\beta)=0,\ \forall\alpha\in I$. In the same way we see that $(\alpha,\beta)=0,\ \forall\alpha\in J$, so

$$(\alpha,\beta)=0,\ \forall\alpha\in I\cup J. \tag{2.86}$$

So $\alpha\in I\cup J\neq\varnothing\Rightarrow(\alpha,\alpha)=0$ and $\alpha=0$, which is a contradiciton. □

Lemma 2.4.5. There is an isomorphism:

$$W\xrightarrow{\cong}N_{G_{\mathrm{ad}}}(\mathfrak{h})/C_{G_{\mathrm{ad}}}(\mathfrak{h}). \tag{2.87}$$

Proof: We will construct this isomorphism in several steps:

- We can transpose the action of W to $\mathfrak{h}$ vi the contragradient representation:

$$\langle w\cdot h,\alpha\rangle=\langle h,w^{-1}(\alpha)\rangle. \tag{2.88}$$

Write s_α for the action of r_α transposed to $\mathfrak{h}\cong\mathfrak{h}^{**}$. Then we have

$$\begin{aligned}\langle s_\alpha(H),\lambda\rangle&=\langle H,r_\alpha(\lambda)\rangle\\&=\langle H,\lambda-2\frac{(\alpha,\lambda)}{(\alpha,\alpha)}\alpha\rangle\\&=\langle H,\lambda-2\frac{\lambda(\nu^{-1}(\alpha))}{(\alpha,\alpha)}\alpha\rangle\\&=\langle H,\lambda\rangle-2\frac{1}{(\alpha,\alpha)}\langle H,\alpha\rangle\langle\nu^{-1}(\alpha),\lambda\rangle\\&=\langle H-2\frac{\langle H,\alpha\rangle}{(\alpha,\alpha)}\nu^{-1}(\alpha),\lambda\rangle\\&=\langle H-\langle H,\alpha\rangle H_\alpha,\lambda\rangle.\end{aligned} \tag{2.89}$$

So $s_\alpha(H)=H-\alpha(H)H_\alpha$, defines the action of $W\cong\langle s_\alpha:\alpha\in\Delta\rangle$ on $\mathfrak{h}$.

- Define $\theta_\alpha=\exp(\mathrm{ad}(E_\alpha))\exp(\mathrm{ad}(-F_\alpha))\exp(\mathrm{ad}(E_\alpha))\in\mathrm{Aut}(\mathfrak{g})$, where $\{F_\alpha,H_\alpha,E_\alpha\}$ is the standard $\mathfrak{sl}_2-$triple, i.e., $(E,F)\in\mathfrak{g}_\alpha\times\mathfrak{g}_{-\alpha}$ is such that $K(E,F)=2/K(\alpha,\alpha)$ and $H_\alpha=2\nu^{-1}(\alpha)/K(\alpha,\alpha)$. We can show that $\theta_\alpha|_{\mathfrak{h}}=s_\alpha$. So each reflection s_α can be extended to an automorphism of $\mathfrak{g}$.

Remark 2.4.6. If $\delta\in\mathrm{Der}(\mathfrak{g})$ is a nilpotent derivation, then $\exp(\delta)\in\mathrm{Aut}(\mathfrak{g})$.

- It's a fact that $\theta_\alpha \in \mathrm{Ad}(G) \subset \mathrm{Aut}(\mathfrak{g})$. This follows by the identity $\mathrm{Ad} \circ \exp(Z) = \exp \circ \mathrm{ad}(Z)$. Also, we have $\langle \theta_\alpha : \alpha \in \Delta \rangle = N_{G_{ad}}(\mathfrak{h})$. So we have a surjective map

$$W \to N_{G_{ad}}(\mathfrak{h}) : r_\alpha \mapsto \theta_\alpha \tag{2.90}$$

that induces the desired isomorphism.

We need to be more careful here. See section (2.8)

Now lets do an example in $\mathfrak{sl}_n$. In this case, the root space is

$$\Delta = \{L_i - L_j : i \neq j\} \subset V = \mathrm{Span}(L_i)_{1 \leq i \leq n} = \mathbb{R}^n. \tag{2.91}$$

with respect to the decomposition

$$\mathfrak{sl}_n = \bigoplus_{i>j} \mathbb{C}E_{ij} \oplus \mathfrak{h} \oplus \bigoplus_{i<j} \mathbb{C}E_{ij} \tag{2.92}$$

where $\mathfrak{h} = \mathfrak{sl}_n \cap \{\text{diagonal matrices}\}$. Of course $\mathfrak{g}_{L_i - L_j} = \mathbb{C}E_{ij}$ as can be seen by the equation $\mathrm{ad}(H)E_{ij} = (L_i - L_j)(H)E_{ij}$.

We define a linear functional $f : V \to \mathbb{R} : L_i \mapsto -i$. So, the positive roots with respect to this choice of functional is $\Delta_+ = \{L_i - L_j : i < j\}$. The Cartan triples are $\{E_{ji}, H_{i,j}, E_{ij}\}$ where $H_{i,j} = E_{ii} - E_{jj}$. So, the Cartan subalgebra $\mathfrak{h}$ lies in the borel set of upper triangular matrices in $\mathfrak{sl}_n$. Said this, the set of simple roots is

$$\Pi = \{L_1 - L_2, L_2 - L_3, \ldots, L_{n-1} - L_n\}. \tag{2.93}$$

In fact, all other positive roots can be writen as a linear combination of these roots. The fact that this spanning set has $n - 1 = \mathrm{Rank}(\mathfrak{sl}_n)$ elements, shows that they must be the simple roots. Now, $H = \mathrm{Diag}(\lambda_1, \ldots, \lambda_n) \in D_\Pi$ only if $\alpha(H) \geq 0 \ \forall \alpha \in \Pi$, i.e,

$$\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_n. \tag{2.94}$$

This define the $\Pi-$dominant set in $\mathfrak{h}$. Remember that $[e, f] = K(e, f)\nu^{-1}(\alpha)$ for $(e, f) \in \mathfrak{g}_{-\alpha} \times \mathfrak{g}_\alpha$. So, if $h_\alpha = K(e, f)\nu^{-1}(\alpha)$ we get

$$[h_\alpha, e] = K(e, f)(\alpha, \alpha)e = 2e \iff \frac{2}{(\alpha, \alpha)} = K(e, f). \tag{2.95}$$

So, the fact that $\{E_{ji}, H_{i,j}, E_{ij}\}$ forms a $\mathfrak{sl}_2-$triple shows that $H_{i,j} = H_{L_i - L_j}$ in the sence of the definition of a coroot $h_\alpha = \frac{2}{(\alpha,\alpha)}\nu^{-1}(\alpha)$. Now, we can use the known expression for the action of the Weyl group on coroots:

$$\begin{aligned} s_{L_i - L_j}(H) &= H - (\lambda_i - \lambda_j)H_{L_i - L_j} \\ &= H - (\lambda_i - \lambda_j)H_{i,j} \\ &= \mathrm{Diag}(\lambda_1, \ldots, \lambda_k - \lambda_k + \lambda_l, \ldots, \lambda_l - \lambda_l + \lambda_k, \ldots \lambda_n) \\ &= \mathrm{Diag}(\lambda_1, \ldots, \hat{\lambda}_l, \ldots, \hat{\hat{\lambda}}_k, \ldots, \lambda_n), \\ &\quad \hat{} = k\text{th position}, \hat{\hat{}} = l\text{th position } (k < l). \end{aligned} \tag{2.96}$$

So the transposed action of the weyl group on $\mathfrak{h}$ permutes elements of the diagonal. We know that this action comes from elements of G_{ad}, so we are not changing the orbit of H by applying these maps. So, reordering the diagonal elements of H, we take its conjugate in the domain D_Π as a diagonal matrix with decreasing elements on the diagonal.

For $H = \text{Diag}(\lambda_1, \ldots, \lambda_n)$, we have the weighted Dynkin diagram:

$$L_i - L_{i+1} \in \Pi \mapsto \lambda_i - \lambda_{i+1} \in \{0, 1, 2\}. \tag{2.97}$$

For example, let $n = 3$. We have the orbit $\mathcal{O}_{[2,1]}$ with representative $X_{[2,1]}$. Remember that

$$X_{[2,1]} = \rho_1(X) \oplus \rho_0(X) = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \oplus (0) = E_{1,2} \in \mathfrak{sl}_3. \tag{2.98}$$

The corresponding semisimple element constructed explictly is

$$H_{[2,1]} = \rho_1(H) \oplus \rho_0(H) = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \oplus (0) = H_{1,2}. \tag{2.99}$$

Reordering the elements of the diagonal in a decreasing order we get $\text{Diag}(1, 0, -1)$. Then the associated weighted Dyinkin diagram is

$$\begin{cases} L_1 - L_2 \mapsto 1, \\ L_2 - L_3 \mapsto 1. \end{cases}$$

2.5 Regular, subregular and minimal orbits

We have a partial ordering for the set of orbits defines as follows:

$$\mathcal{O} \leq \mathcal{O}' \iff \overline{\mathcal{O}} \subset \overline{\mathcal{O}'} \tag{2.100}$$

in the Zariski topology. The reflexive property is a consequence of the fact that in the Zariski topology, every orbit is open in its closure (see [2]). In fact, suppose that $\mathcal{O} \leq \mathcal{O}'$ and $\mathcal{O}' \leq \mathcal{O}$, then $\overline{\mathcal{O}} = \overline{\mathcal{O}}'$. Now, there exist Zariski open set U such that $\mathcal{O} = \overline{\mathcal{O}} \cap U = \overline{\mathcal{O}}' \cap U$. However, by definition of the closure, there exist an element $x \in \mathcal{O}' \cap U \subset \mathcal{O}$ and the orbits intersect. So they are equal.

2.5.1 Regular and subregular orbit

The regular (or principal) orbit is the unique orbit of dimension $\dim(\mathfrak{g}) - \text{Rank}(\mathfrak{g})$. We construct it as follows, let $\mathfrak{h} \subset \mathfrak{g}$ be a Cartan subalgebra. Then we have a triangular decomposition $\mathfrak{g} = \mathfrak{h} \oplus \bigoplus_{\alpha \in \Delta} \mathfrak{g}_\alpha$. Choose a set of positive roots and let Π be the associated set of simple roots. Furthermore, let $H \in \mathfrak{h}$ be such that $\alpha(H) = 2$, $\forall \alpha \in \Pi$ (we can do it because $\text{Span}(\Pi) = \mathfrak{h}^*$). So we can write $H = \sum_{\alpha \in \Pi} a_\alpha H_\alpha$. Now define

$$X = \sum_{\alpha \in \Pi} X_\alpha \text{ and } Y = \sum_{\alpha \in \Pi} a_\alpha Y_\alpha. \tag{2.101}$$

Then $\{Y, H, X\}$ forms a $\mathfrak{sl}_2-$triple. The only nontrivial relation is

$$[X, Y] = \sum_{\alpha,\beta\in\Pi} a_\beta [X_\alpha, Y_\beta] = \sum_{\alpha\in\Pi} a_\alpha H_\alpha = H. \tag{2.102}$$

In the last equation we only used the fact that $\alpha - \beta$ is never a root for simple roots $\alpha, \beta \in \Pi$. Since $\alpha(H) = 2,\ \forall\alpha \in \Pi$, we see that $\alpha(H) \in 2\mathbb{Z},\ \forall\alpha \in \Delta$ and $\mathfrak{g}_1 = 0$. Therefore $\mathfrak{g}_0 = \mathfrak{h}$ and the dimension formula follows. For uniqueness, it can be proven that $\overline{\mathcal{O}}_X = \mathcal{N}$ the nilpotent cone. This proves uniqueness.

The subregular orbit is the unique orbit of dimension $\dim(\mathfrak{g}) - \mathrm{Rank}(\mathfrak{g}) - 2$. See [2] for its existence and uniqueness.

2.5.2 Minimal orbit

The singleton $\{0\}$ is the unique orbit of 0 dimension, because $\mathfrak{g}^X = \mathfrak{g} \iff X \in Z(\mathfrak{g}) = (0)$, because $\mathfrak{g}$ is semisimple. There exist an unique orbit of minimal dimension (above 0) lying over $\{0\}$. Let θ be the highest roots of Δ. Remember that on $\Delta \cup \{0\}$, we have a partial relation defined by $\gamma \geq \beta \iff \gamma - \beta = \sum_{\alpha\in\Pi} m_\alpha\alpha,\ m_\alpha \geq 0\ \forall\alpha$. The minimal orbit is the one containing $X = X_\theta$.

2.5.3 Dimension formula for the orbits

There is a nice formula for the dimension of the orbits in the case of $\mathfrak{sl}_n$. Let $\lambda = [d_1, \ldots, d_r]$ with multiplicities $r_i = \#\{j : d_j = i\}$ and associated conjugated partition with parts $s_i = \#\{j : d_j \geq i\}$. We have

$$\dim \mathcal{O}_\lambda = \sum_i s_i^2 - 1. \tag{2.103}$$

The idea is the following, let $V = \mathbb{C}^n$ be the standard representation of $\mathfrak{sl}_n$ and let $\{Y, H, X\}$ be the $\mathfrak{sl}_2-$triple associated do the nilpotent element $X = X_\lambda$ (see equatio 2.48). Decompose

$$V = \bigoplus_{d\geq 0} M(d) \tag{2.104}$$

where $M(d)$ is the isotypical component containing all the irreducible $\mathfrak{sl}_2-$modules V_d of highest weight d. Also, let $L(d) \subset M(d)$ be the smaller weight subspace, so $\dim L(d)$ is the multiplicity of V_d in $M(d)$. For example, let $\lambda = [2, 1^2]$, so we have

$$X_\lambda = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \tag{2.105}$$

and we have the following decomposion of V:

$$e_3 \xrightarrow{X} 0$$

$$e_4 \xrightarrow{X} 0$$

$$e_2 \xrightarrow{X} e_1 \xrightarrow{X} 0$$

So $V = M(0) \oplus M(1)$, where $M(0) = L(0) = \mathbb{C}e_3 + \mathbb{C}e_4$ and $M(1) = \mathbb{C}e_1 + \mathbb{C}e_2$. Now, take a general partition λ. We define a map

$$T_{d,e} : \mathfrak{gl}_n^X \to \mathrm{Hom}(L(d), M(e)) : Z \mapsto \pi_e \circ Z \circ \iota_d \tag{2.106}$$

where $\pi_e : V \to M(e)$ is the projection and $\iota_d : L(d) \to V$ is the inclusion. Now, we have $M(d) = \bigoplus_{k\geq 0} X^k L(d)$ and if $Z \in \mathfrak{gl}_n^X$, then it's completely determined by its values on $L(d)$. In particular, we have an injective map

$$\mathfrak{gl}_n^X \hookrightarrow \bigoplus_{d\geq 0} \mathrm{Hom}(L(d), V). \tag{2.107}$$

So we construct another monomorphism

$$\mathfrak{gl}_n^X \hookrightarrow \bigoplus_{d,e\geq 0} \mathrm{Hom}(L(d), M(e)) : Z \mapsto (T_{d,e}(Z))_{d,e\geq 0}. \tag{2.108}$$

The idea, is that now we have to calculate $\dim(\mathfrak{gl}_n^X) = \sum_{d,e\geq 0} \dim T_{d,e}(\mathfrak{gl}_n^X)$. Now, for each Jordan Block of size $d+1$ (d being a part of λ), we have an irreducible module $V_d \subset M(d)$. Therefore $\dim L(d)$ is the number of parts in λ equal to $d+1$, i.e., $\dim L(d) = r_{d+1}$. Now, let $e \geq d$, then an irreducible $\mathfrak{sl}_2-$module inside $M(e)$ consist of a string:

$$v_e \xrightarrow{X} v_{e-1} \xrightarrow{X} \cdots \xrightarrow{X} v_d \xrightarrow{X} \cdots v_1 \xrightarrow{X} v_0 \xrightarrow{X} 0 \quad (*) \tag{2.109}$$

Let $v \in L(d)$, then since $X^{d+1}v = 0$, we see that Zv has zero component in the subspace $\mathrm{Span}\{v_e, \ldots, v_{d+1}\}$ for any irreducible string $(*)$ inside $M(e)$. This proves that matrices of $T_{d,e}(Z)$ projected to a string like $(*)$ are of the form

$$\begin{pmatrix} \cdots & a_{0j} & \cdots \\ \vdots & \vdots & \vdots \\ \cdots & a_{dj} & \cdots \\ \cdots & 0 & \cdots \\ \vdots & \vdots & \vdots \\ \cdots & 0 & \cdots \end{pmatrix} \quad (**) \tag{2.110}$$

for $1 \leq j \leq \dim L(d) = r_{d+1}$. Furthermore, since $Z \in \mathfrak{gl}_n^X$, then it's completely determined by the collection of matrices $(**)$. So, we have $(d+1)r_{d+1}$ choices for the coefficients for each

one of the r_{e+1} string $(*)$ in $M(e)$. So we conclude that $\dim T_{d,e}(\mathfrak{gl}_n^X) = (d+1)r_{d+1}r_{e+1},\ d \leq e$. In a similar way, we prove that $\dim T_{d,e}(\mathfrak{gl}_n^X) = (e+1)r_{d+1}r_{e+1},\ d > e$. Now write $S_{d,e} = \dim T_{d,e}(\mathfrak{gl}_n^X)$, then we see that $S_{d,e} = S_{e,d}$ and

$$\begin{aligned}
\sum_{d,e\geq 0} S_{d,e} &= \sum_{d\geq 0} S_{d,d} + 2\sum_{e>d\geq 0} S_{d,e} \\
&= \sum_{d\geq 0}(d+1)r_{d+1}^2 + 2\sum_{e>d\geq 0}(d+1)r_{d+1}r_{e+1} \\
&= \sum_{d\geq 0}(d+1)r_{d+1}(r_{d+1} + 2r_{d+2} + 2r_{d+3} + \cdots) \\
&= \sum_{d\geq 0}(d+1)(s_d - s_{d+1})(s_d - s_{d+1} + 2s_{d+1}) \\
&= \sum_{d\geq 0}(d+1)(s_d - s_{d+1})(s_d + s_{d+1}) \\
&= \sum_{i\geq 0} s_i^2
\end{aligned} \tag{2.111}$$

As an example, for $\lambda = [2, 1^2]$ and $X = X_\lambda$, we have the four images $T_{d,e}(\mathfrak{gl}_4^X),\ d, e \in \{0, 1\}$:

$$\begin{pmatrix} \left(\begin{smallmatrix} a & b \\ 0 & a \end{smallmatrix}\right) & 0 \\ 0 & 0 \end{pmatrix},\ \begin{pmatrix} 0 & 0 \\ \left(\begin{smallmatrix} a & b \\ 0 & a \end{smallmatrix}\right) & 0 \end{pmatrix},\ \begin{pmatrix} 0 & \left(\begin{smallmatrix} a & b \\ 0 & 0 \end{smallmatrix}\right) \\ 0 & 0 \end{pmatrix} \text{ and } \begin{pmatrix} 0 & 0 \\ 0 & \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \end{pmatrix}. \tag{2.112}$$

2.6 Bala - Carter theory

Here, if not stated otherwise, we assume $\mathfrak{g}$ to be a semisimple Lie algebra.

Definition 2.6.1. A parabolic subalgebra $\mathfrak{p} \subset \mathfrak{g}$ is a subalgebra containing a borel subalgebra.

Here we describe a recipe to construct representatives of conjugacy classes of parabolic subalgebras. Choose a Cartan subalgebra $\mathfrak{h} \subset \mathfrak{g}$ and a set of positive roots Δ_+. Let $S \subset \Pi$ be any subset. We define the corresponding parabolic subalgebra

$$\mathfrak{p}_S = \langle \mathfrak{h} \oplus \bigoplus_{\alpha\in\Pi} \mathfrak{g}_\alpha \oplus \bigoplus_{\alpha\in S} \mathfrak{g}_{-\alpha} \rangle. \tag{2.113}$$

The map $S \mapsto \mathfrak{p}_S$ is injective. Since there exist $2^{\mathrm{rank}(\mathfrak{g})}$ possibilities for the choice of subsets $S \subset \Pi$, we see that there exist $2^{\mathrm{rank}(\mathfrak{g})}$ different conjugacy classes of parabolic subalgebras of a semisimple Lie algebra [2].

Example 2.6.2. Let $\mathfrak{sl}_4$ and $\Pi = \{\alpha_1, \alpha_2, \alpha_3\},\ \alpha_i = L_i - L_{i+1}$ and $S = \{\alpha_1\}$. We have

$$\mathfrak{p}_S = \mathfrak{h} \oplus \mathfrak{g}_{\alpha_1} \oplus \mathfrak{g}_{\alpha_2} \oplus \mathfrak{g}_{\alpha_3} \oplus \mathfrak{g}_{-\alpha_1}. \tag{2.114}$$

In fact, the difference of two simple roots is never a root, so the above direct sum is a subalgebra.

Lemma 2.6.3. A parabolic subalgebra $\mathfrak{p}$ has a decomposion $\mathfrak{p} = \mathfrak{l} \oplus \mathfrak{n}$ into a reductive subalgebra and a nilpotent one. The reductive subalgebras $\mathfrak{l}$ is called a Levi subalgebra. If $\mathfrak{p} = \mathfrak{p}_S,\ S \subset \Pi$, then

$$\mathfrak{l} = \mathfrak{h} \oplus \bigoplus_{\alpha \in \langle S \rangle} \mathfrak{g}_\alpha \tag{2.115}$$

where S is the root system generated by S. For example, if $S = \{\alpha\}$, then $\langle S \rangle = \{\pm\alpha\}$.

2.7 Standard slices

Lemma 2.7.1. Let $f : M \to \mathbb{R}^k,\ g : N \to \mathbb{R}^k$ be an imersion and an embedding respectively. Suppose that $x = f(p) = g(q)$ and

$$\mathbb{R}^k = f_{*,p} T_p M \oplus g_{*,q} T_q N. \tag{2.116}$$

Then, there exist a neighborhoods $U \ni p, V \ni q$ such that $f(U) \cap g(V) = \{x\}$.

Proof: Remember the smooth strucutre of $M \times N$: If (φ, U) and (ψ, V) are charts around p and q respectively. Then $(\pi_M^* \varphi, \pi_N^* \psi), U \times V$ is a chart around (p, q). If $\varphi = (x_1, \ldots, x_m)$ and $\psi = (y_1, \ldots, y_n)$, then we will write $\bar{x}_j = \pi_M^*(x_j)$ and $\bar{y}_j = \pi_N^*(y_j)$. So $(\bar{x}_1, \ldots, \bar{x}_m, \bar{y}_1, \ldots, \bar{y}_n)$ is a coordinate chart around (p, q). Under this notation it's clear that

$$(\pi_M)_*(\frac{\partial}{\partial \bar{x}_j}) = \frac{\partial}{\partial x_j},\ (\pi_N)_*(\frac{\partial}{\partial \bar{y}_j}) = \frac{\partial}{\partial y_j} \tag{2.117}$$

and that the map

$$\begin{cases} \dfrac{\partial}{\partial \bar{x}_j} \mapsto (\dfrac{\partial}{\partial x_j}, 0) \\ \dfrac{\partial}{\partial \bar{y}_j} \mapsto (0, \dfrac{\partial}{\partial y_j}) \end{cases} \tag{2.118}$$

defines an isomorphism $T_{(p,q)}(M \times N) \to T_p M \times T_q N$.

Now, let go to the proof. Define the map $\phi : M \times N \to \mathbb{R}^k : (p, q) \mapsto f(p) - g(q)$, that

is $\phi = f \circ \pi_M - g \circ \pi_N$. We have $\phi_*(\frac{\partial}{\partial \bar{x}_j}) = f_*(\frac{\partial}{\partial x_j})$ and $\phi_*(\frac{\partial}{\partial \bar{y}_j}) = -g_*(\frac{\partial}{\partial y_j})$. So

$$\begin{aligned}
&\phi_{*,(p,q)}(\sum_i a_i \frac{\partial}{\partial \bar{y}_i} + \sum_j b_j \frac{\partial}{\partial y_j}) = 0 \iff \\
&f_{*,p}(\sum_i a_i \frac{\partial}{\partial x_i}) - g_{*,q}(\sum b_j \frac{\partial}{\partial y_j}) = 0 \iff \\
&f_{*,p}(\sum_i a_i \frac{\partial}{\partial x_i}) = g_{*,q}(\sum_j b_j \frac{\partial}{\partial y_j}) = 0 \text{ (by transversality)} \iff \\
&a_i = 0,\ \forall i \text{ and } b_j = 0,\ \forall j \quad (f_*, g_* \text{ are injective})
\end{aligned} \tag{2.119}$$

This proves that ϕ_* is injective and by dimensionality it must be an isomorphism as well. Therefore ϕ is a local diffeomorphism around (p,q). So there exist $U \times V \ni (p,q)$ such that $\phi|_{U\times V}$ is a diffeomorphism onto its image. In particular $(\phi|_{U\times V})^{-1}(0) = (p,q)$ and $f(U) \cap g(V) = \{x\}$.

Theorem 2.7.2. Let $\mathfrak{g}$ be a semisimple Lie algebra, $X \in \mathfrak{g}$ a nilpotent element and $\mathfrak{a} = \mathbb{C}\langle Y, H, X\rangle$ a $\mathfrak{sl}_2-$triple containing X as its nilpositive element. Then we have

$$\begin{aligned}
&(1) \quad T_X\mathcal{O}_X \oplus T_X(X + \mathfrak{g}^Y) = [\mathfrak{g}, X] \oplus \mathfrak{g}^Y = \mathfrak{g} \\
&(2) \quad \mathcal{O}_X \cap (X + \mathfrak{g}^Y) = \{X\}
\end{aligned} \tag{2.120}$$

Proof: It's a fact that the inclusion $\mathcal{O}_X \hookrightarrow \mathfrak{g}$ is an imersion and $T_X\mathcal{O}_X = \{\mathrm{ad}(X)Z : Z \in \mathfrak{g}\} = [\mathfrak{g}, X]$ with the identification $T_X\mathcal{O}_X \cong i_*(T_X\mathcal{O}_X)$. The equation $[\mathfrak{g}, X] \oplus \mathfrak{g}^Y = \mathfrak{g}$ is a consequence of the representation theory of $\mathfrak{sl}_2$. Now, by exponentiating the relation $\mathrm{ad}(zH)X = 2zX$ (viewed as elements of $\mathfrak{sl}_2$) we get

$$\exp(\mathrm{ad}(zH))X = (\exp z)^2 X,\ z \in \mathbb{C}. \tag{2.121}$$

which is equivalent to

$$\mathrm{Ad}_{\left(\begin{smallmatrix} t & 0 \\ 0 & t^{-1} \end{smallmatrix}\right)} X = t^2 X,\ t \in \mathbb{C}. \tag{2.122}$$

The last equation, can be seen as an equation in $\mathfrak{g}$. To do this integrate the Lie algebra morphism $\mathfrak{sl}_2 \to \mathfrak{a} \subset \mathfrak{g}$ to a Lie group homomorphism $\gamma : SL_2(\mathbb{C}) \to G$. Finally, restricts γ to the maximal torus $T = \{\left(\begin{smallmatrix} t & 0 \\ 0 & t^{-1} \end{smallmatrix}\right) : t \in \mathbb{C}\}$. Since γ_* is a Lie algebra morphism, we can do all the calculations made before in $\mathfrak{g}$. The last equation is writen as $\mathrm{Ad}_{\gamma(t)}X = t^2X$ provided we identify $T \cong \mathbb{C}^*$, that is we write $\gamma(t)$ for $\gamma\left(\begin{smallmatrix} t & 0 \\ 0 & t^{-1} \end{smallmatrix}\right)$.

We observe that $t^2\mathrm{Ad}_{\gamma(t^{-1})}X = X$ by construction. Moreover, if $zX \in \mathcal{O}_X$ for any $z \in X$ because $\exp(z\mathrm{ad}(H))X = \exp(z)^2 X$. So $t^2\mathrm{Ad}_{\gamma(t^{-1})}$ preserves $\mathcal{O}_X$ as well. Furthermore if $Z \in \mathfrak{g}^Y$, then

$$[Y, t^2\mathrm{Ad}_{\gamma(t^{-1})}Z] = \mathrm{Ad}_{\gamma(t^{-1})}[t^2\mathrm{Ad}_{\gamma(t)}Y, Z] = \mathrm{Ad}_{\gamma(t^{-1})}[Y, Z] = 0. \tag{2.123}$$

The last equation can be seen by exponentiating the relation $\mathrm{ad}(H)Y = -2Y$. In conclusion, we see that the one parameter group $\mathbb{C}^*$ preserves $\mathcal{O}_X \cap (X + \mathfrak{g}^Y)$ through the action $t^2\mathrm{Ad}_{\gamma(t^{-1})}$. Now, observe that $\mathfrak{g}^Y \subset \bigoplus_{i\leq 0} \mathfrak{g}_i$, $\mathfrak{g}_i = E_i^{\mathrm{ad}(H)}$. For $Z \in \mathfrak{g}_i$ we have $t^2\mathrm{Ad}_{\gamma(t^{-1})}Z = t^{2-i}Z$. These equations shows that we can send

$$\lim_{t\to 0} t^2\mathrm{Ad}_{\gamma(t^{-1})}(X+Z) = X + \lim_{t\to 0} t^2\mathrm{Ad}_{\gamma(t^{-1})}Z = X. \tag{2.124}$$

So, if there exist $Z \in \mathfrak{g}^Y \setminus 0$ such that $X+Z \in \mathcal{O}_X$, this will contradict the lemma (2.7.1), because the family $\{t^2\mathrm{Ad}_{\gamma(t^{-1})}(X+Z)\}_{t\to 0}$ will be arbitrarily close, but different to X. $\square$

Example 2.7.3. Let $\mathfrak{g} = \mathfrak{sl}_3$ with $X = E_{12}$. We have the $\mathfrak{sl}_2-$triple $\{E_{21}, H_{1,2}, E_{12}\}$ and $\mathfrak{sl}_3$ decomposes as $V_{(0)} \oplus V_{(1)}^{\oplus 2} \oplus V_{(2)}$ where

$$\begin{aligned} V_{(0)} &= \mathrm{Ker}(L_1 - L_2) = \mathbb{C}\mathrm{Diag}(1,1,-2) \\ V_{(1)} &= \mathbb{C}E_{23} \oplus \mathbb{C}E_{13} \\ V_{(1)} &= \mathbb{C}E_{31} \oplus \mathbb{C}E_{32} \\ V_{(2)} &= \mathbb{C}E_{21} \oplus \mathbb{C}H_{(1,2)} \oplus \mathbb{C}E_{12} \end{aligned} \tag{2.125}$$

Now we observe that

$$X + \mathfrak{g}^Y = \left\{ \begin{pmatrix} a & 1 & 0 \\ b & a & c \\ d & 0 & -2a \end{pmatrix} : a,b,c,d \in \mathbb{C} \right\}. \tag{2.126}$$

Now, let $W \in X + \mathfrak{g}^Y$, then $W \in \mathcal{O}_X$ only if it's conjugate to $X = E_{12}$. So W must be a rank 1 one matrix. This means that all order 2 minors must vanish. Therefore

$$\begin{aligned} \det \begin{pmatrix} a & 1 \\ b & a \end{pmatrix} &= a^2 - b = 0 \\ \det \begin{pmatrix} a & 0 \\ 0 & -2a \end{pmatrix} &= -2a^3 = 0 \\ \det \begin{pmatrix} 1 & 0 \\ a & c \end{pmatrix} &= c = 0 \\ \det \begin{pmatrix} a & 1 \\ d & 0 \end{pmatrix} &= d = 0 \end{aligned} \tag{2.127}$$

So $a = b = c = d = 0$ and $W = E_{12} = X$.

2.8 An isomorphism of Weyl groups

Notation for this section:

- $\mathfrak{g}$ a complex semisimple Lie algebra and $\mathfrak{g}_0$ its real form, i.e., $\mathfrak{g} = \mathfrak{g}_0 \otimes \mathbb{C}$.
- $\mathfrak{h} \subset \mathfrak{g}$ a Cartan subalgebra.
- G is the compact Lie group associated to the real form $\mathfrak{g}_0$.
- $T \subset G$ a maximal torus and $X^*(T) = \{T \overset{C^\infty}{\to} \mathbb{C}^*\}$ the set of its complex characters.
- $N(T)$ is the normalizer of T in G.
- We will write $W_T = N(T)/T$ and W for the Weyl group associated to the root system of $\mathfrak{g}$.

We have the following monomorphism:

$$X^*(T) \hookrightarrow V_{\mathbb{R}} = \mathrm{Hom}_{\mathbb{R}}(\mathrm{Lie}(T), i\mathbb{R}) : \alpha \mapsto d\alpha, \tag{2.128}$$

where we identify $\mathrm{Lie}(\mathbb{S}^1) = i\mathbb{R}$.

Remark 2.8.1. A maximal torus T, in a Lie group, is a maximal (with respect to inclusion) compact, connected, abelian subgroup of G. We can show that in this circunstance we have identifications:

$$\begin{aligned} &T = (\mathbb{R}/\mathbb{Z})^r \cong \underbrace{\mathbb{S}^1 \times \cdots \times \mathbb{S}^1}_{r \text{ times}}, \\ &\mathbb{Z}^r \overset{\cong}{\to} X^*(T) : (k_1, \ldots, k_r) \mapsto [(e^{i\theta_1}, \ldots, e^{i\theta_r}) \mapsto \prod_j e^{i\theta_j k_j}]. \end{aligned} \tag{2.129}$$

Later we will see that $r = \mathrm{Rank}(\mathfrak{g})$.

By the previous equation (2.128), we get an isomorphism

$$\psi : \mathbb{R} \otimes_{\mathbb{Z}} X^*(T) \overset{\cong_{\mathbb{R}}}{\longrightarrow} V_{\mathbb{R}} : \alpha \mapsto d\alpha. \tag{2.130}$$

Now we build the diagram, where the dashed line represents the passage to the "dual".

$$\begin{array}{ccc} T & \xleftarrow{\exp} & \mathrm{Lie}(T) \\ \vdots & & \vdots \\ \downarrow & & \downarrow \\ \mathbb{R} \otimes X^*(T) & \xrightarrow{\psi} & V_{\mathbb{R}} \end{array}$$

There is a natural action of W_T on this diagram. Let $w \in W_T$, then we have:

- Conjugation $c_w : T \to T : t \mapsto wtw^{-1}$
- The associated dual c_w^* on $X^*(T)$, which takes a character χ and gives $c_w^*(\chi) = \chi \circ c_{w^{-1}}$
- The adjoint action Ad_w on $\mathrm{Lie}(T)$

- The co-adjoint action Ad^*_w on $V_{\mathbb{R}}$

We know that $c_w(\exp(H)) = \exp(\mathrm{Ad}_w(H))$. In addition, we verify

$$d(c_w^*\alpha) = d(\alpha \circ c_w^{-1}) = d\alpha \circ \mathrm{Ad}_{w^{-1}} = \mathrm{Ad}^*_w(d\alpha) \tag{2.131}$$

which shows that ψ intertwines the actions of W_T on $V_{\mathbb{R}}$ and $\mathbb{R} \otimes X^*(T)$. Therefore, the maps on the previous diagram intertwines all the actions of W_T. Now we observe that $\mathrm{Lie}(T) \subset \mathfrak{g}_0$ is maximal abelian subalgebra. Moreover, in the complexficiation, we get that $\mathfrak{h} = \mathrm{Lie}(T)_{\mathbb{C}} \subset \mathfrak{g}$ is a maximal toral subalgebra (MTS), so it's a Cartan subalgebra.

Lets $\mathfrak{p}$ be a complementary subspace to $\mathrm{Lie}(T)$ in $\mathfrak{g}_0$. It's a fact that Ad_T restricts to $\mathfrak{p}$, so the decomposition $\mathfrak{g} = \mathfrak{h} \oplus \mathfrak{p}_{\mathbb{C}}$ is fixed by the complexfied action of Ad_T on $\mathfrak{g}$. Let

$$\mathfrak{p}_{\mathbb{C}} = \bigoplus_{\alpha \in \Phi} V_\alpha,\ \Phi = \{\alpha \in X^*(T) : V_\alpha \neq 0\} \tag{2.132}$$

be the corresponding decomposition into $T-$irreducible modules for its adjoint action. Now, consider the complexified map $\psi_{\mathbb{C}} : \mathbb{C} \otimes_{\mathbb{Z}} X^*(T) \to \mathrm{Hom}_{\mathbb{C}}(\mathfrak{h}, \mathbb{C}) = \mathfrak{h}^*$. We claim that $\psi_{\mathbb{C}}(\Phi) = \Delta$, where Δ is the set of roots induced by the Cartan subalgebra $\mathfrak{h}$. In fact, if $X_\alpha \in V_\alpha$ and $H \in \mathrm{Lie}(T)$, then

$$\mathrm{Ad}_{\exp(tH)}X_\alpha = \exp(\mathrm{ad}(tH))X_\alpha \overset{\frac{d}{dt}\big|_0}{\Longrightarrow} \mathrm{ad}(H)X_\alpha = d\alpha(H)X_\alpha \tag{2.133}$$

and $V_\alpha \subset \mathfrak{g}_{d\alpha}$. The right hand side of the last equation is true for $H \in \mathfrak{h}$ by linear extension. The converse inclusion we get by exponentiation of the adjoint action of $\mathfrak{h}$ on $\mathfrak{g}$. It follows that $V_\alpha = \mathfrak{g}_{d\alpha}$ and $\mathfrak{p}_{\mathbb{C}} = \bigoplus_{\alpha\in\Phi} \mathfrak{g}_{d\alpha}$. For simplicity of notation, we will write $\mathfrak{g}_\alpha$, instead of $\mathfrak{g}_{d\alpha}$.

Remember that if $f : \Delta \to \mathbb{R}^*$ is a linear functional, then we have an induced set of positive roots Δ_+. Now if $\alpha \in \Delta_+,\ X_\alpha \in \mathfrak{g}_\alpha,\ Y_\alpha \in \mathfrak{g}_{-\alpha}$ and $K(X_\alpha, Y_\alpha) = 2/(\alpha,\alpha)$, then

$$H_\alpha = [X_\alpha, Y_\alpha] = \frac{2}{(\alpha,\alpha)}\nu^{-1}(\alpha) \tag{2.134}$$

and we get a $\mathfrak{sl}_2-$triple $\mathfrak{sl}_\alpha = \mathbb{C}\langle Y_\alpha, H_\alpha, X_\alpha\rangle$. This defines morphisms $\mathfrak{sl}_2\mathbb{C} \hookrightarrow \mathfrak{g}$ and $di_\alpha : \mathfrak{su}(2) \hookrightarrow \mathfrak{g}_0$, of real forms. Since $SU(2)$ is simple connected, we have an associated map of Lie groups $i_\alpha : SU(2) \to G$.

Remark 2.8.2.

$$\begin{aligned} SU(2) &= \{X \in GL_2(\mathbb{C}) : A\overline{A}^T = I,\ \det(A) = 1\} \\ &= \left\{ \begin{pmatrix} a & b \\ -\overline{b} & \overline{a} \end{pmatrix} : |a|^2 + |b|^2 = 1 \right\} \\ \mathfrak{su}(2) &= \{X \in M_2(\mathbb{C}) : X + \overline{X}^T = 0,\ \mathrm{Tr}(X) = 0\} \\ &= \mathbb{R}\begin{pmatrix} & -1 \\ 1 & \end{pmatrix} \oplus \mathbb{R}\begin{pmatrix} i & \\ & -i \end{pmatrix} \oplus \mathbb{R}\begin{pmatrix} & i \\ i & \end{pmatrix} \end{aligned} \tag{2.135}$$

Notation 2.8.3. Write $T_\alpha = \mathrm{Ker}(\alpha) \subset T$ and $\mathfrak{su}_\alpha = di_\alpha(\mathfrak{su}(2))$.

Lemma 2.8.4. We have

1. $\mathrm{Ad}_w(H) = H$, $\forall H \in \mathrm{Lie}(T_\alpha) = \mathrm{Ker}(d\alpha)$ and $w \in i_\alpha(SU(2))$
2. $\mathrm{Ad}_{w_\alpha}(H_\alpha) = -H_\alpha$
3. $w_\alpha = i_\alpha(\begin{smallmatrix} & -1 \\ 1 & \end{smallmatrix}) \in N(T)$

Proof: Consider the exponential map $\mathfrak{su}(2) \twoheadrightarrow SU(2)$, then if $w \in i_\alpha(SU(2))$, we have $w = \exp(X)$ for some $X \in \mathfrak{su}_\alpha$. Therefore $\mathrm{Ad}_w(Y) = \exp(\mathrm{ad}(X))Y = Y$ provided we show that $[X, Y] = 0$. But this is the case, since $[\mathfrak{su}_\alpha, \mathrm{Ker}(d\alpha)] = 0$. This concludes the first assertion.

For the second assertion, we observe that $A_\alpha = X_\alpha - Y_\alpha = (\begin{smallmatrix} & 1 \\ -1 & \end{smallmatrix})$, and $\mathrm{Ad}_{w_\alpha} = \exp(\frac{\pi}{2}\mathrm{ad}(A_\alpha))$. Define $S_\alpha = X_\alpha + Y_\alpha$, then $\mathrm{ad}(X_\alpha)$ acts on $U = \mathbb{R}\langle H_\alpha, S_\alpha\rangle$ as the matrix $(\begin{smallmatrix} & 2 \\ -2 & \end{smallmatrix})$. Therefore

$$\mathrm{Ad}_{w_\alpha} = \exp(\frac{\pi}{2}\mathrm{ad}(A_\alpha))|_U = -\mathrm{Id}. \tag{2.136}$$

Finally, we observe that $\mathrm{Lie}(T) = \mathbb{R}iH_\alpha \oplus \mathrm{Ker}(d\alpha)$ because $d\alpha(H_\alpha) \neq 0$. Therefore

$$T = \langle \underbrace{\{e^{siH_\alpha}\}_{s\in\mathbb{R}}}_{S}, T_\alpha \rangle. \tag{2.137}$$

However

$$\begin{cases} \text{1th assertion } \Rightarrow c_w(e^H) = e^H,\ \forall H \in \mathrm{Lie}(T_\alpha) \\ \text{2th assetion } \Rightarrow c_{w_\alpha}(e^{siH_\alpha}) = e^{-siH_\alpha} \end{cases} \\ \Longrightarrow \begin{cases} w \in C_G(T_\alpha),\ \forall w \in i_\alpha(SU(2)) \\ w_\alpha \in N(S) \end{cases} \tag{2.138}$$

This proves that w_α normalizes T, that is $w_\alpha \in N(T)$. □

For the first assertion it follows that if $(d\alpha, d\beta) = 0$, then $\mathrm{Ad}^*_{w_\alpha} d\beta = d\beta$. In fact, if $(d\alpha, d\beta) = 0$, then $d\alpha(H_\beta) = 0$ and $H_\beta \in \mathrm{Lie}(T_\alpha)$. Therefore

$$\mathrm{Ad}_{w_\alpha}(H_\beta) = H_\beta \Rightarrow \mathrm{Ad}^*_{w_\alpha} d\beta = d\beta \tag{2.139}$$

follows by the definition of a coroot and the fact that the isomorphism $\nu : \mathfrak{h} \to \mathfrak{h}^*$, induced by the Killing form, intertwines the adjoint and coadjoint actions of G.

For the second assertion of the previous lemma we see that $\mathrm{Ad}^*_{w_\alpha} d\alpha = -d\alpha$. In fact, by applying the Killing form we see that

$$\mathrm{Ad}_{w_\alpha}(H_\alpha) = -H_\alpha \Rightarrow d\alpha(\mathrm{Ad}_{w_\alpha^{-1}} H) = -d\alpha(H),\ \forall H \in \mathfrak{h} \tag{2.140}$$

and the assertion follows.

In conclusion, the two previous observations shows that $\mathrm{Ad}^*_{w_\alpha} = r_\alpha$. In fact, for any $\lambda \in V_{\mathbb{R}}$, we have $\lambda - \frac{(\lambda,\alpha)}{(\alpha,\alpha)}\alpha \in \mathbb{R}\alpha^\perp$. Therefore

$$\mathrm{Ad}^*_{w_\alpha}(\lambda - \tfrac{(\lambda,\alpha)}{(\alpha,\alpha)}\alpha) = \lambda - \tfrac{(\lambda,\alpha)}{(\alpha,\alpha)}\alpha, \tag{2.141}$$

which proves that $\mathrm{Ad}^*_{w_\alpha}(\lambda) = \lambda - 2\frac{(\lambda,\alpha)}{(\alpha,\alpha)}\alpha$. Now let $W' = \langle w_\alpha : \alpha \in \Phi\rangle \subset W_T$, then we see that we get an isomorphism

$$W' \longrightarrow W : w_\alpha \mapsto \mathrm{Ad}^*_{w_\alpha}. \tag{2.142}$$

The next lemma will finnish the proof of the isomorphism.

Lemma 2.8.5. We have $W' = W_T$.

Proof: First we observe that

$$X_{c^*_w\alpha} = \mathrm{Ad}_w(X_\alpha) \neq 0, \text{ if } \alpha \in \Phi, w \in W_T. \tag{2.143}$$

So W_T acts on Φ. In particular, this proves that W_T is finite. So there exist $g_i \in N(T)$ $(i = 1, \ldots, k)$ such that $N(T) = \coprod_{i=1}^k g_i T$ and we see that the left cosets $g_i T$ are the connected components of $N(T)$. Moreover, since T is the connected component that contains the unity, we see that $T = N(T)^\circ$. Now define the Weyl vector $\rho = \frac{1}{2}\sum_{\alpha\in\Delta_+}\alpha$. It's a fact that it belongs to the positive Weyl chamber

$$\mathcal{C}^\circ_+ = \{\lambda \in V_{\mathbb{R}} : (\lambda,\alpha) > 0,\ \forall \alpha \in \Delta_+\}. \tag{2.144}$$

Therefore ρ is not fixed by $\mathrm{Ad}^*_{w_\alpha}$ by any $\alpha \in \Phi$. This shows that

$$H_0 = \nu^{-1}(\rho) = \frac{1}{2}\sum_{\alpha\in\Delta_+}\nu^{-1}(\alpha) \in \mathfrak{h}. \tag{2.145}$$

is not fixed by any Ad_{w_α}, $\alpha \in \Delta$.

Now let $\tilde{w} \in W_T$, we know that $\mathrm{Ad}_{\tilde{w}}(\Phi) = \Phi$, so there exist $w \in W'$ such that $\mathrm{Ad}^*_{w^{-1}}\mathrm{Ad}^*_{\tilde{w}} = \mathrm{Ad}^*_{w^{-1}\tilde{w}}$ permutes the elements of Δ_+. Write $n = w^{-1}\tilde{w}$, then we see by the previous equation that Ad_n fixes H_0. Therefore

$$\begin{aligned} &H_0 \subset \mathrm{Lie}(T) \setminus \bigcup_{\alpha\in\Delta}\mathrm{Lie}(T_\alpha) \\ &\quad\Longrightarrow S = \{e^{tH_0}\}_{t\in\mathbb{R}} \subset T \text{ contains a regular element } e^{H_0} \notin \bigcup_{\alpha\in\Delta} T_\alpha \end{aligned} \tag{2.146}$$

Then we see that $C_G(S) \subset N(T) \overset{C_G(S)\text{ connected}}{\Longrightarrow} C_G(S) \subset N(T)^0 = T$. However $c_n(e^{H_0}) = \exp(\mathrm{Ad}_n(H_0)) = e^{H_0}$ and $n \in C_G(S) \Rightarrow n \in T$. Therefore $\tilde{w} = w \in W_T$, i.e., $W_T = W'$. □

Chapter 3

Springer correspondence

I am following the third chapter of [1].

3.1 Springer Fibers

3.1.1 Variety of Borel subalgebras

As usual, $\mathfrak{g}$ is a finite dimensional complex semisimple Lie algebra and G its associated compact Lie group.

Notation 3.1.1. We write $\mathcal{B}$ for the set of all Borel subalgebras of a Lie algebra $\mathfrak{g}$. It's a subvariety of the Grassmanian $\mathrm{Gr}(\dim(\mathfrak{g}, \mathfrak{b}))$. In fact, the vanishing of the derived series

$$\mathfrak{b} \supset \mathfrak{b}^{(1)} = [\mathfrak{b}, \mathfrak{b}] \supset \cdots \supset \mathfrak{b}^{(N)} = 0 \tag{3.1}$$

is a polynomial condition.

- (Variety of Borel subalgebras). Let $\mathfrak{b} \subset \mathfrak{g}$ be a Borel subalgebra, i.e., a maximal solvable subalgebra of $\mathfrak{g}$. We have a bijection

  $$G/B \to \mathcal{B} : gB \mapsto \mathrm{Ad}_g(\mathfrak{b}) \tag{3.2}$$

  where $B = \mathrm{Stab}_G(\mathfrak{b})$ through the adjoint action. In fact, B is just the compact form of $\mathfrak{b}$, that is $\mathrm{Lie}(B) = \mathfrak{b}$.

- (Bruhat decomposition). Now write $\mathfrak{b}_0 = \mathrm{Lie}(B)$. By equation (3.2), we see that there exsit an unique gB such that $\mathrm{Ad}_g(\mathfrak{b}_0) = \mathfrak{b}$, so $\mathrm{Ad}_B(\mathfrak{b}) = \mathrm{Ad}_{BgB}(\mathfrak{b}_0)$. This gives a surjection:

  $$B\backslash G/B \to \{B - \text{orbits on } \mathcal{B}\} : BgB \mapsto \mathrm{Ad}_{BgB}(\mathfrak{b}_0). \tag{3.3}$$

  Remark 3.1.2. If G is a group and H is a subgroup, then we have a bijection

  $$\phi : \{H - \text{orbits on } G/H\} \overset{\cong_{\text{set}}}{\to} \left\{ \begin{matrix} G - \text{diagonal orbits} \\ \text{on } G/H \times G/H \end{matrix} \right\} : [gH] \mapsto [(gH, H)] \tag{3.4}$$

Furthermore, we have the Bruhat decomposition, which states that

$$W \to B\backslash G/B : w \mapsto BwB \tag{3.5}$$

is a bijection. Here we are viweing W_T in the sense of section (2.8).

Example 3.1.3. If $G = GL_n$, then $B = \begin{pmatrix} a & b \\ 0 & d \end{pmatrix}$. Now, if $c \neq 0$, then

$$\begin{pmatrix} a & b & 1 & 0 \\ c & d & 0 & 1 \end{pmatrix} \sim \begin{pmatrix} c & d & 0 & b - dac^{-1} \\ 0 & 1 & 1 & -ac^{-1} \end{pmatrix} \tag{3.6}$$

by row reduction. So

$$\begin{aligned} \begin{pmatrix} a & b \\ c & d \end{pmatrix} &= \begin{pmatrix} 0 & 1 \\ 1 & -ac^{-1} \end{pmatrix}^{-1} \begin{pmatrix} c & d \\ 0 & b - dac^{-1} \end{pmatrix} \\ &= \begin{pmatrix} 1 & ac^{-1} \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} c & d \\ 0 & b - dac^{-1} \end{pmatrix}. \end{aligned} \tag{3.7}$$

So

$$B \begin{pmatrix} a & b \\ c & d \end{pmatrix} B = BsB, \; s = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \tag{3.8}$$

and $|B\backslash G/B| = 2$.

Theorem 3.1.4. All of the above maps are bijections:

$$W \to B\backslash G/B \to \{B - \text{orbits on } \mathcal{B}\} \to \left\{ \begin{matrix} G - \text{diagonal orbits} \\ \text{on } G/H \times G/H \end{matrix} \right\} \tag{3.9}$$

Remark 3.1.5. Aparently, show that the composition of the first two maps

$$W \to \{B - \text{orbits on } \mathcal{B}\} \tag{3.10}$$

is a bijection, is highly nontrivial (see [1]).

We want to describe the Variety of Borel subalgebras of $\mathfrak{g} = \mathfrak{sl}_n$.

Lemma 3.1.6. We have a bijection

$$\begin{aligned} &\mathcal{F} = \{(F_0, F_1, \ldots, F_n) : F_i \subset F_{i+1}, \dim(F_i) = i\} \to \mathcal{B} \\ &F \mapsto \mathfrak{b}_F = \{x \in \mathfrak{g} : x(F_i) \subset F_i\} \end{aligned} \tag{3.11}$$

Here we have $F_i \subset \mathbb{C}^n$. For $F = (F_0, F_1, \ldots, F_n), g \in G$, we will write $gF = (gF_0, \ldots, gF_n)$.

Proof: We need to show that $\mathfrak{b}_F$ is a Borel subalgebra. We can choose $A \in GL_n$ such that $A(\mathbb{C}^i) = F_i$, $\forall i$, where $\mathbb{C}^i = \langle e_1, \ldots, e_i \rangle$. To see this, write $F_i = \langle v_1, \ldots, v_i \rangle$, $\forall i$. If $v_i = \sum_j a_{ji} e_j$, then $Ae_j = v_j$. Therefore, we get $A(0, \mathbb{C}, \ldots, \mathbb{C}^n) = F$, the standard flag. Observe that since $AxA^{-1} = (A/\sqrt[n]{\det(A)})x(A/\sqrt[n]{\det(A)})^{-1}$, we can assume $A \in SL_n$. Of course $\mathfrak{b}_{(0,\mathbb{C},\ldots,\mathbb{C}^n)} = \mathfrak{b}_0$, the standard Borel subalgebra of upper triangular matrices. So $\mathfrak{b} = \mathrm{Ad}_A(\mathfrak{b}_0)$ is a Borel subalgebra. Finally, sujectivity follows from Lie's theorem (Lie algebra course). $\square$

TODO: prove that this bijection is in fact, an isomorphism of varieties.

Bibliography

[1] Neil Chriss and Victor Ginzburg. *Representation theory and complex geometry*. Vol. 42. Springer, 1997.

[2] David H Collingwood and William M McGovern. *Nilpotent orbits in semisimple Lie algebra: an introduction*. Routledge, 2017.